BEYOND INDIVIDUAL CLAIMS: PROBABILISTIC LOGICAL COHERENCE FOR LONG-FORM LLM EVALUATION

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ABSTRACT

Long-form factuality assessment decomposes a response into atomic claims and verifies each against retrieved evidence. The paradigm is effective and now standard, but it embeds an assumption — that the truth of each claim can be settled in isolation — which discards the structure that makes prose an argument. A response can therefore pass per-claim verification while contradicting itself, and two responses built from *identical* claims can differ in coherence through the order in which they assert them. We formalize *logical coherence* as a property of the joint distribution over claim truth values induced by a Markov random field whose pairwise factors are the relations the response itself draws. The relation vocabulary is a two-level taxonomy: twelve interpretable discourse senses, grounded in PDTB 3.0 and RST, compile through a deterministic map onto five inferential couplings, each fixing a factor table by naming the joint assignment it penalizes. We characterize the tables coherence requires by a row-stochasticity property and show that those used for claim–evidence factuality are wrong here, collapsing the measure's dynamic range; claim priors come from an existing factuality pipeline, making coherence and factuality two stages over one decomposition. On a corpus of response families whose coherence ordering is fixed by construction, the measure recovers 82.7% of declared orderings from gold relation graphs and 34.7–37.1% from graphs mined out of the response text — and since the two conditions differ only in the relation graph, that gap localizes the open problem in relation extraction rather than in the probabilistic model. Against five families of evidence-free baselines it is the only measure that both orders responses correctly ($49/50$ for the headline score, every satisfiable assertion, against the best baseline's $30/50$) and stays exactly invariant under meaning-preserving edits ($20/20$), the latter *by construction*: reordering sentences leaves the Markov network identical, a guarantee no surface statistic can offer and one that an LLM judge violates while swinging 42 points between judge models on identical prose. The score also decomposes into per-claim marginals, so an incoherence is located rather than merely detected.

1 INTRODUCTION

The dominant paradigm for assessing the factuality of long-form generation is decompose-then-verify: split a response into atomic claims, check each against retrieved evidence, and aggregate the verdicts (Min et al., 2023; Wei et al., 2024; Song et al., 2024). It is effective, it is now standard, and it embeds an assumption. Treating each claim as independently checkable discards precisely the structure that distinguishes an *argument* from a *list*.

Two failure modes make this concrete. A report may assert both that no passengers were harmed and that a wire service reported three deaths; per-claim verification reports one claim true and one false, but not the more basic defect — that the response contradicts *itself*. Separately, the inferential links a response draws may fail even when their endpoints are individually true: a chain of accurate statements joined by a spurious "therefore" is not a sound argument, and every atom in it verifies. Xie et al. (2026) quantify how often this matters: on logic-related samples, SAFE, VeriScore and FactCheck-GPT omit the logical dimension on 60%, 78% and 48% of cases respectively. The two axes are orthogonal — a response can be a bag of true facts that is incoherent, or internally impeccable while entirely false — and our sharpest fixture is a pair of summaries of one appellate judgment built from the *same* eighteen claims, one a permutation of the other, differing only in whether self-serving

assertions precede or follow the findings that qualify them: every per-atom check returns an identical verdict, yet our measure separates them (0.95 against 0.90).

A motivating example. Figure 1 shows the failure in the smallest instance we can enumerate exactly, and it is worth reading in full because every later section is an answer to something visible in it. Five claims about Voyager 1 are fixed once and for all: three true of the world and two false. *Both* responses assert all five, so the two are indistinguishable to any method that scores claims one at a time — every retrieval, every NLI check, every per-atom verdict is byte-identical across the pair, and a precision-style aggregate returns $3/5$ for both. What differs is the argument the response builds out of them. Response A names the two falsehoods, attributes them, and refuses each on a stated ground; Response B asserts them as fact, offers the false a_4 ("the plasma wave instrument had failed") as the *reason* the true a_3 ("therefore in interstellar space") holds, and in the same breath denies a_3 outright — then denies a_4 and a_5 in its closing sentence. Read as prose, B is fluent; read as an argument, it is incoherent, and its incoherence is not located in any one claim but in the wiring between them.

Two changes in the relation graph carry the entire difference (callouts **1** and **2** in the figure): the support edge $a_2 \to a_3$ becomes a CONTRADICTION, and the edge between a_3 and a_4 *reverses* — what A uses to reject a falsehood, B uses to lean on one. Our measure scores A at $\mathrm{LCS}_{\mathrm{mm}} = 0.590$ against B's 0.494 (exact, by 2^5 enumeration), and because the priors enter only through unary factors that are identical in the two panels, the whole drop is attributable to arrangement alone. Mentioning a falsehood is not what costs B its score; asserting it and reasoning from it is. The score also localizes the damage, which an aggregate cannot: the *true* claim a_3 finishes at 0.513 under B against 0.992 under A, dragged below its own 0.9 prior because its only support now arrives from a claim the response itself denies — and a premise the text refuses cannot transmit support. A claim true of the world is demoted to a coin flip by the argument built around it. §3.7 works this pair formally once the model is in place, and App. G gives both responses verbatim alongside three further examples.

We therefore ask: how much of what a response asserts does its argument, taken as a whole, uphold? That is answered by a *marginal* in a graphical model whose variables are claim truth values and whose factors are the asserted relations — not by counting violated rules, which cannot propagate a conflict from its endpoints to what they support, and which yields a measure a response maximizes by asserting no relations at all (Prop. 3).

Our contributions are as follows. (i) A **coherence MRF and a formal two-level relation taxonomy** (§3.1–§3.2): twelve discourse senses compile through a deterministic map onto five inferential couplings, each defined by the joint assignment it penalizes, which fixes its factor table — separating the interpretable layer from the inferential one lets the annotation vocabulary be rich while the model stays a closed three-parameter pairwise family. (ii) A **characterization of why factuality's factor tables are wrong for coherence** (§3.3): they charge a premise for being a premise, and the alternative collapses the measure's dynamic range by more than half. (iii) A **selected readout and a two-stage composition with factuality** (§3.4, §3.6): of four candidates evaluated by exact inference against five criteria, only two are monotone across a coherence-ordered family and none on every network, so we select a *balanced* score that rewards support and penalizes conflict against the priors alone and is provably monotone in every relation edit (Thm. 2); on gold relation graphs it satisfies every satisfiable ladder assertion. (iv) **An evaluation separating relation-extraction error from scoring error** (§5), by holding the readouts byte-identical and swapping only the relation graph: the measure satisfies 82.7% of 202 declared constraints from gold graphs against 34.7–37.1% from mined ones, localizing the open problem in extraction; and against five families of evidence-free baselines it is the only one that both orders responses correctly ($49/50$ for the headline score, against the best baseline's $30/50$) and is exactly invariant under meaning-preserving edits ($20/20$) — the latter *by construction*, a guarantee no surface statistic can offer.

2 Related Work

Atomic factuality. FActScore (Min et al., 2023) introduced per-atom precision against a knowledge source; SAFE (Wei et al., 2024) added search-augmented verification and an $F_1@K$ metric, VeriScore (Song et al., 2024) restricted attention to *verifiable* claims, and FLEEK (Bayat et al., 2023) retrieves evidence to detect and correct errors. All aggregate independent per-claim verdicts and are therefore blind to inter-claim structure by construction. FactReasoner (Marinescu et al., 2025) replaced

Query *Where is Voyager 1 now, and has it left the Solar System?*

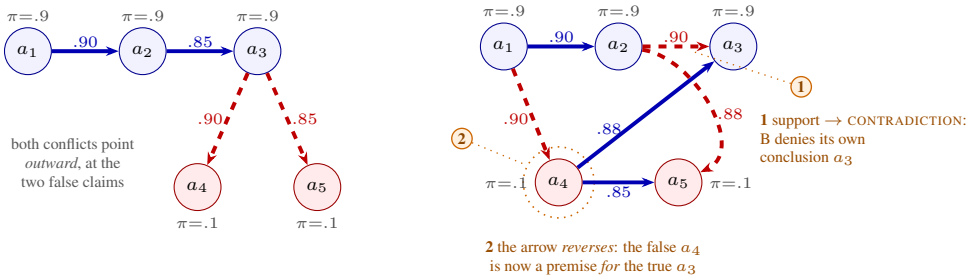
Response A *coherent*

Voyager 1 crossed the heliopause in August 2012, $[a_1]$ and in the days that followed its instruments measured a sharp rise in plasma density $[a_2]$ — *the measurement that established the boundary had been passed.* Voyager 1 is therefore in interstellar space. $[a_3]$ Two persistent misconceptions are worth clearing up. The first is that the plasma wave instrument had failed permanently before the crossing; $[a_4]$ *it cannot have, since it is precisely that instrument which returned the density profile the conclusion rests on.* The second is that Voyager 1 has left the Solar System altogether. $[a_5]$ *Being in interstellar space is not the same thing: the Sun's gravitational reach extends through the Oort cloud.*

Response B *incoherent*

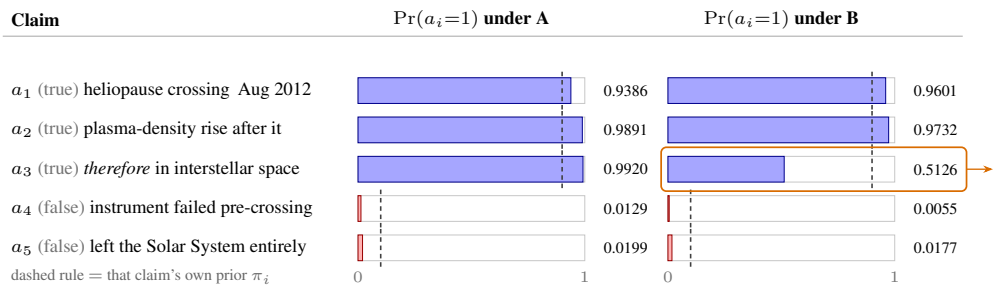
Voyager 1 crossed the heliopause in August 2012, $[a_1]$ and after the crossing its instruments measured a sharp rise in plasma density. $[a_2]$ *That said, the probe cannot really be counted as being in interstellar space.* $[a_3]$ The plasma wave instrument had **in fact** failed permanently before the crossing, $[a_4]$ *and it is precisely that failure which tells us the probe reached interstellar space. The instrument failure also means* Voyager 1 has left the Solar System entirely. $[a_5]$ *Of course, the crossing rules out any such failure, and the density measurement is inconsistent with the probe having left the Solar System.*

The same five claims $a_1, \ldots, a_5$ ($[a_i]$ marks each first mention) ⇒ **identical per-claim factuality verdicts.** Only the italicized *relation-bearing* spans differ.



A: a support chain over the true claims; each false claim *attributed, then refused* from a_3.

B: the same five claims, *rearranged*. Half its conflicts point *inward*, at what it asserts.



The diagnostic. Although a_3 is **true**, B lowers its probability from 0.992 to near chance (0.513), well below its own 0.9 prior because its only remaining support comes from a_4, a claim the contradicted by the response itself. This failure is invisible to aggregate scores but immediately exposed by per-claim marginals.

Coherence score	$\mathrm{LCS_{mm}}$: 0.590 (A) ⟶ 0.494 (B)	**Δ = −0.097**
exact, by 2^5=32-world enum.	$\mathrm{LCS_{cn}}$: 0.963 → 0.414 • log Z: −0.95 → −3.78 • prior-only baseline 0.580 (A *above* it, B *below*)	

The priors enter *only* through the unary factors, which are identical in the two responses. The entire 0.590 → 0.494 drop is produced by the pairwise relation factors — that is, by **arrangement alone**.

Figure 1: **Two responses, the same five claims, different coherence.** Five claims about Voyager 1 are fixed once and for all: a_1, a_2, a_3 true of the world (priors π_i=0.9, blue) and a_4, a_5 false (π_i=0.1, red). Both responses answer the *same* query and assert *all five* claims, so every per-claim factuality check returns the same verdict on both and no decompose-then-verify score can separate them. **A** attributes the two falsehoods and then refuses each on a stated ground; **B** asserts them and reasons *from* them, offering the false a_4 as the reason the true a_3 holds while denying a_3 outright — then denying a_4 and a_5 in its closing sentence. Middle: the relation graphs those spans induce (solid blue ENTAILMENT, dashed red CONTRADICTION, line width growing with the mined weight p), drawn at one shared scale; the two numbered orange callouts **1** and **2** mark the only two changes, which carry the whole difference. Bottom: the posterior marginals of the coherence MRF, and the score. Mentioning a falsehood is not what costs B its score; asserting it and reasoning from it is. §3.7 works this pair formally and App. G gives both responses in full.

verdict counting with probabilistic inference over a factor graph of claim and evidence variables with pairwise factors from NLI relations; it is the closest antecedent — we reuse its decomposition, relation extractor and inference stack, and take its marginals as our priors (§3.6) — but its factors run between a claim and the *evidence* retrieved for it, so its graph has no claim–claim edges and its marginals cannot express internal consistency. Xie et al. (2026) states the gap directly, classifying inter-atom dependencies as a labelling task over claim pairs; our departure is to treat them as *factors in a joint distribution*, which lets a local conflict propagate to distant claims and makes the score a functional of the whole graph rather than a per-edge average.

Discourse coherence. Entity-grid models (Barzilay & Lapata, 2008) and their neural successors (Li & Jurafsky, 2017) score coherence from entity-mention distributions, evaluated against *synthetic perturbations* — a methodology we extend in §5.1 by making the perturbation an operator on a declared relation graph rather than on raw text, so its intended effect is known per item. DiscoScore (Zhao et al., 2023) uses BERT-based discourse representations. What these share is that coherence is read off surface statistics or learned representations: none exposes an inspectable relation graph, and none yields per-claim attributions. Our inventory takes its senses from PDTB 3.0 (Prasad et al., 2019) and RST (Mann & Thompson, 1988) but treats them as an interpretable layer compiling to a smaller inferential vocabulary. Closest to our mining stage is Liu & Strube (2025), who label sentence pairs with PDTB 3.0 senses via a discourse parser (App. F).

Consistency checking, judges, and relational models. SelfCheckGPT (Manakul et al., 2023) detects hallucination by pairwise NLI over sampled responses — the natural ablation isolating what global inference adds — and ROSCOE (Golovneva et al., 2023) scores reasoning chains with a self-consistency term. G-Eval (Liu et al., 2023) is the practical incumbent, an LLM judge scoring coherence from a rubric; judges are strong on aggregate correlation and weak on per-claim attribution and on any guarantee that a meaning-preserving edit cannot change the score. Steen & Markert (2022) argue coherence measures must be checked against system-level confounders, which we adopt in our ladder-blind controls. Markov logic (Richardson & Domingos, 2006) is the natural alternative encoding; App. I shows the obvious one-clause-per-relation encoding is *not* equivalent to ours and is degenerate at its mode, while a three-clause expansion reproduces it exactly on the three-coupling fragment. We keep the pairwise MRF because its factor family is closed, and the Markov logic view because the beyond-pairwise rules addressing our main limitation live there.

3 Probabilistic Logical Coherence

This section builds the measure in six steps. We define the probabilistic object — a Markov random field over claim truth values whose factors are the relations a response draws (§3.1) — then fill in the two things that definition leaves open: *which* relations the vocabulary admits (§3.2) and *what factor table* each contributes (§3.3). With the model fixed, §3.4 reads a score off it, §3.5 estimates the relations from text, §3.6 couples coherence to factuality through the free prior slot, and §3.7 works a five-claim pair end to end. Each step closes a choice the previous one left open, and the two that matter most — the coupling vocabulary and the factor tables — are exactly where a coherence model diverges from a factuality one.

3.1 The coherence Markov random field

Definition 1 (Coherence MRF). *Let $\mathcal{A} = \langle a_1, \ldots, a_n \rangle$ be the atomic claims of a response in assertion order, each binary with $a_i = 1$ read as "the claim holds". Let $\mathcal{R} = \{(s_r, t_r, \tau_r, p_r)\}_{r=1}^{|\mathcal{R}|}$ be a set of* relations*, where $s_r \neq t_r$ index claims, τ_r is an inferential coupling (Def. 2) and $p_r \in [0,1]$ is the relation's weight. Given priors $\pi \in [0,1]^n$, the coherence MRF induced by $(\mathcal{A}, \mathcal{R}, \pi)$ is*

$$\mathbb{P}(\mathbf{a}) \;=\; \frac{1}{Z} \prod_{i=1}^{n} \phi_i(a_i) \prod_{r \in \mathcal{R}} \psi_{\tau_r, p_r}(a_{s_r}, a_{t_r}), \qquad Z \;=\; \sum_{\mathbf{a} \in \{0,1\}^n} \prod_i \phi_i(a_i) \prod_r \psi_r, \tag{1}$$

where $\phi_i(a_i) = [1 - \pi_i,\ \pi_i]$ over the states $(0, 1)$ and $\psi_{\tau,p}$ is given by §3.3.

Three structural properties are deliberate commitments, each load-bearing later.

Remark 1 (The factor family is closed). *Every $\psi_{\tau,p}$ is determined by p, the family's fixed source-prior parameter π_s and the penalized set, so the whole inferential vocabulary is a three-parameter pairwise family — a guarantee about what the model* cannot *do: no relation type, however elaborately named at the discourse level, can smuggle in expressiveness beyond a pairwise potential.*

Remark 2 (Order-sensitivity enters only through the relations). *Nothing in Eq. 1 references claim position; assertion order enters exclusively through* which *pairs are related and in* which *direction. An edit preserving the relation set — reordering claims whose relations are symmetric, or exchanging a* Precedence *for a* Succession*, which compiles to no factor — therefore provably cannot change any readout, a falsifiable prediction the invariance ladders of §5.1 test.*

Remark 3 (Coherence is evaluated without evidence). *The graph contains claim variables only, and with uniform priors $\pi_i \equiv \frac{1}{2}$ the model consults no external knowledge whatever: it scores how the response hangs together, not whether it is true, which is what makes the two axes separable.*

3.2 Two levels: inferential couplings and discourse senses

Definition 2 (Inferential couplings). *A coupling τ is specified by the set $V(\tau) \subseteq \{0,1\}^2$ of joint assignments (a_s, a_t) it* penalizes *— the configurations at which the relation is violated:*

Coupling	*Reading*	*Directed*	$V(\tau)$
ENTAILMENT	$a_s \Rightarrow a_t$	*yes*	$\{(1,0)\}$
CONTRADICTION	$a_s \Rightarrow \neg a_t$	*yes*	$\{(1,1)\}$
EQUIVALENCE	$a_s \Leftrightarrow a_t$	*no*	$\{(0,1),(1,0)\}$
EXCLUSIVE	*exactly one of a_s, a_t holds*	*no*	$\{(0,0),(1,1)\}$
CO_NECESSITY	*at least one of a_s, a_t holds*	*no*	$\{(0,0)\}$
NONE	*no coupling*	—	$\varnothing$ *(no factor)*

Write $\mathcal{T}$ for the five edge-producing couplings and $\mathcal{T}_{\text{conf}} = \{\text{CONTRADICTION}, \text{EXCLUSIVE}\}$ for the conflict *couplings, whose penalized set contains the both-true world.*

The first three are exactly the NLI edge types of an existing factuality pipeline (App. B), which lets a mined relation reuse the existing graph machinery unchanged. The last two are additions required by argumentative prose: CONTRADICTION forbids only the both-true world, whereas *exhaustive* alternatives also forbid the both-false one — "no one was harmed" and "three people died" can neither both hold nor both fail, so treating them as mere opposition leaves a world in which *neither* is true and under-penalizes — and CO_NECESSITY is its dual (App. C).

Couplings are the right vocabulary for a factor table and the wrong one for annotation, since what is observable in the text is a discourse fact rather than a logical one. We therefore mine an interpretable layer and compile it down through a total, deterministic map $\mathcal{C}$ assigning each sense a coupling, a strength prior ($\perp$ meaning "use the estimator's own value") and flags:

Level-2 sense	$\mathcal{C}$ coupling	prior	note
Cause-Effect*, Effect-Cause*, Evidence*	ENTAILMENT	$\perp$	directed
Condition, Instantiation	ENTAILMENT	$\perp$	directed; inadmissible (§3.5)
Restatement*	EQUIVALENCE	0.90	definitional identity
Contrast*	CONTRADICTION	$\perp$	opposed, *not* exhaustive
Concession*	CONTRADICTION	$\perp$	a tension the text may resolve
Alternative*	EXCLUSIVE	$\perp$	exhaustive: exactly one holds
Disjunction*	CO_NECESSITY	$\perp$	at least one holds
Precedence*, Succession	NONE	—	**ordering-only: no factor**

Five senses collapse onto ENTAILMENT, a deliberate loss: a pairwise potential over truth values cannot distinguish causation from evidence. What survives compilation is the inferential force, the sense being retained for interpretability. The two ordering-only senses draw a relation with no truth-functional content — exactly what makes an order-only edit provably score-invariant (Rem. 2).

A Concession asserts a tension and then overrides it, and treating that as a live contradiction would penalize a response for doing what careful argument requires. When a resolving *holding* claim h is

present we discount the conflict's weight,

$$p_r^{\text{eff}} \;=\; p_r\,\big(1-\lambda\,\pi_h\big), \tag{2}$$

with $\lambda = 0.45$; absent a resolver the tension counts at full strength. Holdings are found by adjudicative cue phrases on a non-endpoint claim maximizing lexical overlap with the endpoints — a heuristic which, with λ, is the one place hand-set constants enter.

3.3 The pairwise potentials, and why factuality's tables are wrong here

A coupling names the world it penalizes; it does not say by how much. Filling that in completes Def. 1, and it is where a coherence model parts company with the factuality model it otherwise reuses.

The reason is that the two settings put different objects at the *source* end of an edge. In factuality the source is a retrieved passage, and its own reliability is exactly what is in doubt, so a factor that revises belief in the source is doing its job. In coherence the source is another claim of the same response. We are not asking whether that claim is reliable; we are asking what the response's own inferences do with it. A table built for the first job does the wrong thing in the second.

The tables. Each $\psi_{\tau,p}$ is a 2×2 table, written here as a list of four numbers in the order $(0,0),(0,1),(1,0),(1,1)$, with the penalized set $V(\tau)$ in bold. Read it as two rows: the first two numbers are the case where the source is false, the last two the case where it is true. The constant π_s is a *fixed parameter of the factor family*, set by the edge's link type rather than read off either endpoint: $\pi_s = \frac{1}{2}$ on the claim–claim edges coherence uses (App. B). It is emphatically *not* the source claim's own prior — a source at $\pi_i = 0.9$ still contributes the row $[1-\pi_s, \pi_s] = [\frac{1}{2}, \frac{1}{2}]$ — and it is independent of the unary priors π_i of Def. 1, which remain free. At $\pi_s = \frac{1}{2}$ that row is *flat*, hence a constant multiplier that cancels under normalization; this is what Prop. 1 means when it says a false source returns the target to its prior, and the target's *own* prior is what it returns to.

ENTAILMENT	CONTRADICTION	CO_NECESSITY	EQUIVALENCE	EXCLUSIVE
$[1-\pi_s, \pi_s, \mathbf{1-p}, p]$	$[1-\pi_s, \pi_s, p, \mathbf{1-p}]$	$[\mathbf{1-p}, \pi_s, \pi_s, p]$	$[p, \mathbf{1-p}, \mathbf{1-p}, p]$	$[\mathbf{1-p}, p, p, \mathbf{1-p}]$

Proposition 1 (Row-stochasticity of the asymmetric couplings)**.** *For* $\tau \in \{\text{ENTAILMENT}, \text{CONTRADICTION}\}$ *every row of* $\psi_{\tau,p}$ *sums to one, so the row indexed by* a_s *is the conditional* $\mathbb{P}(a_t \mid a_s)$*:* $\mathbb{P}(a_t{=}1 \mid a_s{=}0) = \pi_s$*, and* $\mathbb{P}(a_t{=}1 \mid a_s{=}1) = p$ *for* ENTAILMENT *and* $1-p$ *for* CONTRADICTION*. Such a factor therefore constrains its target only when the source is true; when the source is false it contributes the constant row* $[1-\pi_s, \pi_s]$*, which at the* $\pi_s = \frac{1}{2}$ *coherence uses is flat and so cancels under normalization, handing the target back to its* own *prior. This fails for the symmetric couplings, and is not claimed of them: the rows of* CO_NECESSITY $= [1{-}p, \pi_s, \pi_s, p]$ *sum to* $(1-p)+\pi_s$ *and* $\pi_s + p$*, which are both one only in the degenerate case* $p = \pi_s = \frac{1}{2}$*.*

Proof. The $a_s{=}0$ row is $[1-\pi_s, \pi_s]$ in both tables, and the $a_s{=}1$ row is $[1-p, p]$ for ENTAILMENT and $[p, 1-p]$ for CONTRADICTION; each sums to one, so dividing by the row sum is the identity. For CO_NECESSITY, row-stochasticity needs $\pi_s = p$ and $\pi_s = 1-p$ at once, hence $p = 1-p$. □

One caution about what the proposition does not say. It is a property of the *table*, not of the marginals. Each row being normalized means $\psi(a_s, \cdot)$ can be read directly as $\mathbb{P}(a_t \mid a_s)$, and the flat $a_s{=}0$ row is the formal content of "an implication says nothing when its antecedent fails". It does not mean the source stays put: once all the factors are multiplied together and the result renormalized, both endpoints can move.

At $p = 0.9$ and $\pi_s = 0.5$ the two asymmetric tables are ENTAILMENT $= [.5, .5, .1, .9]$ and CONTRADICTION $= [.5, .5, .9, .1]$. Both rows sum to one, and reading the rows off gives $\mathbb{P}(a_t{=}1 \mid a_s{=}0) = 0.5 = \pi_s$ in each case, with $\mathbb{P}(a_t{=}1 \mid a_s{=}1)$ equal to $0.9 = p$ for ENTAILMENT and $0.1 = 1-p$ for CONTRADICTION.

The symmetric couplings behave differently, and CO_NECESSITY shows why the proposition is stated only for $\{\text{ENTAILMENT}, \text{CONTRADICTION}\}$. Its table $[.1, .5, .5, .9]$ has rows summing to 0.6 and 1.4, so neither is a conditional. Its *informative* row is the one where the source is false (0.833, against

0.643 where the source is true) — a disjunction constrains its target when the source *fails*, which no source-conditioned normalization can express. Example 1 works the joint through: both endpoints rise together, $0.5 \to 0.7$, because the factor is symmetric.

Where the factuality tables differ. The factuality literature uses a *no-priors* family whose ENTAILMENT table is $[p, p, \mathbf{1} - \mathbf{p}, p]$. The whole disagreement sits in one row. Both families treat a *true* source the same way, charging the world where it leaves its target false. They differ when the source is *false*: our row is $[\frac{1}{2}, \frac{1}{2}]$, flat and uninformative, while theirs is $[p, p]$ — also flat, but scaled by p. That extra factor of p is the problem. It rewards worlds in which the source holds, which turns the factor into evidence *for* the source instead of a constraint acting *from* it.

What that costs, in one edge. Strip everything else away and the whole effect can be computed by hand. Take a single ENTAILMENT edge $a_1 \to a_2$ with $p = 0.90$, over two claims that both start at prior $\pi_1 = \pi_2 = 0.9$, and nothing else in the graph. Two distinct quantities meet here and must not be conflated: the claims' unary priors, $\pi_1 = \pi_2 = 0.9$, which enter through $\phi_i = [0.1, 0.9]$; and the factor family's source-prior parameter, $\pi_s = \frac{1}{2}$, which enters through ψ alone. The source's being at 0.9 does *not* put 0.9 into the table — the with-priors $(0, \cdot)$ row is $[1 - \pi_s, \pi_s] = [0.5, 0.5]$ regardless. Eq. 1 then has four worlds, each weighted by two unary factors and one pairwise factor, $w(a_1, a_2) = \phi_1(a_1)\,\phi_2(a_2)\,\psi(a_1, a_2)$:

(a_1, a_2)	$\phi_1\phi_2$ (from π_i=0.9)	ψ (with / no priors) (from $\pi_s = \frac{1}{2}$, p)	w with	w no
$(0,0)$	$0.1 \times 0.1 = 0.01$	$1 - \pi_s = 0.5$ / $p = 0.9$	0.005	0.009
$(0,1)$	$0.1 \times 0.9 = 0.09$	$\pi_s = 0.5$ / $p = 0.9$	0.045	0.081
$(1,0)$	$0.9 \times 0.1 = 0.09$	$\mathbf{1} - \mathbf{p} = 0.1$ (both)	0.009	0.009
$(1,1)$	$0.9 \times 0.9 = 0.81$	$p = 0.9$ (both)	0.729	0.729
$Z = \sum w$			0.788	0.828

The source marginal is $\mathbb{P}(a_1{=}1) = \big(w(1,0) + w(1,1)\big)/Z$. Its *numerator is* $0.009 + 0.729 = 0.738$ *in both families* — the two rows where the source is true are identical, as observed above. Only Z differs, because the no-priors table puts more weight on the two worlds where the source is *false*: $0.005 \to 0.009$ and $0.045 \to 0.081$. So

$$\mathbb{P}(a_1{=}1) \;=\; \frac{0.738}{0.788} \;=\; \frac{369}{394} \;=\; 0.9365 \qquad \text{against} \qquad \frac{0.738}{0.828} \;=\; \frac{41}{46} \;=\; 0.8913.$$

Our tables lift the source above its 0.9 prior: its target is probably true, and an entailment whose target holds is mild corroboration. The no-priors tables push the same source *below* its prior — with nothing contradicting it and every relation touching it satisfied — and the derivation shows exactly why. Nothing was charged against a_1; the source-false worlds were simply made more attractive, which is the same thing after normalization. The claim loses credence for one reason only: it was cited as a premise. In factuality that is right, since staking a claim on a passage should move belief in the passage. In coherence it is a defect, because being the thing an argument reasons *from* should not by itself make a claim less credible.

What that costs on the running example. One edge is a toy; the effect survives at scale. Return to the five-claim Voyager 1 pair of Figure 1, worked formally in §3.7. Everything is held fixed — the same five claims, the same priors $(0.9, 0.9, 0.9, 0.1, 0.1)$, the same relation graphs (four edges for A, six for B) — and only the potential family is swapped. Each score is the mean of the five posterior marginals, computed by enumerating all $2^5 = 32$ worlds of Eq. 1 exactly, so the comparison isolates the tables and nothing else:

	tables	a_1	a_2	a_3	a_4	a_5	$\mathrm{LCS}_{\mathrm{mm}}$	below own prior
A *(coherent)*	with priors	0.939	0.989	0.992	0.013	0.020	**0.590**	a_4, a_5
	no priors	0.890	0.976	0.975	0.014	0.021	0.575	$\mathbf{a_1}, a_4, a_5$
B *(incoherent)*	with priors	0.960	0.973	0.513	0.006	0.018	**0.494**	a_3, a_4, a_5
	no priors	0.855	0.890	0.545	0.004	0.025	0.464	all five

The *ranking* survives: the coherent response A still beats the incoherent B, 0.575 against 0.464. What breaks is the per-claim reading, which is where a coherence score earns its keep. Response A asserts

three true claims in an unbroken support chain and contradicts none of them, yet its chain root a_1 falls from 0.939 to 0.890 — below its own 0.9 prior — by the same mechanism the four-world table above makes explicit, since a_1 is the source of A's $a_1 \to a_2$ edge. Under our tables the only claims A leaves below their priors are a_4 and a_5, the two it explicitly refuses, which is the right answer.

That matters because §3.4 uses the count of below-prior claims to *localize* an incoherence. The no-priors tables report a support chain's own root as damage in a response with no conflict in it at all. In B they flag all three true claims instead of the single broken one, so B's real defect — the true a_3 collapsing because its only support is a premise the text itself denies — no longer stands out from two artefacts of the table.

And in the aggregate. At sixteen claims the same mechanism costs the measure its dynamic range. The recall report of App. G has a twelve-edge entailment backbone, one EQUIVALENCE edge and two conflicts, with all sixteen priors uniform at $\pi_i = \frac{1}{2}$. Here the effect compounds along the backbone: every claim on it is the source of a downstream entailment, so under the no-priors tables every one of them is charged, and the root — the source of the longest chain — absorbs the most, falling to $\mathbb{P}(a_1{=}1) = 0.19$ from its 0.5 prior.

The consequence is measured by the same procedure in both families: score the graph as given, then score it again with *every* conflict edge deleted, and read the gap. That gap is the range the measure has available to separate a response with conflicts from the otherwise identical response without them. Under the no-priors tables the mean marginal moves $0.490 \to 0.498$, a range of 0.008; under ours it moves $0.607 \to 0.620$, a range of 0.013 sitting an order of magnitude higher on the scale. All four values are exact, by $2^{16} = 65\,536$-world enumeration. A coherence measure built on the factuality tables spends its range penalizing premises and has almost none left for conflicts.

In brief. A relation's coupling says *which* joint assignment of two claims it penalizes; the potential $\psi_{\tau,p}$ says *by how much*. Read as two rows — source false, then source true — our ENTAILMENT table is $[1{-}\pi_s, \pi_s, \mathbf{1}{-}\mathbf{p}, p]$ with $\pi_s = \frac{1}{2}$, against the $[p, p, \mathbf{1}{-}\mathbf{p}, p]$ used throughout the factuality literature. The two agree wherever the source is true, and differ only where it is false: a flat, uninformative $[\frac{1}{2}, \frac{1}{2}]$ against a $[p, p]$ that is also flat but scaled up by p. That scaling makes the factor evidence *for* the source rather than a constraint acting *from* it, which is correct when the source is a retrieved passage whose reliability is in doubt and wrong when it is another claim of the same response. The cost is visible in four worlds: for one ENTAILMENT edge at $p = 0.9$ between two claims of prior 0.9, the source's marginal has numerator 0.738 under both families, but $Z = 0.788$ under ours against 0.828 under theirs, so the source moves to 0.9365 in one and 0.8913 in the other — below its own prior, penalized for nothing but having been cited as a premise. On the five-claim pair of Figure 1 this leaves the A-over-B ranking intact (0.575 against 0.464) while corrupting the per-claim diagnostic: the coherent response's own chain root drops below its prior, so the count of below-prior claims, which §3.4 uses to localize an incoherence, now fires on a response that has none. On a sixteen-claim report the same charge compounds along the entailment backbone, driving the root to 0.19 from a 0.5 prior and leaving a 0.008 range between the graph and its conflict-free variant, against 0.013 an order of magnitude higher up the scale under ours. Hence Def. 1 takes the with-priors family: a coherence model must not spend its dynamic range penalizing premises.

3.4 THE COHERENCE SCORE

The model is now fully specified: Def. 1 gives the joint, §3.2 the admissible relations, §3.3 their tables. What remains is to summarize the resulting distribution in a single number — and, as it turns out, not every natural way of doing so is admissible.

We report two readouts of the model, a headline score with a guarantee and a diagnostic that locates. The diagnostic is the *mean posterior marginal* $\mathrm{LCS}_{\mathrm{mm}}(y) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{P}(a_i = 1)$, read from a single MAR query: the expected fraction of the response's claims that the argument, taken as a whole, upholds. A conflict suppresses the marginal of its target (Prop. 5), a satisfied entailment chain propagates support along itself, and a claim no relation touches stays at its prior. Three diagnostics come free with the same query: the per-claim marginals, which *localize* an incoherence; the count of claims dragged *below their own prior*; and the mean normalized binary entropy.

App. D defines three alternatives — a two-term consistency probability, a reified coherence node and a normalized log-partition — states five criteria, and evaluates all four by exact inference. Only the mean marginal and the normalized log-partition are monotone across a coherence-ordered family, and the two failures share one cause: they measure whether a conflict is *active* rather than what should be *believed*, and softening a resolved concession by Eq. 2 necessarily *increases* the probability of its both-true world (Prop. 4). A fifth candidate — the probability that every asserted relation is satisfied — is maximized by asserting no relations whatever (Prop. 3). The mean marginal alone decomposes into per-claim contributions, but monotonicity under conflict removal is not a theorem for it: it provably holds for a contradiction's own target (Prop. 5), and deleting a contradiction can lower the mean marginal elsewhere (Prop. 6).

The headline score. Monotonicity is a theorem for a score that reads every relation against the claims' priors alone, rather than against a distribution the relations themselves shape. Let $\pi(\mathbf{a})$ be the product of the priors, let each conflict multiply the worlds that violate it by $1 - p_r$, and let each support relation multiply the worlds where it does work — both claims true for ENTAILMENT and EQUIVALENCE, exactly one for CO_NECESSITY — by $1 + p_r$. Writing $G(\mathbf{a})$ for the resulting product, the headline is

$$\mathrm{LCS}_{\mathrm{bl}}(y) \;=\; \sigma\Big(\tfrac{2}{n} \log \mathbb{E}_{\mathbf{a}\sim\pi}\big[G(\mathbf{a})\big]\Big), \tag{3}$$

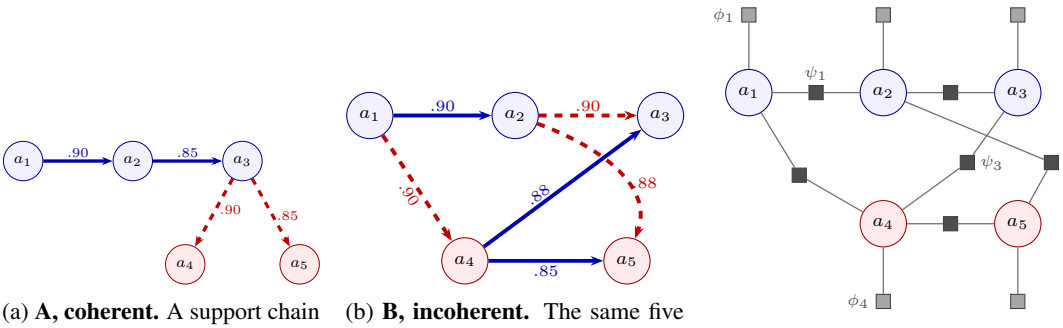
one PR query, with σ the logistic function. It is monotone in every relation edit (App. D.9, Thm. 2): removing or softening a conflict, and adding or strengthening a support, never lowers it, for any priors, because the weights π depend on no relation and every edit raises G in every world. A response with no relations scores exactly $\frac{1}{2}$. The factor 2 is a fixed display scale read off the model, not fitted: no relation factor exceeds 2, so one support adds at most one bit of evidence, and at scale 2 each bit per claim multiplies the odds by 4. On gold relation graphs it satisfies every satisfiable ladder assertion on both corpora (Table 30). Its price is stated with it: padding a response with support relations raises the score, and it does not localize — which is what the mean marginal is kept for. App. D.8 gives the conflict-only special case.

3.5 ESTIMATING RELATIONS FROM THE RESPONSE

Everything so far has taken the relation set $\mathcal{R}$ as given, which is what makes the model's properties provable. In deployment it is not given.

In use $\mathcal{R}$ must be estimated from the text, and it is the weakest link in the pipeline (§5). A relation's weight factors into a type term and a strength term, $p_r = \mathbb{P}(\tau \mid a_s, a_t, y) \cdot \mathbb{P}(a_t \mid a_s, \tau, y)$, estimated by two separate prompts (verbatim in App. K); conflating them is a mistake, since a confidently-identified weak link and a doubtfully-identified strong one are different epistemic situations one number cannot distinguish. *Prompt A* elicits the discourse sense with chain-of-thought *preceding* the label, so the token logprobs measure a decision already made, and the type confidence is the per-token geometric-mean probability of the label span. *Prompt B* elicits the conditional strength as a surrogate token, the renormalized $\mathbb{P}(\texttt{Yes}) / (\mathbb{P}(\texttt{Yes}) + \mathbb{P}(\texttt{No}))$ (Kadavath et al., 2022), asked in graded form so a weak but real link is not forced to answer No; a verbalized variant is kept only for comparison, verbalized confidence being weakly calibrated (Xiong et al., 2024).

Grounding is mandatory, not optional. Prompt A always receives the full response and asserts a coupling only where the response *itself* draws the connection — the single most consequential decision in the estimator. Judging a pair in isolation invites any abstractly plausible relation, and the failure is not mild: relation density reaches six to nine per claim, including spurious contradictions on paragraphs containing none. Since Eq. 1 is a product over relations, many soft mutually-constraining edges drive every marginal toward the prior, so an over-connected graph destroys the measure's ability to distinguish coherent from incoherent rather than merely adding noise. Two mechanisms add precision: the *admissibility filter* restricts output to the nine senses marked $\star$ above, excluding Condition and Instantiation as broad enough that a model reaching for "related somehow" lands on them readily (13% of one relation source's edges, zero in gold); and the *evidence-span requirement* obliges the model to quote the linking words verbatim, with the parser verifying the quotation really is a substring — the only precision mechanism that is *checked* rather than asked for.



(a) **A, coherent.** A support chain over the true claims; each false claim is attributed and then *rejected* from a_3. $\mathrm{LCS}_{\mathrm{mm}} = 0.590$.

(b) **B, incoherent.** The same five claims. $a_2 \to a_3$ has become a conflict and the *false* a_4 is now a premise for the true a_3. $\mathrm{LCS}_{\mathrm{mm}} = 0.494$.

(c) **The MRF of (b).** Grey squares are the unaries ϕ_i carrying the priors, black squares the pairwise ψ_r, one per relation.

Figure 2: The five-claim example of §3.7. Blue nodes are claims true of the world ($\pi_i = 0.9$), red nodes false ($\pi_i = 0.1$); *the same five claims appear in both graphs*, so every per-claim factuality check returns the same verdict on both responses. Solid blue edges are ENTAILMENT, dashed red CONTRADICTION; line width grows with the relation weight p. The unary factors ϕ_i of panel (c) are identical for the two responses, so the entire $0.590 \to 0.494$ drop is produced by the pairwise ψ_r — by how the response arranges its claims, which is the whole of what coherence measures.

Candidate pairs. Querying all $n(n-1)$ ordered pairs is quadratic and actively harmful, so a policy restricts which are scored: ALL_PAIRS; WINDOWED, forward pairs with $0 < j - i \leq w$ ($w = 4$); GATED, the window plus similar long-range forward pairs; and BIDIRECTIONAL, both arc directions within $|j - i| \leq w$. The last exists because of a structural limitation, not a tuning preference: a forward-only policy *cannot emit* a backward arc, so a gold relation that is both directed and backward is unreachable however the model answers — and 105 of our corpus's 603 edge-producing gold relations are of that kind, capping forward-only recall at 82.6%. Each policy is then refined against the response, promoting discourse-adjacent out-of-window pairs and demoting in-window pairs that are not (App. E).

3.6 COMPOSING COHERENCE WITH FACTUALITY

The priors π are a free slot, and the natural thing to put in them is what a factuality pipeline already computed: $\pi_i \leftarrow q_i = \mathbb{P}(a_i = 1 \mid \mathcal{C})$ for retrieved evidence $\mathcal{C}$. A claim is then credited when it is both externally supported and internally upheld, and an unsupported claim enters the coherence graph already doubted, so an argument resting on it is penalized twice. Setting $q_i \equiv \frac{1}{2}$ recovers the pure coherence score of Rem. 3, so the composition is an option rather than a commitment. We keep two stages rather than one fused network for reasons of cost, ablation and caching that App. E sets out.

3.7 EXAMPLE

The pieces are easiest to see at work on a case small enough to enumerate exactly. This is the pair of Figure 1, revisited now that the model is defined: the same five claims and the same two arrangements, with the relation graphs read against §3.2's vocabulary and the MRF they induce drawn explicitly. It also makes the orthogonality claim of §1 concrete, since it holds the claim set — and therefore every factuality verdict — fixed while varying only the arrangement.

Five claims about Voyager 1 are fixed once and for all, three true of the world (priors 0.9) and two false (0.1): a_1, it crossed the heliopause in August 2012; a_2, after the crossing its instruments measured a sharp rise in plasma density; a_3, it is therefore in interstellar space; and, false, a_4, its plasma wave instrument failed permanently before the crossing, and a_5, it has left the Solar System entirely.

Two responses assert *these same five claims* and differ only in how they arrange them, so every per-claim factuality check returns the same verdict on both. **Response A** reports the density rise as what established the crossing, concludes the probe is in interstellar space, then names the two

false claims and refuses each on a stated ground. **Response B** instead denies that the probe counts as being in interstellar space, asserts the instrument failure as fact, offers *that failure* as the reason the probe reached interstellar space, chains it into having left the Solar System, and finally denies both falsehoods in its closing sentence.

Figure 2 shows the resulting graphs and the MRF that B induces. Two relation changes carry the difference: the support edge $a_2 \to a_3$ in A becomes a CONTRADICTION in B, which denies its own conclusion, and the edge between a_3 and a_4 *reverses* — A rejects the false a_4 from a_3, while B makes a_4 a premise *for* a_3. Mentioning a falsehood is not what costs B its score; asserting it and reasoning from it is. Exactly, by 2^5 enumeration, A scores $\mathrm{LCS}_{\mathrm{mm}} = 0.590$ against B's 0.494: in A the true claims sit at 0.939, 0.989 and 0.992, each *above* its 0.9 prior because the chain supplies mutual support, and the false ones at 0.013 and 0.020, whereas in B the *true* a_3 is dragged to 0.513 because its only support arrives from a_4, which the response itself denies — a premise the text refuses cannot transmit support. The priors enter only through the unary ϕ_i, identical in both responses, so the entire drop is attributable to arrangement alone. App. G gives both responses in full and two further examples.

4 THE LOCOBENCH BENCHMARK

Validating a coherence measure requires responses that differ in coherence *while holding content fixed*, each accompanied by the relation graph a correct extractor would recover. Natural corpora supply neither. They offer no controlled variation in coherence at constant claim content, so an observed score difference cannot be attributed to coherence rather than to topic or length; and they carry no gold relation graph, without which *scoring* error cannot be separated from *estimation* error — the one distinction our evaluation is built to make. We therefore evaluate on a synthesized corpus, LoCoBench, and because the corpus is generated the burden is on us to state its construction completely. This section does so: the object generated (§4.1), the pipeline that generates it (§4.2), how each rung's gold graph is derived (§4.3), the validation that gates admission (§4.4), and what the released corpus contains (§4.6).

4.1 COHERENCE LADDERS

Definition 3 (Coherence ladder)**.** *A* ladder *is a tuple* $(\mathcal{P}, \langle \rho_0, \ldots, \rho_4 \rangle, \Gamma)$ *in which* $\mathcal{P}$ *is a base* relation plan *— a selected claim set together with a declared relation graph over it — each rung* ρ_k *is a sequence of perturbation operator calls applied to* $\mathcal{P}$*, realized as a response text and carrying* its own *gold relation graph, and* Γ *is a set of constraints on the relative ordering of the rungs' scores. A ladder is* admitted *only if all five rungs pass every validation gate of §4.4.*

Two properties of Def. 3 carry the experimental design. First, all five rungs share one claim set and one topic, so a measure that orders them is responding to relational structure and not to content. Second, each rung carries its own gold graph rather than inheriting the base plan's, which is what makes a rung a genuine test of a readout rather than a restatement of the corpus; §4.3 gives the derivation.

Whole-family admission. A ladder is admitted entire or rejected entire. Four of the five rungs carry no standalone claim — only a claim relative to their neighbours — so a family missing a rung is not a partial success but a broken comparison. A family whose fourth rung fails validation is therefore rejected together with the four rungs that passed.

Table 1 gives the four types, and the mix is load-bearing rather than incidental. CONFLICT is the reference ladder: only the conflict structure varies, which is what makes an ordering attributable to coherence. CHAIN tests a different failure mode — breaking N links of an entailment chain and repairing them, so the defect is a *missing* relation rather than a present conflict. The distinction matters because a measure built on contradiction detection alone will do well on the first ladder and fail the second. ORDER and CONTROL are the negative controls, and a corpus containing only increase-type ladders cannot distinguish a coherence measure from an edit-count proxy: every rung of these two applies only meaning-preserving edits, so they assert that the score does *not* move. §5.3 reports that this is precisely where mined relation graphs fail, a result no increasing-order ladder could have exposed.

Table 1: The four family types. Each holds the claim set, the topic and every unperturbed relation fixed, varying one structural dimension. ORDER and CONTROL apply only meaning-preserving edits and so assert *invariance*: CONTROL is the false-positive check, since a measure showing any monotone trend across it is tracking *having been edited* rather than coherence. The final column is the count in the released corpus (§4.6).

Type	Dimension varied	Rungs, $\rho_0 \to \rho_4$	Required pattern	n
CONFLICT	conflict structure	worse, base, concession_resolved, fix_one_conflict, coherent	strictly increasing	8
CHAIN	entailment-chain integrity	break_3, break_2, break_1, base, coherent	strictly increasing	2
ORDER	assertion order	shuffle_full, shuffle_block, shuffle_adjacent, base, ordering_only	invariant	3
CONTROL	nothing (meaning-preserving)	ordering_only, direction_reversal, base, direction_reversal, ordering_only	flat throughout	1

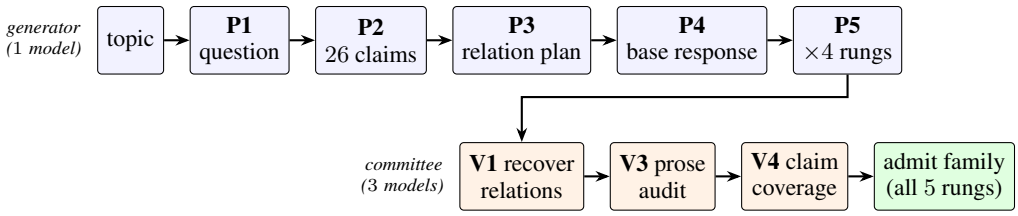


Figure 3: The generation pipeline. Five stages produce a family of five rungs from one topic; three validators, run by a committee that *excludes* the generating model, gate admission. A family is admitted whole or rejected whole (Def. 3).

Ordering constraints. A family declares its constraints in three classes, asserted per readout: **C1** a strict increase between consecutive rungs; **C2** a *predicted* decrease or invariance, which is where the concession dip of Prop. 4 becomes falsifiable and where an ordering-only edit must be exactly flat by Rem. 2; and **C3** endpoint separation between ρ_0 and ρ_4. A missing score fails rather than passes. C2 is the only class that predicts the internal behaviour of a particular readout, which is why the baseline comparison of §5.5 excludes it: a single-score baseline makes no claim a C2 constraint could test.

4.2 THE GENERATION PIPELINE

Generation is a five-stage prompted pipeline followed by validation. The stages are separated so that a defect can be attributed to one of them and repaired there, and each stage's prompt is fixed and versioned, with its numbered-instruction count and placeholder contract declared in Table 3 and asserted by a test against the prompt strings themselves. The mechanism matters because the alternative failure is silent: a corpus generated by prompts that no document describes.

The five stages, in order (Fig. 3), are as follows.

P1: topic to question. Given a topic from an inventory of 36 topics across four domains, emit one open-ended question. The prompt enumerates the target relations, so the question is chosen with the downstream plan in mind rather than for topical interest, and it prefers adjudicative domains — incident investigations, litigation, diagnostic reasoning, attribution disputes, competing hypotheses — because those are the settings in which exhaustive alternatives and resolving holdings arise naturally.

P2: question to claims. Emit 26 atomic claims answering the question, each a single proposition, tagged with a truth status. A surplus is generated deliberately: P3 selects a subset, and selection from a surplus is what lets the plan choose claims that support the required relation structure instead of forcing relations onto whatever claims exist.

P3: claims to relation plan. Select 14–16 claims (aiming for 15) and plan exactly 10 relations over them using the two-level scheme of §3.2, with at least one relation of each required sense, together with 4–6 pairs *explicitly declared unrelated*. Relations are kept local, preferring pairs within four positions, which is what makes a windowed candidate policy a reasonable baseline rather than a

Table 2: The six perturbation operators and the ten calls realizing them. Each call has a declared effect on the rung's *gold relation graph* as well as on its text, which is what makes §4.3's derivation well defined. The two O6 calls are factor-invariant by design: they are what the ORDER and CONTROL ladders exist to check.

Op	Calls	Effect on the text
O1 relabel	wrong_sense, direction_reversal, exhaustiveness_flip	rewrite the connective so the text draws a different sense, reverses the relation, or flips Contrast↔Alternative
O2 add conflict	inject_contradiction, spurious_relation	add one sentence directly contradicting a named claim, at least two sentences later; or assert a link between two claims the response kept separate
O3 break chain	break_chain	remove or negate the link realizing an edge so the chain no longer connects, keeping both endpoints
O4 drop relation	drop_relation	remove the connective realizing an edge, leaving the two claims as adjacent assertions
O5 resolve	add_resolution, remove_resolution	add or delete the holding sentence that settles a conceded tension
O6 reorder	shuffle_order, ordering_only	permute sentences at full, block or adjacent granularity; or exchange Precedence↔Succession preserving chronology — *no coupling changes*

straw man. The declared non-relations are the corpus's labelled negative set, and they are what makes a precision figure meaningful: precision against "every pair not in gold" is dominated by unlabelled pairs of unknown status, whereas precision against pairs the item asserts are unrelated is a real number. Each relation also carries an intended strength band and an intended *validity* — valid when the relation genuinely holds, invalid when the prose draws it unsoundly — targeting a valid fraction near 0.55, so a scorer cannot succeed by assuming every relation it finds is sound. The shipped items carry 7–11 relations each and 2–3 false claims, the spread arising because a rung's perturbations add and remove edges relative to the plan.

P4: plan to response. Realize the plan as continuous prose of 550–650 words, asserting every selected claim and drawing every planned relation, with no headings or lists. The floor exceeds the default completion-token grant of some gateways; a model that hits the cap returns prose stopped mid-sentence, which every parser rejects but for the wrong stated reason, so the ceiling is raised explicitly rather than left to the backend default.

P5: response to rung. Apply exactly one perturbation to the base response, changing the minimum number of words and preserving everything else verbatim. Table 2 gives the six operators and the ten parameterized calls realizing them. The stage also emits a structured diff naming the operator, its target, the sentences changed and the relations added, removed or relabelled; length drift is capped at 15% except for the three calls that add or remove a sentence by definition.

The failure policy. What happens when a stage's output is defective is a specification decision, and the policy is per stage with four outcomes. A **retry** handles a transient parse failure where the inputs are sound; because a byte-identical prompt re-sent to a temperature-zero backend cannot produce a different parse, retries are issued at successively higher sampling temperatures, which is what makes a retry budget worth more than one attempt. A **regenerate** handles an artifact that is wrong while its inputs are fine. A **drop edge** applies to V1 alone: an unrecovered relation must never be retained as gold, since gold the prose does not realize is exactly the defect that gate exists to catch. A **reject** applies when a family cannot carry its ranking claim, and stores the family with its reason rather than discarding it, so rejection rates remain auditable.

4.3 DERIVING EACH RUNG'S GOLD RELATION GRAPH

A rung's gold graph is the base plan's graph with that rung's operator calls applied. Three mechanisms make this well defined, and each closes a way the derivation could otherwise produce a rung that does not match its own prose.

Table 3: Prompt specification. Each prompt's numbered-instruction count and required placeholder set are declared here and asserted by a test against the prompt strings, so a prompt cannot drift from the document describing it without the test failing. Instructions are counted by matching lines beginning at column one with a digit, period and space, so indented sub-items — P3's per-sense glosses, P5's per-call definitions — do not inflate the count. The validators carry no numbered instructions because their prompts specify an output schema rather than a procedure.

Stage	Role	Instructions	Required placeholders
P1	topic → question	9	`TOPIC`
P2	question → claims	7	`QUESTION`
P3	claims → relation plan	10	`CLAIMS`, `QUESTION`
P4	plan → base response	8	`PLAN`, `QUESTION`
P5	response → rung	5	`OPERATOR`, `PLAN`, `RESPONSE`
V1	recover relations	0	`ATOMS`, `RESPONSE`
V3	prose audit	0	`RESPONSE`
V4	claim coverage	0	`ATOMS`, `RESPONSE`

Target assignment. Each call in a rung is assigned a distinct eligible edge, so a rung composing two drop_relation calls drops two *different* edges. Eligibility is call-dependent: a drop_relation in a conflict ladder targets an unresolved conflict edge, an add_resolution a resolvable one. A CONFLICT plan therefore needs three unresolved conflict edges to support its full ladder — one for the resolution and two for the drops — and a plan with fewer is rejected at generation time rather than silently emitting a rung that duplicates its neighbour.

Derived-field re-derivation. When a call changes an edge's sense, the coupling, directedness, ordering-only status and exhaustiveness are re-derived from the new sense through the compile map $\mathcal{C}$ rather than carried over, so a transformed edge cannot contradict $\mathcal{C}$.

The adjacency gate. Deriving a rung's labels this way makes one failure possible that no textual check can see: two adjacent rungs can come out with the *same* edge set, in which case they build the same network and nothing can order them. That is an admission gate rather than a property of the derivation, so §4.5 states it with the rest of the mechanism.

4.4 VALIDATION

An item is admitted only if an independent committee certifies it, and three properties of the arrangement matter more than the individual gates.

The committee excludes the generator. Validation is run by a panel of models, and the model that generated a family is removed from its own panel before voting. Without this, the dominant failure mode of an LLM-generated benchmark — the generator certifying its own idiosyncrasies as ground truth — is structurally unaddressed. The gates are then combined by three different rules — V1 by best rater, V3 and V4 by majority — for reasons §4.5 gives, and a V3 facet counts as failed only when a majority objects, so one unusually strict auditor cannot reject a sound family.

Every threshold lives in one place. All acceptance thresholds are declared in Table 4 and nowhere else. A benchmark whose thresholds are scattered across the code applying them cannot be replicated from its description, nor re-run at a different operating point without a search for the constants.

Generation is rejected, never repaired. A family failing a gate is stored with its reason and discarded. No stage patches a defective artifact into an acceptable one, because a repaired item's gold labels no longer describe a process we can state.

The three gates test different things. **V1**, the load-bearing one, hands an independent model the response and its claims and asks it to recover the relations the response *itself* draws, emitting for each ordered pair a sense, a coupling, a strength band and a resolved flag; admission requires ≥ 0.80 of valid planned relations returned with the correct coupling and ≥ 0.70 with the correct sense. This is what certifies that the gold graph describes the prose rather than the plan, and it is the gate that makes the GOLD relation source of §5.1 meaningful. **V3** audits the prose for fluency, formality and organization, and requires the leakage and hedging span lists to be empty — leakage being any

Table 4: Every acceptance threshold. V1 rates are computed over the *valid* planned relations only, since an invalid planned relation is one the prose is not required to draw soundly; with P3's six valid relations both V1 rates quantize to 5 of 6 (§4.5). The rightmost column is the decision rule over the committee, and the three gates use three different ones on purpose: V1 is satisfied by the best rater, V3 rejects a facet only on a majority of auditors, and V4 needs a majority to call a claim lost. Every gate is boolean and admission is their conjunction at *family* granularity — see §4.5 for the mechanism, including the two per-rung gates and the adjacent-pair edge-effect gate, which are not per-item thresholds and so have no row here.

Gate	Quantity	Threshold	Rule
V1	planned relations recovered with correct coupling	≥ 0.80	any-of
V1	… and with correct sense	≥ 0.70	any-of
V3	fluency, formality, organization (each, of 5)	≥ 4	majority (facet)
V3	leakage and hedging span lists	must be empty	majority (facet)
V4	claim coverage over gating statuses	$= 1.00$	majority (atom)
P3	claims selected	14–16	—
P3	relations planned (accepted band; 10 instructed)	8–12	—
P3	declared non-relations	4–6	—
P3	fraction of relations valid	0.55 ± 0.15	—
P3	false claims selected	≥ 2	—
P5	length drift	$\leq 15\%$	—

text that names the annotation scheme or the perturbation, which would let a scorer detect the rung without reading for coherence. **V4** checks that every selected claim is actually asserted, gating on the missing and merged statuses, since a claim the response never makes cannot stand in a relation.

4.5 How a family is admitted

§4.4 names the gates and their thresholds; this section states the *mechanism* that combines them, because the arithmetic of admission is not a scoring rule and reporting it as one would misdescribe the corpus. There is no score, no average and no threshold on how many rungs cleared: every gate returns a boolean and a family is admitted when the conjunction of all of them is true.

The unit of admission is the family, not the rung. A ladder's claim is *relational* — that rung $k{+}1$ is more coherent than rung k — so four rungs out of five is not four fifths of a benchmark item. It is nothing: the four surviving rungs carry no ranking claim that any readout could be held to, and the pair that failed is precisely the comparison the family existed to license. A family is therefore admitted whole or rejected whole, and a rejection is stored with the gate and reason that caused it rather than repaired. Concretely, admission requires that the family reach the final stage with all five items present and every recorded gate passing; one failing gate anywhere discards all five.

Gates apply at four different granularities. Reading Table 4 as a per-item checklist would miss that its rows are evaluated against different objects. *Per plan*, once per family: the P3 structural quotas, before any prose exists. *Per base response*, once per family: V1, V3 and V4, on the committee of §4.4. *Per rung*, on each of the four derived rungs: the perturbation gates below. *Per adjacent rung pair*: the edge-effect gate below. The four derived rungs are never validated by V1/V3/V4 individually — those run on the base prose, and the derived rungs inherit its certification through the perturbation gates, which bound how far an edit may move the text.

The three committee gates combine by three different rules, deliberately. They measure different things, so a single rule would be wrong for at least two of them. **V1 is any-of**: recoverability asserts that a careful reader *can* recover the plan from the prose, so one competent rater succeeding is sufficient evidence, and the best rater's verdict is the one reported while every rater's rates are recorded. Requiring each rater to clear the threshold independently would be a materially harsher gate than the one we intend. **V3 is majority per facet**: a facet counts as failed only when a majority of auditors objects, so one unusually strict auditor cannot reject sound prose. **V4 is majority per**

atom: no single rater can declare a claim absent. Majority throughout means a *strict* majority, and a tie escalates rather than resolving silently.

V1's thresholds are integer gates in disguise. Both V1 rates are computed over the *valid* planned relations only, and P3 mandates six valid relations of the ten it plans, so the denominator is 6 and the thresholds quantize: 0.80 requires 5 of 6 and 0.70 also requires 5 of 6, since $4/6 = 0.667$ clears neither. A single relation therefore decides admission. This also bounds how far the coupling threshold can be raised: at any value above 0.834 it becomes unanimity-of-six, a gate no rater panel reliably clears, so the nominal headroom between 0.80 and 1.00 is not usable.

Per-rung gates: the edit must happen, and must stay small. Each derived rung is produced by one or two perturbation calls, and each call is checked twice. *Text-changed* requires the prose to differ from what that call was handed — not from the base — so a composed rung whose second call is a no-op is caught even though its first call already moved the text away from the base. *Length drift* requires the rung to stay within 15% of the base length, so a perturbation cannot smuggle in a coherence change by rewriting at volume. A failure of either aborts the family immediately.

The gate that is not about prose: adjacent rungs must build different graphs. A ladder can satisfy every textual gate and still be unusable, if two adjacent rungs happen to induce the *same* gold edge set. Then the two rungs build the identical Markov network, no readout can separate them, and the ranking assertion across that pair has no label behind it. The edge-effect gate compares the edge signatures of each adjacent pair and rejects the family when they coincide. Two details of its scope are load-bearing. It compares *adjacent* rungs rather than each rung against its parent: every CONFLICT rung has the base as its parent, so a parent comparison admits a rung that duplicates the one immediately below it. And it *exempts* any pair in which either rung's calls are all edge-invariant — sentence reordering and swapping Precedence for Succession move no edge by construction, which is exactly the invariance the ORDER and CONTROL ladders exist to test, so flagging those pairs would reject the negative controls for doing their job.

Why the yields in §4.6 are what they are. Because admission is a conjunction, rejections are single-point rather than cumulative: the first binding gate is the only one visible in a rejection record, and a campaign that fails on a structural quota reports nothing about the prose gates that never ran. In development this produced several successive regimes in which the admitted count was zero for a different single reason each time. It is the honest behaviour for a conjunction, but it means a low yield localizes one gate and says nothing about the rest, and a reader comparing yields across configurations should read the gate attribution rather than the count.

4.6 THE RELEASED CORPUS

Table 5 states what the corpus contains. It was produced by `claude-opus-5` and admitted by a three-model cross-vendor committee (`gpt-5.6-terra`, `gemini-3.1-pro`, `claude-sonnet-5`), none of which is the generator. The 14 families cover 14 distinct topics spanning all four domains — four Natural Sciences, four Engineering & Technology, three Humanities, three Social Sciences — so no single topic or domain carries a result. All four family types are represented, which is what allows §5.3 to report the increase and invariance axes separately; the 202 ordering constraints decompose as 112 over CONFLICT, 30 over CHAIN, 45 over ORDER and 15 over CONTROL.

Two properties of the corpus bound what can be concluded from it, and we state them here rather than in the limitations so that the results below are read correctly. It is LLM-generated throughout, so an idiosyncrasy shared by the generator and all three validators would be invisible to this construction. And at 70 items it is small in absolute terms: the per-source comparisons of §5.3 rest on 202 assertions, which is enough to separate sources differing by tens of points but not enough to resolve differences of a few.

Table 5: Composition of the released corpus. Every figure is computed from the shipped items rather than from the generation targets. *Edge-producing* excludes the 67 ordering-only relations, which compile to NONE and contribute no factor; *backward* counts edge-producing relations whose target precedes its source in assertion order, the relations a forward-only candidate policy cannot reach (§5.2). A single ordering constraint asserts one direction for one readout on one pair of rungs, so the three graded readouts multiply the 4 adjacent pairs and the one endpoint pair of each family; the by-type figures are CONFLICT / CHAIN / ORDER / CONTROL.

Quantity	Value	Quantity	Value
Families	14	Gold relations, total	670
Items (rungs)	70	edge-producing	603
Claims	1,115	valid 408	(0.61)
Claims per item	15–16	backward	105
Words per item	609–698	Declared non-relations	342
Distinct topics	14	Ordering constraints per source	202
CONFLICT / CHAIN	8 / 2	by ladder type	112 / 30 / 45 / 15
ORDER / CONTROL	3 / 1	by class (C1 / C2 / C3)	96 / 64 / 42
Generator / committee	1 / 3	readouts asserted per pair	3

5 EXPERIMENTS

The measure of §3 has two components that can fail independently: the probabilistic model, and the estimator that supplies its relation graph. An evaluation conflating them cannot say which to fix, so our experiments are built around one device — holding the readouts byte-identical and swapping only the relation graph (§5.1) — which attributes any shortfall to one component or the other by construction. We then ask three questions in turn: how accurate is the estimator (§5.2), how well does the model order responses given a correct graph (§5.3), and how does the measure compare against evidence-free alternatives (§5.5). §5.9 draws the answers together and §5.10 bounds them. The experiments were run before the headline readout (f) was defined. Wherever a cached run stored the relation graph it was scored on, (f) has been scored on exactly that graph and added to the table: the ladder tables, on gold and mined graphs, and the uniform-prior factuality runs (Tables 11 and 16). Runs that stored scores only — the factuality-prior factuality runs and the human study — report (a)–(d), and are marked where (f) is absent. The 202 ladder constraints are declared per readout for (a)–(d), so (f)'s ordering is read from the readout-independent axes of Table 8.

5.1 PROTOCOL

A *relation source* fixes where the relation graph comes from while holding claims, priors, readouts and scorer settings byte-identical — the central experimental device of the paper. GOLD takes each item's own labelled relations, GOLD_VALID only those labelled valid, and a MINED source runs the estimator of §3.5 on the response text; because the readouts are identical across sources, any shortfall below gold is attributable to relation estimation and not to the probabilistic model. Mined relations are compared to gold at three nested levels, defined with the measurement in §5.2.

The corpus is the one specified in §4: 14 families, 70 items, 1,115 claims and 670 gold relations, declaring 202 ordering constraints per source (Table 5). Mining is sampled once per cell rather than averaged, so we measure the resulting variability instead of estimating it: two *independent* runs of the same configurations moved every pair, coupling and sense F1 by at most 0.014, while the gold sources reproduced bit-for-bit. Per-source ladder agreement is the noisier statistic, moving by four to six of 202 assertions between those runs, so a single source's percentage should not be read to the point and differences below roughly six assertions are not resolvable here. Exact values are by 2^{16} enumeration, corpus-scale ones by weighted mini-bucket (Liu & Ihler, 2011) at i-bound 6.

Five baseline families are chosen to fail in informative ways; four are implemented and share the LCS's own decomposition, so each sees the same atoms and no retrieved evidence: ladder-blind controls (claim count, relation count, length) that catch a measure secretly tracking edit count or graph density (Steen & Markert, 2022); pairwise NLI with no global model (Manakul et al., 2023); ROSCOE's Self-Consistency (Golovneva et al., 2023) in max, mean and symmetric variants; LLM

Table 6: Mined-versus-gold agreement, micro-averaged over the seventy items. The denominator is the 603 *edge-producing* gold relations (670 total, less the 67 ordering-only edges, which compile to NONE and contribute no factor). Agreement is scored at three increasingly strict levels: recovering the **pair**, then its **coupling** (which selects the factor table), then its **sense**. **By direction** decomposes coupling recall according to whether the gold relation runs forward or backward in assertion order — 105 of the 603 run backward, and no forward-only candidate set can reach them. **Errors** counts violations of the 342 declared-unrelated pairs (**non-rel**) and unordered pairs receiving two factors (**dup**). All figures are percentages except the three counts (edges, non-rel, dup). **Bold marks the best F1, not the best recall**: BIDIRECTIONAL wins every recall column under either miner and is nonetheless the weaker policy, which is the trade-off this table exists to show.

		Pair			Coupling			Sense	Recall by dir.		Errors	
Policy	Edges	P	R	F1	P	R	F1	F1	fwd	bwd	non-rel	dup
mined by `llama-3.3-70b`												
WINDOWED	1,106	44.8	82.3	58.0	34.1	62.5	44.1	37.6	71.1	21.9	34	0
BIDIRECTIONAL	3,172	28.5	91.7	43.5	17.9	79.3	29.3	24.7	79.9	76.2	119	1,231
mined by `gpt-oss-120b`												
WINDOWED	1,094	46.0	83.4	59.3	40.2	73.0	**51.9**	**47.0**	82.1	29.5	38	0
BIDIRECTIONAL	2,973	29.8	96.5	45.5	22.4	88.4	35.8	32.3	89.0	85.7	147	1,020

judges — G-Eval (Liu et al., 2023) with the SummEval rubric (Fabbri et al., 2021) and a direct rating prompt, each a mean over five runs; and the three single-document DiscoScore metrics (Zhao et al., 2023) with an entity grid (Barzilay & Lapata, 2008).

5.2 RELATION ESTIMATION IS THE BOTTLENECK

We begin with the estimator, since every mined relation source inherits its errors. Measuring it against gold separates a model that scores a graph badly from an extractor that supplies the wrong graph.

How the numbers were obtained. Each source of Table 6 is one miner paired with one candidate-pair policy, run over the same seventy items. The policy fixes which claim pairs the miner is ever asked about, and the two differ structurally rather than by degree: WINDOWED proposes only *forward* arcs within an order radius of $w = 4$ (source before target in assertion order), yielding 1,693 candidate pairs over the corpus, while BIDIRECTIONAL proposes *both* arcs of every pair and applies no window, yielding 7,520 — a $4.4\times$ larger question set from identical inputs. Each surviving candidate then costs two LLM calls, Prompt A for the discourse sense and its coupling and Prompt B for the conditional strength (App. E); the measured rate is 1.65 calls per pair under WINDOWED and 1.41 under BIDIRECTIONAL, below two because Prompt B is skipped whenever Prompt A returns none — which it does for 593 and 4,447 pairs respectively. We then compare the surviving relations to the item's gold graph at three *nested* levels, because they answer different questions: **pair** asks only whether the miner saw the two claims as related at all, isolating candidate selection from labelling; **coupling** additionally requires the correct Level-1 coupling, which is the headline because the coupling is what selects the factor table; and **sense** additionally requires the Level-2 discourse sense, which diagnoses taxonomy confusion rather than anything the MRF does. A mined edge matches a gold one on the *ordered* pair when the coupling is asymmetric, where direction is part of the claim, and on the *unordered* pair when it is symmetric, where scoring direction would charge a policy's arc ordering as a labelling error. Counts are micro-averaged over the seventy items against the 603 edge-producing gold relations.

What they mean. The decomposition matters more than any single F1, because it says *which* step fails. Take `llama-3.3-70b` under WINDOWED: of the 603 gold relations it finds 496 as pairs, types 377 of those with the right coupling, and gets the sense right on 321. So the estimator is not mostly failing to notice that two claims are related — it recovers four fifths of the pairs — it is failing to *type* what it found, losing 119 relations to the wrong coupling and 56 more to the wrong sense. That is what makes "extraction is the bottleneck" a claim about labelling rather than about recall, and it is why coupling F1, not pair F1, is the column to read. The second half of the story is precision. The same source emits 729 spurious couplings against its 377 correct ones, and under BIDIRECTIONAL

Table 7: **Left:** declared ordering constraints satisfied, by ladder type — 202 assertions per source. ORDER and CONTROL assert *invariance*, so a readout that trends there is tracking surface form rather than coherence; GOLD_VALID is near-flat on CONFLICT by construction, so its failures carry no information about the readouts and the GOLD source is what the constraints test. **Right:** each readout's mean over the seventy items, with the standard deviation in grey. Averaging across a ladder's rungs *deliberately collapses the ordering the ladder exists to test*, so the right-hand block is a **level comparison only** and carries no evidence about ordering — that is entirely the left block's job; the spreads are given because a mean alone cannot say whether a source is stable or merely centred. The reified readout is omitted here; App. D compares all six. The 202 constraints are declared per readout for (a)–(d) and none for the headline $\mathrm{LCS}_{\mathrm{bl}}$, so the left-hand block covers those four; $\mathrm{LCS}_{\mathrm{bl}}$'s ordering is scored on the readout-independent axes of Table 8, where on GOLD it satisfies $49/50$ and $20/20$. Its mean sits near $\frac{1}{2}$ by construction and is not comparable in level to the marginal-based readouts.

	Constraints satisfied					Mean readout value (sd)			
Source / policy	Cfl	Chn	Ord	Ctl	%	$\mathrm{LCS}_{\mathrm{bl}}$	$\mathrm{LCS}_{\mathrm{mm}}$	$\mathrm{LCS}_{\mathrm{cn}}$	$\mathrm{LCS}_{\mathrm{lp}}$
GOLD	89/112	18/30	45/45	15/15	**82.7**	0.503 ±0.026	0.795 ±0.022	0.515 ±0.135	0.410 ±0.090
GOLD_VALID	37/112	12/30	45/45	15/15	54.0	0.491 ±0.021	0.794 ±0.020	0.576 ±0.151	0.485 ±0.047
mined by `llama-3.3-70b`									
WINDOWED	53/112	17/30	0/45	0/15	34.7	0.592 ±0.076	0.796 ±0.089	0.676 ±0.239	0.224 ±0.158
BIDIRECTIONAL	57/112	18/30	0/45	0/15	37.1	0.772 ±0.116	0.723 ±0.171	0.542 ±0.196	0.037 ±0.026
mined by `gpt-oss-120b`									
WINDOWED	59/112	12/30	0/45	0/15	35.1	0.565 ±0.058	0.825 ±0.074	0.500 ±0.138	0.129 ±0.096
BIDIRECTIONAL	53/112	17/30	0/45	0/15	34.7	0.756 ±0.100	0.808 ±0.128	0.462 ±0.095	0.033 ±0.059

that becomes 2,187 against 478. Because Eq. 1 is a *product* over relations, a spurious factor is not a diluted signal but an active one pulling marginals toward the prior, so precision is not tradeable against recall here in the way an F1 suggests; the 1,231 duplicate unordered pairs are the same defect seen structurally — two factors on one variable pair, which the model has no principled way to combine — and declared-non-relation violations rise from $34/342$ to $119/342$ in step. The two effects together explain the table's central inversion. Forward-only mining is *structurally* capped, reaching 21.9–29.5% of backward-running gold relations against 76.2–85.7% for BIDIRECTIONAL, because 105 of the 603 relations run backward and no forward-only candidate set can emit them; but buying that reach costs more in precision than it returns in recall, so coupling F1 *falls* from 44.1 to 29.3. Reach and accuracy are separate axes, and on this corpus the wider policy is the weaker one. Finally, `gpt-oss-120b` is the better miner at every level — by 7.7 points of coupling F1, 9.5 of sense F1 and 7.6 of backward recall under WINDOWED, while emitting slightly *fewer* edges — but it is also the less informative one: 99% of its strength estimates sit at a clamp boundary against 59% interior for `llama`, and it returns just nine distinct strength values across 1,094 relations against `llama`'s 600. It decides confidently and supplies almost no graded uncertainty for the model to use.

5.3 THE MODEL ORDERS CORRECTLY WHEN THE GRAPH IS CORRECT

Holding the readouts fixed and varying only the graph now isolates the model's own behaviour.

How a constraint is scored. A *relation source* fixes only where the relation graph comes from; the claims, the priors, the readouts and every scorer setting are byte-identical across sources (§5.1), so any difference between two rows of Table 7 is caused by the graph and nothing else. A single *constraint* is one assertion about one readout on one pair of rungs: we score both rungs, take the difference, and ask whether it has the declared sign, counting a difference below 10^{-6} as flat. The assertions come in three classes — **C1** a strict increase between consecutive rungs (96 of them), **C2** a *predicted* decrease or exact flatness (64), and **C3** separation between the first and last rung (42) — for 202 per source, distributed over the four ladder types as the table's columns show. The percentage is simply the fraction satisfied, and a missing score counts as a failure rather than being skipped. Because a constraint is a claim about a *difference*, the left-hand block of Table 7 is the only part that tests ordering; the mean readout values on the right average that difference away by construction and are reported for level comparison alone.

On gold relations the readouts satisfy 82.7% of these constraints — 167 of 202 — against 34.7–37.1% for every mined source. Since sources differ only in the graph, that 46–48 point gap is attributable to relation estimation alone, and its size is what §5.2 predicts: at 29–52% coupling F1 roughly half the factors in a mined network are wrong or missing, and a ladder whose consecutive rungs differ by one or two relations cannot survive that. This is the paper's central empirical claim, and it is a claim about *where* the remaining problem is — the probabilistic model orders responses correctly when handed a correct graph, and extracting one is the open problem.

The per-ladder breakdown localizes the failure. On gold, ORDER and CONTROL are perfect ($45/45$, $15/15$) since those rungs move no factor, CONFLICT is strong at $89/112$, and CHAIN the weak ladder at $18/30$. But *every mined source scores* $0/45$ *on* ORDER *and* $0/15$ *on* CONTROL — the most consequential number in the table, and not a scoring artefact: those ladders hold the claim set fixed and permute the prose, yet on one ORDER family the `gpt-oss` WINDOWED miner emitted 9, 16, 20, 20 and 17 edges across five rungs differing *only* in sentence order. LLM relation extraction is therefore not reorder-invariant, and any measure built on it inherits that sensitivity however sound its probabilistic layer. Mined sources are nonetheless *not* uniformly worse (CONFLICT 53–59 of 112, CHAIN 12–18 of 30), so a measure evaluated only on increase-type ladders would have called mined extraction far closer to gold than it is — precisely the false positive the CONTROL ladder was added to catch. Mined ladder agreement also does *not* track coupling F1 — `llama`'s BIDIRECTIONAL source has the worst coupling F1 (29.3) and the best agreement (37.1%) — so these sources are not separated by ladder agreement at all, and candidate policies should not be ranked on it.

5.4 WHAT THE BASELINES ARE

Table 8 sets the LCS against ten baseline columns, and reading it needs to know what each one computes. All are evidence-free and all consume the LCS's own atom decomposition, so every row sees the same claims and no retrieved passages; the only thing varying across the table is how a measure turns those claims into a number. App. F gives the design rationale — why each family is chosen to fail informatively rather than to be a weak competitor — and what follows is the operational content.

Discourse representations (RC, LC, ENTITY GRAPH). Three single-document metrics from DiscoScore (Zhao et al., 2023). DiscoScore is presented as reference-based and most of its variants consume a reference a standalone response does not have; these three ignore the reference argument entirely, and the last *is* an entity grid (Barzilay & Lapata, 2008), so the classical baseline comes from the same definition. RC is repeated noun mentions over distinct nouns; LC is nouns occurring in more than one sentence over distinct nouns; ENTITY GRAPH is the mean sentence-adjacency weight, an edge of weight $1/(j-i)$ wherever sentences i and j share a noun lemma. The reported values are computed by our own implementation of these three definitions rather than by DiscoScore's code, which is not installable as a dependency — it is absent from PyPI and downloads a model at import time — and §F.7 states exactly what our noun approximation does and in which direction it errs; the comparison rests on the definitions, which are theirs. All three therefore reduce to noun-lemma repetition, with no representation of entailment, relation type or contradiction. A response can contradict itself in every sentence and still score at ceiling provided its nouns recur, which is what makes them a test of whether coherence is separable from *cohesion* rather than a weak competitor — and RC is nonetheless the strongest baseline in the table at $30/50$.

Local contradiction detection (CONTRADICTION RATE). One minus the fraction of claim pairs the NLI extractor labels contradictory, in the spirit of SelfCheckGPT's NLI variant (Manakul et al., 2023). It runs the *same* extractor over the *same* atoms as the LCS and is deliberately left ungrounded, so the only differences from our measure are relation *typing* and *propagation*: it isolates what joint inference adds over local detection. The prediction, which the table bears out at $12/50$, is that it does creditably where the planted defect is a *present* contradiction and poorly where it is a *missing* entailment, since a broken support chain leaves no contradictory pair to find.

Structured reasoning consistency (ROSCOE-SC, max and mean). ROSCOE's Self-Consistency (Golovneva et al., 2023), $1 - \max_i \max_{j<i} p_{\text{contr}}(h_i, h_j)$. The two rows are one *ablation*, not two baselines, because the published measure differs from the LCS in three separable ways: it is a maximum, so a second conflict changes nothing; it is forward-only ($j < i$), so a relation running

backward in claim order is unreachable, which is not hypothetical here — §3.5 measures 105 of the 603 scorable gold relations as both directed and backward; and it is untyped, so entailment, equivalence, exclusivity and co-necessity are invisible. Reporting a mean-aggregated variant alongside the faithful one says *which* difference is doing the work, and the answer is the aggregation. Under `llama-3.3-70b` the published max variant is exactly constant across all seventy items; under `gpt-oss-120b` it does vary, but every value it returns lies below 5×10^{-7}, so the maximum is saturated and the score is pinned at the floor either way. The mean variant, on the same items and the same pairwise probabilities, takes 57 and 63 distinct values respectively. ROSCOE targets step-by-step reasoning chains and a long-form response is not a chain, so we label it adapted.

LLM judges (JUDGE, G-Eval and direct rating). G-Eval's coherence dimension (Liu et al., 2023) with the SummEval coherence rubric (Fabbri et al., 2021), and a direct "rate the logical coherence from 1 to 5" prompt as the no-rubric control. These are the practical incumbent and the comparison a reader will want. Each is reported as a mean over five runs, because the determinism of Rem. 2 is only an advantage next to a measurement of the judge's instability, and because a spread wider than the gap under discussion bounds what the comparison can conclude. The ladder is a demanding test for them: consecutive rungs differ by a few words, so a judge must be sensitive at very small edit distances while staying flat where meaning is preserved.

Ladder-blind controls (CLAIM COUNT, LENGTH). Two quantities that ought *not* to track coherence: the atom count and the response length. They exist because Steen & Markert (2022) argue coherence measures must be checked against system-level confounders, and they catch a measure secretly tracking edit count or graph density. A control that *succeeds* is bad news for the corpus, not good news for the control.

Reading the vacuous rows. Three rows reach $20/20$ on invariance without measuring anything, and the mechanism differs in all three cases — and, for one of them, between the two backends. ENTITY GRAPH reports a single value on all seventy items, but it is not a measure that cannot move: the metric is an unbounded sum of $1/(j-i)$ adjacency weights, and its raw value takes 62 distinct values spanning 1.36 to 4.93 on these items. Every one of them exceeds the $[0, 1]$ reporting range, so all seventy clamp to the ceiling and the reported constancy is saturation *above* the range rather than constancy within it. ROSCOE-SC (max) is the mirror image, saturated at the floor: under `llama` it is exactly constant at zero, while under `gpt-oss` it is *not* constant — it takes 18 distinct values — but every one lies below 5×10^{-7}, so no rung pair separates by the 10^{-6} the axis requires and it passes by tolerance rather than by constancy. CLAIM COUNT is the third mechanism: it takes two distinct values across the corpus (65 items at one, 5 at the other), but it never varies *within* a family, in 0 of 14. Since invariance is a within-family test, that is enough to pass it perfectly while scoring $0/50$ on ordering. Clamped above the ceiling, saturated at the floor, and flat within families though not across the corpus: three ways for a measure to be unable to move *where the test looks*, which is why Table 8 reports the two axes side by side and never combines them — averaged with an ordering score, a vacuous $20/20$ would read as competence.

5.5 COMPARISON WITH EVIDENCE-FREE BASELINES, ON GOLD AND MINED GRAPHS

The gold relation source establishes what the model does with a correct graph; it does not establish that the exercise needs a probabilistic model at all. Answering that requires holding rival measures to the same ladders — measures that see the same atoms, consult no retrieved evidence, and return a single score.

The two axes, and why they are these sizes. A baseline returns one number per response, so only the *readout-independent* content of a constraint transfers to it. That leaves two axes. **Increase** takes the C1 consecutive-increase pairs together with the C3 endpoint separations over the ten increase-type families — eight CONFLICT and two CHAIN — deduplicated across readouts, since a rung pair is one assertion rather than one per readout. That is 50 assertions. The C2 class is excluded by construction: it predicts the internal behaviour of a *particular* readout, and a single-score baseline makes no claim a C2 constraint could test. **Invariance** takes every rung pair of the four invariance-type families — three ORDER and one CONTROL — and counts a pair satisfied when the score does not move, to 10^{-6}. That is 20 pairs. These 70 assertions are therefore a strict subset of the 202 of Table 7, and

the percentages are *not* comparable between the two tables. The design is deliberately two-sided: the first axis asks whether a measure can tell a more coherent response from a less coherent one, the second whether it stays put when nothing about coherence has changed, and a measure is useful only if it does both.

Both axes are scored *per backend*, and for the same reason. A measure that calls an LLM is run once under each of the two, which fixes not only the scores it assigns along an increase ladder but equally the scores it assigns across a permuted one; there is no sense in which the second axis is the more stable of the two, and it is in fact where the two backends diverge most. Table 8 is therefore organized by whether the backend can matter at all rather than by what kind of measure a row is. Blocks **a** and **b** come first, and their cells span each axis: the LCS on gold graphs calls no LLM, and the model-free baselines are bit-identical across the two runs, so in neither case is there a second number to report. Blocks **c** and **d** follow, and there each axis splits into a `llama` and a `gpt-oss` sub-column. Grouping them this way puts the two conditions whose backend dependence is the finding — the mined graphs and the LLM judges — side by side in the lower half of the table, and keeps the backend-independent reference rows above them as a fixed background to read against.

Gold graphs against the baselines (blocks a, b and d). Given a correct relation graph, the LCS is the only measure tested that passes both axes non-vacuously — $49/50$ on ordering for the headline $\mathrm{LCS}_{\mathrm{bl}}$, the one miss being a rung pair with identical gold graphs, and $37/50$ for the diagnostic $\mathrm{LCS}_{\mathrm{mm}}$, with $20/20$ on invariance for both — and no baseline matches it on both at once: the best ordering score any of them reaches is RC's $30/50$ at $16/20$ invariance, and every baseline that beats $16/20$ on invariance scores $0/50$ on ordering. The baselines fail in three distinguishable ways, and the differences between the failures are more informative than the ranking. The split between blocks **b** and **d** cuts across those three: the cohesion metrics and the two controls are model-free and sit in **b**, while the contradiction-based measures and the judges call an LLM and sit in **d**, so the same failure family is read in one block or the other according to whether a backend can move it. The *cohesion* metrics are the strongest and the most instructive: RC reaches $30/50$ and LC $28/50$ while being defined over noun repetition alone, so they cannot separate a response that contradicts itself from one that does not when both repeat the same nouns — a neighbouring construct measured well, not this one measured badly. The *contradiction-based* measures detect conflict and then lose it in aggregation: ROSCOE-SC in its published maximum form scores $0/50$, while the same pairwise probabilities aggregated by mean reach 16–$17/50$, so the loss is attributable to the aggregation rather than to the signal. The *judges* are the least stable: G-Eval moves from $0/50$ to $21/50$ and direct rating from $4/50$ to $18/50$ on a change of backend alone, on identical prose and identical rubric, against a background in which every model-free baseline is bit-identical across those same two runs — the internal check that the harness is deterministic where it should be, and what makes the swing attributable to the judge rather than to the pipeline. Three baselines, finally, reach $20/20$ on invariance without measuring anything, being unable to vary within a family at all — which is why the two axes are reported side by side rather than summarized, a $0/50$ beside a vacuous $20/20$ being a different object from a measure that is mediocre at both.

The two axes move in opposite directions across backends. Reading both axes per backend makes a sharper statement available than either axis alone supports, and it concerns exactly the incumbent measure. `llama-3.3` is uniformly the *better* backend on invariance — $13/20$ against $5/20$ on G-Eval, $14/20$ against $5/20$ on direct rating, $9/20$ against $5/20$ on ROSCOE-SC (mean) — and, on the same three measures, uniformly the *worse* on increase, at $0/50$, $4/50$ and $16/50$ against $21/50$, $18/50$ and $17/50$. The backend that orders responses well drifts under paraphrase, and the backend that stays put cannot order; neither judge is good at both, and which failure a single-judge evaluation reports is decided by the judge it happens to pick. This is the mechanism behind the instability rather than a second instance of it: §F.11 measures `llama`'s G-Eval at three distinct values over seventy items with mean 0.996, so it is nearly constant, and a nearly constant measure passes an invariance test and fails an ordering test for one and the same reason. Reading the two axes side by side is what keeps that trade-off from reading as a strength: in each of the four judge columns one axis is bought with the other, and no column has both above half.

Mined graphs (block c). On the increase axis the mined sources are more competitive than Table 7's aggregate suggests, spanning 40–70%: two of the twelve rows clear the best evidence-free baseline's 60% — `llama`/WINDOWED $\mathrm{LCS}_{\mathrm{lp}}$ at $35/50$ and `llama`/BIDIRECTIONAL $\mathrm{LCS}_{\mathrm{cn}}$ at $34/50$ — three

Table 8: **Everything on the same two axes.** Four blocks in one table, ordered so that the two backend-independent conditions come first: **a** the LCS on *gold* relation graphs and **b** the model-free baselines, neither of which can vary with the backend — **a** calls no LLM, and **b**'s rows are bit-identical across the two runs — then **c** the LCS on graphs *mined* from the response text and **d** the evidence-free baselines that call an LLM, both of which are reported per backend and split into `llama` and `gpt-oss` sub-columns. Every cell is the assertion *count* with the same figure as a *percentage* in parentheses. **Increase** is the 50 readout-independent assertions (C1 + C3 over the ten increase-type families); **Invariance** the 20 rung pairs of the four invariance-type families, satisfied when the score is unchanged to 10^{-6}. A measure is useful only if it does well on *both*, so the two axes are to be read together and neither alone: they are deliberately not combined into a summary score, since the interesting failures are precisely the measures that are strong on one axis and weak on the other. In block **c** the rows are the two candidate policies and the sub-columns the two *miners*, so a mined graph is identified by reading a row label together with a sub-column head. These percentages are *not* comparable to Table 7's 34.7–37.1%, which runs over the fuller 202-assertion set including the C2 class. **The Invariance axis is the consequential one:** every row of block **c** is $0/20$ under *both* miners, except the headline $\mathrm{LCS_{bl}}$ at $1/20$ under `gpt-oss` WIN, while gold is $20/20$. $\mathrm{LCS_{bl}}$ rows are scored on the same cached graphs as the rest.

	Increase (of 50)		**Invariance** (of 20)	
Measure	`llama`	`gpt-oss`	`llama`	`gpt-oss`
a. *LCS on* GOLD *relation graphs — calls no LLM*				
$\mathrm{LCS_{bl}}$	49 (**98**%)		20 (**100**%)	
$\mathrm{LCS_{mm}}$	37 (74%)		20 (100%)	
$\mathrm{LCS_{lp}}$	36 (72%)		20 (100%)	
$\mathrm{LCS_{cn}}$	34 (68%)		20 (100%)	
b. *evidence-free baselines: model-free — bit-identical across the two runs*				
RC (cohesion)	30 (60%)		16 (80%)	
LC (cohesion)	28 (56%)		16 (80%)	
LENGTH (control)	20 (40%)		12 (60%)	
ENTITY GRAPH	0 (0%)		20 (100%)†	
CLAIM COUNT (control)	0 (0%)		20 (100%)†	
c. *LCS on* MINED *relation graphs — rows are the candidate policy, sub-columns the miner*				
WIN / $\mathrm{LCS_{bl}}$	22 (44%)	25 (50%)	0 (0%)	1 (5%)
WIN / $\mathrm{LCS_{mm}}$	20 (40%)	25 (50%)	0 (0%)	0 (0%)
WIN / $\mathrm{LCS_{lp}}$	35 (70%)	29 (58%)	0 (0%)	0 (0%)
WIN / $\mathrm{LCS_{cn}}$	23 (46%)	25 (50%)	0 (0%)	0 (0%)
BID / $\mathrm{LCS_{bl}}$	31 (62%)	22 (44%)	0 (0%)	0 (0%)
BID / $\mathrm{LCS_{mm}}$	28 (56%)	21 (42%)	0 (0%)	0 (0%)
BID / $\mathrm{LCS_{lp}}$	25 (50%)	28 (56%)	0 (0%)	0 (0%)
BID / $\mathrm{LCS_{cn}}$	34 (68%)	24 (48%)	0 (0%)	0 (0%)
d. *evidence-free baselines: call an LLM — both axes move with the backend*				
ROSCOE-SC (mean)	16 (32%)	17 (34%)	9 (45%)	5 (25%)
CONTRADICTION RATE	12 (24%)	12 (24%)	9 (45%)	8 (40%)
JUDGE, G-Eval	0 (0%)	21 (42%)	13 (65%)	5 (25%)
JUDGE, direct rating	4 (8%)	18 (36%)	14 (70%)	5 (25%)
ROSCOE-SC (max)	0 (0%)	0 (0%)	20 (100%)†	20 (100%)‡

WIN/BID abbreviate the WINDOWED and BIDIRECTIONAL candidate policies. † vacuous: the measure never varies *within* a family, so it cannot fail an invariance pair (§5.4). ‡ vacuous by tolerance rather than by constancy: under `gpt-oss` this measure does take 18 distinct values, but every one lies below 5×10^{-7}, so no pair clears the 10^{-6} threshold.

more reach 56–58%, and the remaining seven fall below it. The invariance columns are where the pipeline breaks: *every one of the twelve mined rows scores* $0/20$ *under both miners*, against $20/20$ for all three gold rows. Splitting that axis per backend makes the failure harder to dismiss rather than easier: it is not one miner dragging an average down, but two independently trained models, six readout–policy combinations each, failing every single one of the twenty pairs — twelve cells at the floor with no exception to appeal to. Nor is it a scoring artefact. Those ladders hold the claim set fixed and permute the prose, so a measure reading a correct graph cannot move at all, and the

mined graphs move because neither miner is reorder-invariant (§5.3). The gold rows' advantage over the baselines is therefore concentrated entirely on the second axis, and the honest one-line summary of the pipeline as it stands is that mined relations buy competitive ordering and forfeit invariance completely.

One row invites a misreading worth forestalling. `llama`/WINDOWED $\mathrm{LCS}_{\mathrm{lp}}$ at $35/50$ sits just below the gold $\mathrm{LCS}_{\mathrm{lp}}$ of $36/50$, which suggests mined relations nearly match gold on this readout. They do not: that source's per-rung sequences are not monotone — on one CONFLICT family the five rungs read $0.162, 0.169, 0.138, 0.125, 0.130$ — so it satisfies the consecutive-increase and endpoint assertions the axis counts while wandering in between, and its mean value (0.224) is roughly half gold's (0.410). The readout-independent subset is a coarse instrument by construction, which is exactly why it is paired with the invariance axis rather than reported alone.

Take-away. The gap between the LCS and the baselines is one of *kind* rather than of degree. No evidence-free baseline tested here both orders responses by coherence and stays invariant under meaning-preserving edits, and the measures that come closest on one axis are not the ones that come closest on the other: the cohesion metrics order tolerably and drift, the vacuous three never drift and never order, and the judges trade one axis for the other as the backend changes. Every baseline row — blocks **b** and **d** alike — is weak somewhere, and the table is laid out so that where each one is weak stays visible. Given a correct graph the LCS does both, and the invariance half is structural rather than fitted — permuting sentences leaves the Markov network identical, so it is a property of the model rather than a result that happened to come out well. Given a *mined* graph it does neither reliably, and the table is arranged so that the two statements cannot be confused: what the model buys and what the current pipeline delivers are different claims, and only the first is settled.

From gold graphs to mined ones. The rows above hold the relation graph correct and ask whether a probabilistic model is needed at all. They do not say whether the end-to-end pipeline delivers, and that is a separate question with a different answer, which is why blocks **a** and **c** of Table 8 sit in the same units: only the provenance of the relation graph changes down the table, so a mined source can be read against gold and against a cohesion metric in a single glance.

5.6 DO READERS AGREE? A HUMAN STUDY

Every ordering in this section is declared by construction: the corpus states how consecutive rungs must rank, and a measure is scored against that declaration. A constructed ground truth is only worth having if readers agree it is the truth, so we ran a human study on a subset of the ladders. The falsifiable version of the claim is sharp — if readers cannot reproduce the declared orderings, then 82.7% against those orderings measures agreement with our perturbation recipe rather than with coherence, and every comparison in Table 8 loses its referent.

Design. Four annotators each judged the same 14 screens, a complete design of 56 ratings with no missing data. A screen shows two responses and the query that produced them, and asks which is more logically coherent — A, B, or about equal — with a confidence rating, a free-text reason, and a checkbox for whether the two looked like the same content reordered. The screens cover seven families across all four ladder types and seven topics: 10 *increase* screens, where the corpus declares one side more coherent, and 4 *invariance* screens, where it declares the two equal. Families were chosen on *measured* disagreement between the LCS and the strongest evidence-free baseline rather than on convenience, and one family was kept deliberately because the baseline wins there. Two pairs per family bracket the ends of each ladder: the $(0, 4)$ endpoint assertion, the largest intended gap, and the $(0, 1)$ adjacent step, the hardest. The CONTROL ladder is symmetric by construction, so its $(0, 4)$ pair is byte-identical and would ask a reader to compare a text with itself; it contributes $(0, 2)$ and $(1, 2)$ instead.

Annotators saw only the query and the two responses. This redaction is not incidental: `items.jsonl` states the intended answer in six independent places — the rung name is literally worse or coherent, the item id suffix is monotone in the ladder, the perturbation record names the edit and its call-list *length* alone reveals the rung index, the notes say outright that the items form an ordered ladder, and per-atom flags mark the planted falsehoods — and App. F's export asserts against a forbidden-key list rather than relying on omission. Both invariance outcomes were

Table 9: The four invariance screens, split by ladder type. Both ladders are built from operators that provably move no factor, so the LCS is exactly invariant across every rung of both — to six decimals, $\mathrm{LCS}_{\mathrm{mm}} = 0.784182$ on all five ORDER rungs and 0.788987 on all five CONTROL rungs. Readers separate the two cases sharply. Fisher's exact test on equal-versus-not by ladder gives $p = 0.007$; the ratings within a ladder are four annotators over two screens of one family rather than eight independent trials, so this is a descriptive contrast rather than a family-level significance claim.

Ladder	Screens	"Equal" ratings	LCS	What readers said
CONTROL	s013, s014	$6/8$	invariant	the pair is the same content
ORDER	s011, s012	$0/8$	invariant	the shuffled response is worse

pre-declared publishable before the data arrived, which matters for how the result below should be read: readers calling the reordered rungs equal would validate the invariance requirement, and readers seeing a real difference would say the requirement is too strict. We got one of each, and which one depended on the ladder.

Agreement. Krippendorff's α (nominal) is 0.748 over all screens, 0.744 on the increase screens and 0.667 on the invariance screens. Two screens have no strict majority and are reported as genuinely ambiguous rather than resolved: on a corpus whose ground truth is declared by construction, "readers do not agree here" is a finding about the item, not noise to average away. Per-annotator accuracy against the declared answer on the increase screens is 9, 9, 7 and 7 of 10. The median time per screen is 15.5 minutes; the maximum is 42 hours, which is a browser tab left open rather than reading time, so we report the median only and no mean.

Readers reproduce the increase ladders. Eight of the ten increase screens have a majority matching the declared answer, and 32 of the 40 individual ratings agree with it. At the rating level that is far from chance ($p = 1.8 \times 10^{-9}$ against a uniform choice over the three options, $p = 9.1 \times 10^{-5}$ against a coin flip between A and B). At the screen level, $8/10$ gives $p = 0.003$ against the uniform null but only $p = 0.055$ against a coin flip, which does not clear 0.05: with ten screens the majority-level test is underpowered, and we state that rather than quote the favourable null alone. The free-text reasons say what readers were responding to. Of the 55 ratings of 56 that carry a written reason, 32 name a relational defect — a contradiction, a presupposition used before it is introduced, a broken chain of support — and only 2 mention surface fluency or style. Readers were reasoning about structure, not readability.

One invariance requirement is validated and the other is not. Pooled, the invariance screens look like a failure: 1 of 4 majorities match. Split by ladder type they are two different results, and the pooled fraction is an average of them (Table 9).

On CONTROL, which exchanges Precedence↔Succession while preserving chronology, six of eight ratings say the pair is equal and one screen is unanimous. That invariance requirement is human-validated, and the LCS's exact flatness there is a property readers endorse. On ORDER, which permutes sentences, *not one* of the eight ratings called the pair equal, and all eight preferred the same side — the less aggressively shuffled one. On both screens the rejected side was the shuffle_full rung, and on one of them readers preferred a mere Precedence swap over it, so they were tracking the *degree* of scrambling that the shuffle levels encode.

The reasons name a specific and checkable defect: dangling anaphora. Two annotators independently flagged that the shuffled response "relies on *that deficit* and *that stress* before those concepts are introduced" and "contains summary-level language to concepts before they have been introduced". This is verifiable in the corpus text. In the base response the phrase *that stress* occurs after the sentence introducing drought-induced stress; in shuffle_full it occurs roughly $1{,}500$ characters earlier, with no antecedent anywhere before it. The claim set is identical, every relation is identical, and the perturbation record says so in as many words: *sentence order only; edge set unchanged.*

So the LCS is invariant here for a principled reason and readers disagree for an equally principled one. Atoms are order-free by construction — nothing in Eq. 1 references claim position (Rem. 2) — which is what makes that determinism provable; the cost is that referential cohesion between

Table 10: Agreement with the human majority on the twelve screens that have one, beside agreement with the declared ordering on all fourteen. LCS rows are the mm readout. The gold row is the model given a correct graph; the mined rows are the end-to-end pipeline. Only measures covering all fourteen screens are listed. Generated by `scripts/analyze_human_study.py`.

Measure	vs. human majority	vs. declared
$\mathrm{LCS}_{\mathrm{mm}}$, gold graph	$\mathbf{9}/12$	$\mathbf{14}/14$
$\mathrm{LCS}_{\mathrm{cn}}$, gold graph	$8/12$	$13/14$
$\mathrm{LCS}_{\mathrm{mm}}$, mined (`gpt-oss`, bidir.)	$7/12$	$7/14$
$\mathrm{LCS}_{\mathrm{mm}}$, mined (`llama`, windowed)	$3/12$	$4/14$
JUDGE direct (`gpt-oss`)	$8/12$	$5/14$
LENGTH control	$7/12$	$9/14$
JUDGE G-Eval (`gpt-oss`)	$6/12$	$7/14$
ROSCOE-SC mean (`gpt-oss`)	$6/12$	$6/14$
RC cohesion	$4/12$	$8/14$
CLAIM COUNT control	$2/12$	$4/14$
CONTRADICTION RATE (`gpt-oss`)	$1/12$	$2/14$

sentences is not representable at all, and it is a real component of what readers call coherence. The ORDER ladder's $20/20$ invariance is therefore not a human-referenced success but a measured blind spot, and we count it as a limitation (§5.10) rather than as evidence. The CONTROL result is the human-referenced success, and it is the one the invariance claim should rest on.

Measures against the human majority. Scoring each measure the way a reader was scored — mapping the sign of its two rung scores onto the side that held the higher rung, with the ladder scorer's own 10^{-6} tie tolerance so that float noise reads as "equal" rather than as a preference — gives the only comparison in this paper where a measure is graded against human judgement rather than against a construction. The twelve screens with a majority are the denominator; a measure cannot be credited for agreeing with a majority that does not exist.

Three readings, in descending order of how comfortable they are. First, on a correct graph the LCS leads both columns, at $9/12$ against the best baseline's $8/12$ and $14/14$ against the best baseline's $9/14$; and its three misses are exactly the cases already discussed — the two ORDER screens, plus one screen where four readers unanimously called a single-conflict repair too subtle to matter while the model separated the pair. Second, the baseline ranking is not the ladder ranking: the length control reaches $7/12$, which is a reminder that fourteen screens cannot separate measures finely, and the direct judge at $8/12$ does far better against readers than against the declared orderings ($5/14$) — a judge and a human agree partly because both read prose. Third and least comfortable, the mined arms do not beat the best baseline here. At $7/12$ and $3/12$ they are well below the gold row, which is the same gold-versus-mined gap as finding (2) of §5.9, now measured against readers instead of against the corpus. The human study localizes the bottleneck to relation extraction rather than to the scoring model, and it does not show that the deployable configuration tracks human judgement.

What this does and does not establish. It establishes that the increase ladders track reader judgement, so the declared orderings are not an artefact of our recipe; that the CONTROL invariance requirement is one readers endorse; and that the ORDER invariance requirement is not. It does not establish human correlation at scale. Fourteen screens over seven of the fourteen families, one generator, four annotators and a single judgement per screen is a validation of the corpus, not a benchmark of measures against human coherence ratings; the comparison above should be read as a consistency check with a known answer, and the wide confidence intervals implied by $n = 12$ are why we draw no ranking conclusion from the one-screen differences among the baselines.

Summary. Four annotators rated 14 screens drawn from seven of the corpus's fourteen families, 56 ratings in a complete design, at $\alpha = 0.748$. Three conclusions follow, and together they change how two numbers elsewhere in this paper should be read. *First, the constructed increase orderings are real.* Readers reproduce 8 of 10 of them by majority and 32 of 40 at the rating level ($p = 1.8 \times 10^{-9}$ against a uniform null), and their written reasons name relational defects — contradiction, presupposition,

broken support — in 32 of 55 cases against surface fluency in 2. The 82.7% gold-graph figure of §5.3 is therefore agreement with coherence as readers construe it, not agreement with our perturbation recipe. *Second, the two invariance ladders are not one requirement, and only one of them is validated.* CONTROL, which reverses a relation's direction, draws 6 of 8 "equal" ratings; ORDER, which shuffles sentences, draws 0 of 8, with readers naming anaphora left stranded by the permutation (Fisher $p = 0.007$). Because assertion order reaches Eq. 1 only through which pairs are related, that defect is outside the model's expressive range rather than mismeasured by it, so the $45/45$ ORDER invariance result is a blind spot and the invariance claim rests on CONTROL's $15/15$. *Third, the gold-versus-mined gap is not an artefact of the corpus.* Given a correct graph the measure leads every rival against the human majority, $9/12$ to the best baseline's $8/12$; given a mined one it reaches $3/12$–$7/12$ and leads nothing. Finding (2)'s bottleneck thus reproduces against human judgement, on responses and readers that know nothing of the declared constraints — which is the strongest available evidence that relation extraction, and not the scoring model, is what stands between this measure and use. What the study cannot support is a ranking of measures against human coherence ratings, for the reasons just given; 56 ratings validate a corpus, they do not benchmark a field.

5.7 REAL RESPONSES: THE FACTUALITY CORPORA

LoCoBench establishes what the model does when the ordering is known, at the cost of being synthetic. The complementary question — what the measure does on prose nobody constructed for it — needs a natural corpus, and answering it also exercises the two-stage composition of §3.6 end to end, which the ladder corpus never does because it supplies no evidence.

One corpus of 250 responses answers three questions, and this section takes them in turn. Can the measure tell real responses apart at all, where the incumbent judges are the comparison a reader will want? What does the factuality stage of §3.6 actually contribute, which only an ablation that removes it can say? And what do the miners *extract* — not how well they score, but what relations they find — which turns out to separate them more sharply than any score does. The three share one sample, one pipeline and one set of runs, so they are reported together; each is answerable on this corpus and the first is the only one the corpus can settle on its own terms, since it carries human labels at the atom level but no coherence gold standard.

THE CORPUS, AND WHAT IT CAN AND CANNOT SETTLE

Corpus and cost. We draw a stratified sample of 250 responses, 50 from each of five long-form factuality datasets (Table 12), and run the full pipeline over each: retrieval, NLI, stage-one inference, relation mining under both candidate policies, all four readouts, and every baseline of §5.4. Sampling is not a convenience. The complete corpus is 1,137 items and 28,646 atoms, which prices at roughly 131K NLI calls plus 619K mining calls *per model*; the 250-item sample still took 25.6 and 61.4 hours of wall time for the two models reported here. The sample is a deterministic function of its seed and every item records its source file and line, so it can be regrown or extended without redoing completed work.

How Table 13 is computed. Stage one returns, for each atom a_i, a posterior marginal $\pi_i = \mathbb{P}(a_i{=}1)$ given the passages retrieved for it. Those π_i are the quantity the table averages, and they are also exactly what §3.6 hands to the coherence MRF as unary factors, so the table doubles as a description of the priors every later result inherits. Two choices in the arithmetic are worth stating because they change the numbers. First, the means are *micro*: every atom counts once, pooled across responses, rather than averaging within a response and then across responses. Micro is the right unit here because the claim under test is about atoms — whether a supported atom receives a higher marginal than an unsupported one — and a macro average would weight a three-atom response equally with a forty-atom one. It is not a cosmetic choice: the same check computed macro gives a separation of $+0.319$ rather than $+0.357$. Second, the two blocks of the table have different denominators. The left block runs over all 6,266 atoms of the 250-response sample; the right block runs over the 1,555 atoms of the 50 BIO responses, which are the only atoms in either corpus carrying a human label. The remaining 4,711 atoms are scored by stage one but have nothing to be scored *against*. Alignment is positional: BIO ships one label per atom in assertion order, matched index by index to the marginals, with no length mismatch on any of the 50 items.

Stage one behaves, and one dataset can prove it. On that labelled subset the separation is clean under both backends: supported atoms average 0.933 and 0.887 against 0.576 and 0.523 for unsupported, a gap of $+0.357$ and $+0.364$. The two figures are not identical, and their near-equality is the point — the backends differ by 0.059 on the all-atom mean and by 0.007 on the separation, so retrieval and NLI quality shift the scale while leaving the ordering intact. That is the pattern that licenses using these marginals as coherence priors: the prior slot needs a value that ranks a well-supported claim above a poorly-supported one, not a calibrated probability. What the check cannot establish is calibration of the absolute level, and §5.7 shows that the level is in fact saturated in a way that costs the coherence readouts resolution — so this table licenses the handoff and the ablation prices it.

How Table 14 is computed. Each row is one measure scored on all 250 responses under one backend, and each cell reports three things about that set of 250 numbers: their mean, their standard deviation, and **d**, the count of *distinct* values the measure returned, computed at four decimal places. The last is the one that matters here, and it needs no ground truth to interpret: a measure that returns the same score for every response cannot rank any response above another, whatever that score is and however well it correlates with anything else. So d is a *precondition* for usefulness rather than a measure of accuracy, and it is the only question this corpus can settle, since it carries human labels at the atom level but no coherence gold standard. The LCS rows give all four readouts under both candidate policies, so the effect of the candidate set is read off directly rather than inferred.

The judges collapse where the measure does not. Both LLM judges saturate on natural prose. Under `llama-3.3-70b` the direct rating returns three distinct values across 250 responses, with a standard deviation of 0.000, and G-Eval five: they rate essentially everything at ceiling. ROSCOE-SC in its published maximum form collapses further still, to two distinct values under `gpt-oss-120b` — the max-aggregation failure of §5.4 appearing on real text rather than in an ablation. Against that, every LCS readout discriminates: across all sixteen readout–policy–backend cells the smallest count is 24, so even the least informative configuration separates more responses than the *best* judge configuration does at 16. The candidate policy shifts these counts but never collapses them — BIDIRECTIONAL returns fewer distinct values than WINDOWED in all eight comparisons, consistent with the extra spurious factors of §5.2 pulling marginals toward a common region, yet its worst cell still clears every judge. That the ordering is uniform across readouts, policies and backends is what makes it a property of the measure rather than of one configuration.

This corroborates §5.5's judge result from the opposite direction: on constructed ladders the judges moved 42 points on a change of backend alone, and on natural prose they barely move at all. One measure, two corpora, two incompatible failure modes — and neither corpus alone would have shown both.

The two axes order the corpora differently. The quantity that rises across the five datasets is the per-response *factuality score* — the fraction of a response's atoms stage one judges true, which is a harder read of the same evidence than the marginals of Table 13 and is the column reported in Table 15. It runs 0.67, 0.90, 0.89, 0.96, 0.97 under `llama`, rising across the five with one inversion of 0.006 between ASKHIST and BOOKS, and monotonically under `gpt-oss` (0.60, 0.77, 0.79, 0.88, 0.94). $\mathrm{LCS}_{\mathrm{cn}}$ does not follow either sequence, running 0.92, 0.59, 0.82, 0.76, 0.93 over the same datasets in the same order. ASKHIST is the sharpest case: its responses are well supported by retrieved evidence and among the least internally consistent in the sample. Both backends reproduce that divergence, which is the claim at issue — not the monotonicity of the factuality sequence, which holds exactly on one backend and up to a 0.006 inversion on the other.

A question levels cannot settle. One reading of the divergence just described is that coherence and factuality genuinely disagree about these datasets; another is that the coherence numbers inherited the factuality ones and the divergence is what survived the inheritance. Levels alone cannot choose between them, and the correlations sharpen the difficulty rather than resolving it: Spearman correlation against the stage-one factuality score is $+0.67$ for $\mathrm{LCS}_{\mathrm{mm}}$ and $+0.35$ for $\mathrm{LCS}_{\mathrm{cn}}$, and the first figure is partly expected *by construction*, since §3.6 makes the factuality posteriors the coherence MRF's priors and a mean of posterior marginals must inherit some of them. Only removing stage one can separate agreement from inheritance, which is what the ablation below does.

Table 11: The uniform-prior ablation, on both complete runs. The same 250 responses and the same readouts are scored twice: once with the stage-one factuality posteriors in the prior slot, once with uniform priors $\pi_i \equiv \frac{1}{2}$, which removes stage one entirely. Items are paired by response. **d** counts the distinct values a readout takes over the 250 items — on a corpus with no ordering ground truth it is what separates a measure that can rank responses from one that cannot — and ρ is Spearman correlation against the stage-one factuality score, which is defined in both conditions because it always comes from the priors-on run. The relation graphs were mined separately in the two runs and are not identical item by item (only 18–27 of the 250 items per arm share even a relation count), so each row compares two runs over the same responses rather than one graph under two priors. Two patterns hold in every block. Removing the priors *raises* **d**, in 14 of the 16 readout–policy cells (bold marks the larger of each pair; the exceptions are $\mathrm{LCS}_{\mathrm{re}}$ under `gpt-oss-120b` on both policies). And it *lowers* ρ for the three readouts that inherit priors directly, sharply so for $\mathrm{LCS}_{\mathrm{mm}}$ — which is the inheritance the ablation was built to quantify. $\mathrm{LCS}_{\mathrm{lp}}$ is the exception in the other direction and is discussed in the text.

			factuality priors			uniform priors		
Miner	Policy	Readout	mean	d	ρ	mean	d	ρ
`llama-3.3-70b`	WIN	$\mathrm{LCS}_{\mathrm{bl}}$		*n/a**		0.929	231	+0.333
		$\mathrm{LCS}_{\mathrm{mm}}$	0.976	90	+0.665	0.955	**196**	+0.398
		$\mathrm{LCS}_{\mathrm{cn}}$	0.803	74	+0.347	0.815	**163**	+0.212
		$\mathrm{LCS}_{\mathrm{lp}}$	0.685	45	+0.265	0.772	**86**	+0.339
		$\mathrm{LCS}_{\mathrm{re}}$	0.436	69	+0.393	0.451	**86**	+0.020
	BID	$\mathrm{LCS}_{\mathrm{bl}}$		*n/a**		0.951	212	+0.376
		$\mathrm{LCS}_{\mathrm{mm}}$	0.980	79	+0.626	0.948	**171**	+0.369
		$\mathrm{LCS}_{\mathrm{cn}}$	0.804	59	+0.296	0.805	**147**	+0.184
		$\mathrm{LCS}_{\mathrm{lp}}$	0.661	35	+0.287	0.735	**58**	+0.297
		$\mathrm{LCS}_{\mathrm{re}}$	0.427	51	+0.381	0.441	**63**	+0.204
`gpt-oss-120b`	WIN	$\mathrm{LCS}_{\mathrm{bl}}$		*n/a**		0.828	246	+0.246
		$\mathrm{LCS}_{\mathrm{mm}}$	0.948	148	+0.674	0.908	**224**	+0.169
		$\mathrm{LCS}_{\mathrm{cn}}$	0.711	134	+0.361	0.704	**212**	+0.080
		$\mathrm{LCS}_{\mathrm{lp}}$	0.671	85	+0.165	0.764	**120**	+0.188
		$\mathrm{LCS}_{\mathrm{re}}$	0.478	**37**	+0.407	0.491	32	−0.027
	BID	$\mathrm{LCS}_{\mathrm{bl}}$		*n/a**		0.856	239	+0.181
		$\mathrm{LCS}_{\mathrm{mm}}$	0.960	111	+0.560	0.909	**195**	+0.103
		$\mathrm{LCS}_{\mathrm{cn}}$	0.721	87	+0.266	0.708	**169**	+0.046
		$\mathrm{LCS}_{\mathrm{lp}}$	0.620	58	+0.131	0.726	**85**	+0.173
		$\mathrm{LCS}_{\mathrm{re}}$	0.464	**24**	+0.380	0.481	15	+0.124

WIN is the forward-only WINDOWED candidate policy, BID the BIDIRECTIONAL one. All ρ are against the same quantity, the stage-one `factuality_score` of the priors-on run. One item of the 250 mines no relations under uniform priors and therefore scores at the prior throughout; it is retained rather than dropped. *The headline $\mathrm{LCS}_{\mathrm{bl}}$ was defined after these runs and is scored on the cached graphs. Only the uniform-prior runs stored theirs; the factuality-prior runs kept scores only, and their graphs differ from the uniform-prior ones, so those cells cannot be filled without re-mining. Bold compares the two prior conditions for readouts (a)–(d) only.

Table 12: The factuality sample: 50 responses from each of five long-form datasets, drawn with a fixed seed and stratified so that no dataset dominates. Atom counts are per response. Both models score the *same* 250 responses, so the corpus block is model-independent; what differs by model is the pipeline run over it (Tables 13 and 14).

Dataset	Domain	Items	Atoms/item	Relations/item*
BIO	biographies (human-labelled)	50	31.1	85.3
ASKHIST	history questions	50	23.2	87.5
BOOKS	book questions	50	24.0	89.0
ELI5	explanatory answers	50	21.1	76.0
LFOBJ	long-form objective	50	25.9	111.2
Total		**250**	25.1	89.8

* mined relations per response, `llama-3.3-70b` under the WINDOWED policy.

Table 13: Stage-one factuality marginals, per backend. Every cell is a mean of *per-atom posterior marginals* $\pi_i = \mathbb{P}(a_i{=}1)$ — the values §3.6 hands to the coherence MRF as its priors — and means are taken over atoms, pooling across responses, so a 40-atom response contributes forty times what a one-atom response does. **The two blocks have different denominators.** *All atoms* covers all 6,266 atoms of the 250-response sample, and **sd** is the spread of π_i across those atoms rather than any between-response or run-to-run variation. The right-hand block is the external check and covers only the BIO subset: 1,555 atoms, a quarter of the sample, being the 50 BIO responses, which are the only ones in either corpus carrying human atom-level labels. **sep.** is the supported mean minus the not-supported mean. Both backends separate the two classes by nearly the same margin while disagreeing on the absolute level.

	All atoms (6,266)		BIO only (1,555): vs. human labels		
Backend	mean	sd	supported	not supported	sep.
`llama-3.3-70b`	0.912	0.228	0.933	0.576	**+0.357**
`gpt-oss-120b`	0.853	0.289	0.887	0.523	**+0.364**

The BIO block partitions its 1,555 labelled atoms exactly: 992 supported and 563 not-supported. The remaining 4,711 atoms of the sample carry no human label, so they enter the left block and not the right.

Table 14: Discriminative power on 250 real responses, by backend. **d** is the number of *distinct* values a measure returns over the 250 items (at four decimal places): a measure with d near 1 assigns nearly every response the same score and cannot rank them, whatever its mean. Values $d \leq 10$ are bold, marking collapse. The two judges and ROSCOE-SC (max) collapse; every LCS readout under either candidate policy and both cohesion metrics do not. All four readouts are given under both policies, so the effect of the candidate set can be read off directly: it shifts d but never collapses it. The four model-free baselines are identical across backends by construction and are given once.

	`llama-3.3-70b`			`gpt-oss-120b`		
Measure	mean	sd	d	mean	sd	d
this work, WINDOWED *candidate policy*						
$\mathrm{LCS_{mm}}$	0.976	0.069	90	0.948	0.090	148
$\mathrm{LCS_{lp}}$	0.685	0.450	45	0.671	0.433	85
$\mathrm{LCS_{cn}}$	0.803	0.246	74	0.711	0.253	134
$\mathrm{LCS_{re}}$	0.436	0.154	69	0.478	0.090	37
this work, BIDIRECTIONAL *candidate policy*						
$\mathrm{LCS_{mm}}$	0.980	0.059	79	0.960	0.078	111
$\mathrm{LCS_{lp}}$	0.661	0.465	35	0.620	0.460	58
$\mathrm{LCS_{cn}}$	0.804	0.241	59	0.721	0.251	87
$\mathrm{LCS_{re}}$	0.427	0.166	51	0.464	0.123	24
baselines that call an LLM						
CONTRADICTION RATE	0.997	0.008	48	0.996	0.009	54
ROSCOE-SC (mean)	0.996	0.011	73	0.994	0.012	69
ROSCOE-SC (max)	0.625	0.467	51	0.564	0.496	**2**
JUDGE, G-Eval	1.000	0.003	**5**	0.917	0.101	**10**
JUDGE, direct	1.000	0.000	**3**	0.829	0.162	16
model-free baselines: identical across backends						
RC (cohesion)	0.502	sd 0.180	$d = 224$			
LC (cohesion)	0.184	sd 0.064	$d = 209$			
ENTITY GRAPH	0.972	sd 0.119	$d = 17$			
LENGTH (control)	0.207	sd 0.046	$d = 136$			

WHAT THE FACTUALITY STAGE CONTRIBUTES: THE UNIFORM-PRIOR ABLATION

The condition is a single substitution: the prior slot is filled by $\pi_i \equiv \frac{1}{2}$, which removes stage one entirely — no retrieval, no claim–passage NLI, no factuality assessment of any kind — while leaving the mined graph, the factor tables and the readouts untouched. Because only the priors change, any

Table 15: The two axes order the datasets differently. Every cell is a mean over the 50 items of a dataset, with the standard deviation in grey after it, WINDOWED policy. Factuality rises monotonically down each block under both backends. $\mathrm{LCS}_{\mathrm{cn}}$ and $\mathrm{LCS}_{\mathrm{lp}}$ do not follow it and instead agree with each other, both bottoming out on ASKHIST (bold): responses well supported by retrieved evidence and among the least internally consistent in the sample. $\mathrm{LCS}_{\mathrm{mm}}$ tracks factuality closely, as its construction requires — its priors *are* the factuality posteriors — which is why the divergence is read off the other readouts. $\mathrm{LCS}_{\mathrm{re}}$ is the least informative here, sitting near 0.5 throughout and, under `gpt-oss-120b`, varying by less than the standard deviation of any single dataset.

Backend	Dataset	factuality	$\mathrm{LCS}_{\mathrm{mm}}$	$\mathrm{LCS}_{\mathrm{cn}}$	$\mathrm{LCS}_{\mathrm{re}}$	$\mathrm{LCS}_{\mathrm{lp}}$
`llama-3.3-70b`	BIO	0.665 ±0.211	0.945 ±0.103	0.922 ±0.171	0.498 ±0.008	0.929 ±0.232
	ASKHIST	0.897 ±0.112	0.967 ±0.078	**0.589** ±0.198	0.364 ±0.205	**0.287** ±0.431
	BOOKS	0.891 ±0.153	0.982 ±0.069	0.820 ±0.250	0.442 ±0.145	0.721 ±0.431
	ELI5	0.963 ±0.065	0.995 ±0.013	0.757 ±0.252	0.405 ±0.180	0.603 ±0.472
	LFOBJ	0.974 ±0.033	0.990 ±0.020	0.927 ±0.174	0.469 ±0.107	0.885 ±0.312
`gpt-oss-120b`	BIO	0.602 ±0.194	0.877 ±0.101	0.776 ±0.238	0.472 ±0.100	0.902 ±0.234
	ASKHIST	0.769 ±0.151	0.942 ±0.091	**0.522** ±0.136	0.488 ±0.046	**0.377** ±0.432
	BOOKS	0.789 ±0.196	0.959 ±0.110	0.729 ±0.276	0.482 ±0.075	0.658 ±0.432
	ELI5	0.878 ±0.138	0.982 ±0.042	0.668 ±0.238	0.481 ±0.084	0.544 ±0.471
	LFOBJ	0.935 ±0.067	0.977 ±0.033	0.859 ±0.219	0.464 ±0.123	0.875 ±0.301

difference between the two arms is attributable to the factuality stage and to nothing else. Everything else is held fixed: the same 250 responses, both candidate-pair policies, all four readouts, all twelve baselines, and 500 of 500 runs completed without a failure, on *both* miners. All four arms cover the *same items*, so every comparison in this part is paired — Table 11 gives the full 16-cell grid and Table 16 the two-miner summary at the WINDOWED policy.

The ablation is also substantially cheaper, which is itself informative about where the cost of the two-stage pipeline lies: 0.35× the two-stage arm for `llama-3.3-70b` and 0.52× for `gpt-oss-120b`. Stage one is most of the bill for the former and about half for the latter, whose reasoning-channel output makes relation mining the dominant expense instead.

The three main results do not point the same way. Inheritance is real but miner-dependent; removing the priors *raises* the measure's resolution rather than lowering it; and the two arms rank responses differently, so neither is a proxy for the other. Three shorter findings follow them: the conflict separation survives the ablation untouched, one association we cannot explain, and a harness check the baselines supply for free.

Inheritance is real, partial, and miner-dependent. $\mathrm{LCS}_{\mathrm{mm}}$'s correlation with factuality falls from +0.665 to +0.398 once the priors are uniform under `llama-3.3-70b`, so roughly 40% of it came through the prior slot and the remainder is agreement between two independently computed quantities. Under `gpt-oss-120b` the same fall is far steeper, +0.674 to +0.169, so on that miner the association was inheritance almost in full; $\mathrm{LCS}_{\mathrm{re}}$ behaves the same way on both (+0.393 → +0.020 and +0.407 → −0.027). How much of a coherence–factuality association is real rather than inherited therefore cannot be read off one backend. $\mathrm{LCS}_{\mathrm{lp}}$ is the lone readout whose correlation does *not* fall (+0.265 → +0.339 for `llama`), and the exception is principled: it is the one readout that is a normalized partition function rather than a function of the marginals, so it has no prior mass to inherit in the first place. What does *not* depend on the prior handoff is the per-dataset divergence above — $\mathrm{LCS}_{\mathrm{cn}}$ still takes its minimum on ASKHIST under uniform priors (0.620 against 0.589 with priors) — so the two axes order the corpora differently for reasons of their own.

Removing the priors *increases* the measure's resolution. This is the ablation's central and least expected result. In 7 of the 8 miner–readout cells of Table 16 the uniform-prior arm takes *more* distinct values than the two-stage arm, and the margins are large rather than marginal: 196 against 90 for $\mathrm{LCS}_{\mathrm{mm}}$ under `llama`, 212 against 134 for $\mathrm{LCS}_{\mathrm{cn}}$ under `gpt-oss`. The single exception is $\mathrm{LCS}_{\mathrm{re}}$ under `gpt-oss` (32 against 37). Extending the count to the BIDIRECTIONAL policy as well (Table 11) gives 14 of 16 cells, the two exceptions being that same readout under that same miner

Table 16: The four arms: two miners × two prior sources, over the same 250 responses with the WINDOWED policy. It is the two-miner slice of Table 11, repeated here with both miners adjacent so they can be read against each other directly. **d** counts the *distinct* values a readout takes across the corpus; on a corpus with no ordering ground truth it is the column that separates a measure which can rank responses from one which cannot. Bold marks the higher **d** within each miner–readout pair, for (a)–(d). *Not cached: the headline $\mathrm{LCS}_{\mathrm{bl}}$ is scored on stored graphs, and only the uniform-prior runs stored theirs (Table 11).

		$\mathrm{LCS}_{\mathrm{bl}}$		$\mathrm{LCS}_{\mathrm{mm}}$		$\mathrm{LCS}_{\mathrm{cn}}$		$\mathrm{LCS}_{\mathrm{lp}}$		$\mathrm{LCS}_{\mathrm{re}}$	
Miner	Priors	mean	d	mean	d	mean	d	mean	d	mean	d
`llama-3.3-70b`	factuality	*n/a**		0.9759	90	0.8032	74	0.6850	45	0.4358	69
	uniform	0.9294	231	0.9547	**196**	0.8150	**163**	0.7719	**86**	0.4507	**86**
`gpt-oss-120b`	factuality	*n/a**		0.9475	148	0.7107	134	0.6713	85	0.4775	**37**
	uniform	0.8278	246	0.9084	**224**	0.7037	**212**	0.7641	**120**	0.4914	32

on both policies — so the effect is general across readouts, policies and miners, with one localized counterexample rather than a scattering of them.

The mechanism is prior saturation, and it is measurable. The factuality posteriors are strongly ceiling-weighted — 82.7% of claims above 0.99 for `llama` and 72.1% for `gpt-oss` — so as unary factors they push the coherence marginals toward 1 and *compress* the range the readout can occupy. Replacing them with 0.5 returns that range to the relations. Consistent with this, the more saturated miner shows the larger relative gain. The implication for practice is not that priors are harmful but that *saturated* priors cost resolution, and a factuality stage intended to prime a coherence model should be calibrated with that in mind rather than merely accurate.

The two arms are not interchangeable: the priors change *which* responses score low. Rank correlation between the arms over the 250 paired responses is $+0.69$ for $\mathrm{LCS}_{\mathrm{mm}}$, $+0.78$ for $\mathrm{LCS}_{\mathrm{cn}}$ and $+0.91$ for $\mathrm{LCS}_{\mathrm{lp}}$ — substantial but well short of the agreement that would make one arm a proxy for the other. The paired difference is systematic rather than noisy: on $\mathrm{LCS}_{\mathrm{mm}}$ the uniform arm is lower on 87% of items, by -0.021 on average ($t = -5.3$). So the factuality stage does contribute, but what it contributes is a reordering of the low-scoring tail, not a shift in level.

The conflict separation owes nothing to the priors. §5.9 reports that the consistency readout separates responses whose mined graph contains a conflict edge from those whose does not. That result survives the ablation intact. With uniform priors, conflict-bearing responses score 0.519 against 0.976 for conflict-free ones under `llama` (a gap of $+0.457$, Welch $t = 32.9$) and 0.488 against 0.926 under `gpt-oss` ($+0.438$, $t = 32.2$) — statistically indistinguishable from the two-stage arm's $+0.489$ and $+0.438$. The readout is therefore responding to relational structure it mined, not to factuality signal it inherited, and the ablation establishes this directly rather than by the correlational argument available before.

One association we cannot explain. The uniform arm consults no factuality signal whatsoever, yet its $\mathrm{LCS}_{\mathrm{mm}}$ still correlates $+0.379$ with the two-stage arm's mean prior ($t = 6.45$, $n = 250$). Mean-centring within dataset reduces this to $+0.213$ but does not remove it ($t = 3.43$), so a between-corpus offset is part of the story and not all of it. Per domain the association is significant on two of five (BIO $+0.34$, LFOBJ $+0.34$) and not on the other three ($+0.06$ to $+0.14$), and nothing we measured distinguishes those groups — notably *not* the spread of factuality within a domain, which is uncorrelated with the effect ($+0.12$ across the five). Controlling for contradiction count, relation count, claim count or mean relation strength does not account for it either. The plausible reading is a shared upstream cause: a response whose claims are well supported may also be one whose relations mine cleanly. We report the association and decline to assert the mechanism.

The baselines confirm the harness and sharpen the judge result. Under uniform priors every model-free baseline is bit-identical to its two-stage value — RC 0.5024, LC 0.1840, LENGTH 0.2066, claim count 0.3916 — as it must be, since none of them consults a prior; this is the internal check that the two runs differ only where intended. The LLM judges behave exactly as §5.9 describes

Table 17: The mined relation inventory over the same 250 responses, as a percentage of all relations the miner emitted under each policy. Level-1 couplings select the factor table; Level-2 senses compile to them through $\mathcal{C}$. WIN is the forward-only WINDOWED policy, BID the BIDIRECTIONAL one. Senses below 1% under both miners (ALTERNATIVE, PRECEDENCE, SUCCESSION) are omitted from the lower block; the last two are ordering-only and compile to no factor, which is why they should be rare by design.

	`llama-3.3-70b`		`gpt-oss-120b`	
	WIN	BID	WIN	BID
Relations mined	22,417	24,015	14,695	15,656
per response	89.7	96.1	58.8	62.6
contributing a factor	84.8	91.8	58.8	62.6
Level-1 coupling (% of relations mined)				
ENTAILMENT	90.1	88.0	84.9	84.2
CONTRADICTION	3.9	3.9	**10.8**	**11.6**
EQUIVALENCE	4.5	6.0	4.1	3.9
CO_NECESSITY	1.4	2.0	0.1	0.3
EXCLUSIVE	0.1	0.1	0.0	0.0
Level-2 sense (% of relations mined)				
Instantiation	**48.0**	**49.3**	32.9	34.6
Evidence	23.0	22.7	18.5	20.1
Cause-Effect	12.3	9.0	**25.0**	**18.1**
Restatement	4.3	5.7	4.1	3.9
Contrast	3.5	3.6	6.5	7.5
Condition	1.2	1.3	4.5	3.8
Disjunction	1.3	1.9	0.1	0.3
Effect-Cause	0.5	1.6	4.0	**7.6**
Concession	0.4	0.4	4.4	4.1
None	5.3	4.4	0.0	0.0

and the ablation adds a cross-model contrast: `llama`'s direct rating takes 4 distinct values over 250 responses and its G-Eval 5, while `gpt-oss`'s take 17 and 12. As §5.9 notes, that apparent resolution is four rungs of a five-point scale plus seed-averaging noise, not measurement. Against either judge the LCS readouts take 120–224 values on the same responses.

WHAT THE MINERS ACTUALLY EXTRACT: RELATION TYPES AND SENSES

Everything so far has compared miners by the scores they produce and, in §5.2, by their agreement with gold labels. Neither says what a miner *extracts*. That matters here because the previous part attributed a score difference to the priors across two runs over the same responses; the graph itself is the other input, and it has not yet been looked at. (The two runs mined their graphs separately, and those graphs are not identical item by item — Table 11 — so part of that difference may come from mining variation rather than from the priors.) Because the uniform-prior runs record every mined relation with its Level-1 coupling and Level-2 sense rather than merely a count, the two miners can be compared by the content of their output over the same 250 responses (Table 17). This separates them more sharply than any score does.

The two miners disagree about what coherence structure *is*, not merely how much of it there is. The volume difference is the visible one — `llama-3.3-70b` emits 89.7 relations per response against `gpt-oss-120b`'s 58.8 — but the composition difference matters more. `llama` places nearly half of everything it finds in a single sense, Instantiation (48.0%), which is the most generic link in the menu: one claim exemplifies another. `gpt-oss` distributes its output across the causal and adversative senses instead, with Cause-Effect at 25.0% against 12.3%, Concession at 4.4% against 0.4% and Condition at 4.5% against 1.2%. So the miner that finds 53% more relations finds a *narrower* variety of them, and a reader comparing two LCS values computed under different miners is not comparing two measurements of one quantity.

The contradiction gap is systematic, and it explains the score gap. Under `gpt-oss` 10.8% of mined relations are contradictions against `llama`'s 3.9%, and this is not driven by a few responses: computing the rate *per response* and pairing, `gpt-oss` is higher on 133 of 249 responses with a paired $t = +10.5$ (computed below). This is the mechanism behind a score difference the ablation could not explain: `gpt-oss`'s consistency mean is the lower of the two (0.704 against 0.815, Table 11) not because it scores the same graphs more harshly but because it *finds different graphs* — roughly two and a half times the conflict density on identical prose. Which miner is closer to correct is not settled by this corpus, which has no relation ground truth; §5.2 settles it on the corpus that does, where `gpt-oss` is the more accurate of the two at every level.

How the paired comparison is computed, and why the pairing carries the claim. For each response i scored by both miners we form $r_i^{(m)}$, the fraction of that miner's mined relations on that response labelled CONTRADICTION, and take the within-response difference $d_i = r_i^{(\text{GPT-OSS})} - r_i^{(\text{LLAMA})}$. The statistic is $t = \bar{d}/(s_d/\sqrt{n})$ on $n-1$ degrees of freedom, s_d being the sample standard deviation of the differences. Here $\bar{d} = +0.0593$, $s_d = 0.0892$ and $n = 249$ — one of the 250 responses mines no relations at all, leaving its rate undefined — giving $t = 0.0593/0.00565 = +10.50$ on 248 degrees of freedom. Differencing within a response is the point: length, topic, claim count and intrinsic difficulty are properties of the response, so they enter both terms and cancel, and what survives is the miner effect alone. An unpaired comparison would instead have to overcome the between-response variance, which on this corpus is several times the effect being measured.

The sign decomposition is the more robust form of the same evidence, and we prefer it. Of the 249 paired responses `gpt-oss` has the higher contradiction rate on 133 and `llama` on 4, with 112 ties — overwhelmingly responses where both miners find no contradiction at all. The two miners therefore agree exactly on 45% of the corpus and, on the 137 responses where they disagree, `gpt-oss` is higher 133 times. That asymmetry needs no distributional assumption, which matters because d is heavily zero-inflated and the normality premise behind t is correspondingly weak; at $n = 249$ the central limit theorem protects the mean difference, but the 133-to-4 split is the claim we would defend.

What licenses the causal reading is that the scoring model is byte-identical across the two arms — same factor tables, same readouts, same inference, same priors — so a difference in output must originate in the input graph, and the paired test shows the input graphs differ in precisely the way that moves this readout: CONTRADICTION is what the consistency readout penalizes. Two things the test does *not* establish. It says nothing about which miner is correct: on a corpus without relation ground truth a higher contradiction rate is equally consistent with better sensitivity and with over-firing, and §5.8 settles that on the corpus which does carry gold labels. And it does not license comparing absolute LCS values across miners — if two systems report different coherence scores for one response, the difference may lie entirely in what their extractors found, so a reported score must always name its extractor.

Not every mined relation becomes a factor, and the two miners differ there too. The senses that compile to Level-1 NONE — None, Precedence, Succession — record that two claims are related while coupling no truth values, so they contribute nothing to Eq. 1. Under `llama` these account for 5.4% of mined relations (1,208 of 22,417); under `gpt-oss`, one relation in total. The comparison that matters for the network is therefore 84.8 against 58.8 factors per response, narrowing `llama`'s apparent advantage from 53% to 44%. We note this because a relation count is the natural thing to report and is the wrong thing to compare: a miner can inflate it with links the model then ignores.

The sense inventory exposes the forward-only policy's structural limit. §5.2 measures that limit against gold as a backward-recall gap. The sense distribution shows the same defect intrinsically, with no labels required. Effect-Cause is the reversed form of Cause-Effect: it asserts that the target causes the source, so realizing it requires an arc that runs backward in assertion order. Under WINDOWED both miners emit **exactly zero** backward-running relations — 0 of 22,417 and 0 of 14,695 — and Effect-Cause accordingly collapses to 0.5% and 4.0%, against 1.6% and 7.6% under BIDIRECTIONAL, where 46.7% and 45.3% of relations run backward. The raw counts show the asymmetry directly: under `llama`'s windowed policy Effect-Cause appears 106 times forward and 0 times backward, against 73 forward and 303 backward under its bidirectional one (589/0 and

436/760 for gpt-oss). A forward-only candidate window cannot express "B, because A" when B is asserted first, whatever the prose says, and the resulting deficit is visible in the inventory of a single run rather than only in comparison with a gold graph. That is the more useful diagnostic, because it is available on any corpus.

A caveat on scope. The relation inventory of this part — unlike the scores and the ablation above it — is available for the uniform-prior runs only. The two-stage runs predate the change that records each relation's labels rather than their count, so a priors-on comparison of the same kind would require re-mining. Nothing in this part's analysis depends on the prior setting — mining precedes scoring and is unaffected by what fills the prior slot — so we expect it to transfer, but we have not measured that and do not claim it.

5.8 The same inventory on LoCoBench, where the labels adjudicate it

The analysis of §5.7 can be repeated on the ladder corpus, and doing so adds the one thing the factuality corpora cannot supply: gold relations, against which a miner's inventory can be scored rather than merely described. The four mined sources of §5.2 already recorded each relation's coupling and sense, so no new mining was required.

The contradiction gap replicates, at almost exactly the same magnitude. Repeating the paired test of §5.7 on these 70 items gives $\bar{d} = +0.0577$ ($t = +3.98$, WIN) and $\bar{d} = +0.0574$ ($t = +5.26$, BID), against $+0.0593$ on the factuality corpora. Two corpora sharing no responses, no topics and no generation procedure yield the same 5.8-point difference between these two miners, which is better evidence that the effect is a property of the extractors than either corpus could furnish alone. The sign split is 46 to 21 (WIN) and 53 to 17 (BID), so the asymmetry is present but less extreme than on natural prose, where near-half the responses contained no conflict for either miner to disagree about.

Here the gold labels say which miner is right, and it is the one finding more. The corpus declares 27.2% of its edge-producing relations to be contradictions. gpt-oss-120b mines 27.1% under WINDOWED; llama-3.3-70b mines 22.5%. The higher rate is thus the better-calibrated one, and recall settles what a rate alone cannot: of the 164 gold contradictions gpt-oss recovers 70.1% against llama's 48.2%, and 76.2% against 56.7% under BIDIRECTIONAL. The gap is *sensitivity* rather than over-firing — llama misses roughly half the conflicts its corpus contains. This resolves in gpt-oss's favour the question §5.7 had to leave open, and it agrees with Table 6, where gpt-oss leads on every accuracy column at matched policy. The verdict holds at Level 1 and at the level of conflict recall, which is what those measurements cover; the sense block qualifies it in one place, and we take that up below.

The gold sense column shows which senses are over- and under-produced. A Level-1 comparison can say that a miner finds too many entailments; only a gold sense distribution can say *which* discourse relation it is over-reporting, and three patterns in Table 18 are sharp enough to name. Evidence is heavily over-produced by both miners — 21.9–35.5% of mined relations against 13.6% in gold, worst for llama at 2.6× the gold rate. This is the same failure the factuality corpus shows as Instantiation over-use (§5.7): a miner unsure which support relation holds falls back on the most generic one available, and which sense plays that role depends on the corpus. Conversely Concession and Cause-Effect are under-produced by *all four* sources (3.8–7.9% against gold's 12.3%, and 2.6–8.2% against 11.4%), and these are the senses whose factor tables do the discriminating work — a concession is the one construction whose Level-1 coupling depends on whether the response resolves it (§3.2). Missing them is therefore costlier than the raw percentages suggest.

Neither miner dominates at Level 2, which the Level-1 block could not have revealed. On Cause-Effect gpt-oss is the closer of the two (absolute error 5.1 points against llama's 8.8); on Concession llama is closer (4.4 against 7.9). And on one sense llama is clearly the better-calibrated miner: Condition is 0.7% of gold relations and 0.9–1.1% of llama's output, against 5.8–6.1% of gpt-oss's — an 8.7× over-production, a sense it invents where the corpus barely uses it. This is the qualification promised above: gpt-oss is the more accurate miner on coupling F1 and on conflict recall, and it is *not* uniformly more accurate sense by sense. A single accuracy column cannot show that, which is the argument for reporting the inventory at both levels.

Table 18: The mined relation inventory on the ladder corpus (70 items), as a percentage of all relations emitted, at both levels of the taxonomy. *Gold* is the corpus's own distribution over its 603 edge-producing relations — the column the factuality corpora have no counterpart for, and the reason this table can show *over*- and *under*-production per sense rather than merely describing a distribution. Level-1 couplings select the factor table; Level-2 senses compile to them through $\mathcal{C}$, and the sense rows are ordered by their gold share so a mined column can be read against it down the page. Unlike Table 17, no sense is omitted here except by folding the three ordering-only ones into a single line: Alternative and Disjunction are negligible on natural prose but major here, since eight of the fourteen families vary conflict structure by construction. The final block is gold-referenced recall of CONTRADICTION specifically: of the 164 gold contradictions, the fraction each source recovered with the correct coupling. WIN is the forward-only WINDOWED policy, BID the BIDIRECTIONAL one.

	`llama-3.3-70b`		`gpt-oss-120b`		
	WIN	BID	WIN	BID	Gold
Relations mined per item	15.8	45.3	15.6	42.5	8.6
Level-1 coupling (% of relations mined)					
ENTAILMENT	47.2	49.9	43.0	46.9	32.8
CONTRADICTION	22.5	22.4	**27.1**	**28.3**	27.2
EXCLUSIVE	15.1	12.6	12.4	10.2	12.9
CO_NECESSITY	8.0	10.2	6.2	6.6	11.6
EQUIVALENCE	7.1	5.0	11.2	8.0	15.4
Level-2 sense (% of relations mined), ordered by gold share					
Restatement	7.1	4.9	11.2	8.0	15.4
Contrast	14.7	17.1	22.8	24.6	14.9
Evidence	32.6	35.5	21.9	22.2	13.6
Alternative	15.0	12.6	12.4	10.2	12.9
Concession	7.9	5.5	4.4	3.8	12.3
Disjunction	8.0	10.1	6.2	6.6	11.6
Cause-Effect	2.6	2.6	6.3	8.2	11.4
Instantiation	9.9	9.6	7.6	6.9	6.3
Effect-Cause	0.0	0.3	1.4	3.6	0.8
Condition	0.9	1.1	5.8	6.1	0.7
ordering-only†	1.3	0.9	0.0	0.0	0.0
Gold-referenced CONTRADICTION *recall (%)*					
of the 164 gold contradictions	48.2	56.7	**70.1**	**76.2**	—

† Precedence, Succession and None, folded into one line. These compile to Level-1 NONE and contribute no factor (§3.2); the ladder pipeline records their pre-compile coupling, so they are counted under ENTAILMENT in the block above, inflating it by at most 1.3 points for `llama` and not at all for `gpt-oss`, which emits none of them.

Effect-Cause adds a third reading of the forward-only limit, now against a known truth: gold puts the reversed causal sense at 0.8%, and the mined rates are 0.0 and 0.3% for `llama` against 1.4 and 3.6% for `gpt-oss` under the two policies. Both miners recover more of it when the candidate window admits backward pairs, as §5.7 predicts, but only `gpt-oss` overshoots the gold rate in doing so — so the policy relaxes a structural cap and the miner decides what fills the space.

A caution against reading one corpus's inventory through the other. The two inventories differ sharply, and the cause is the corpus rather than the miners. The ladder corpus is built to contain conflict — eight of its fourteen families vary conflict structure by construction — so both miners report contradiction rates above 22% here against below 11% on naturally occurring prose. Conversely Instantiation, which dominates the factuality inventory at 33–48%, accounts for 6.9–9.9% of either miner's output here (Table 18) against 6.3% in gold — so on this corpus it is barely over-produced at all, and the sense that dominates a mined inventory is set by the text as much as by the extractor. Alternative and Disjunction make the same point from the other direction: both fall below the reporting threshold of Table 17 on natural prose and together account for a fifth of everything mined here. A miner's inventory is therefore a joint property of the extractor and the text it reads, and a calibration established on one corpus does not transfer to the other. What does transfer is the

difference between two miners on shared text, which is what both paired tests measure and why both are reported.

5.9 SUMMARY OF FINDINGS

We close by stating what the experiments establish, each finding paired with the measurement that supports it and with the boundary beyond which it does not hold.

(1) The probabilistic model orders coherence correctly when its input is correct. On gold relation graphs the readouts satisfy 167 of 202 declared constraints (82.7%), including $45/45$ and $15/15$ on the two ladder types where exact flatness is required (Table 7). Because sources differ *only* in where the relation graph comes from, this figure is a property of the scoring model in isolation.

(2) Relation extraction, not scoring, is the bottleneck — and the gap is large. The same readouts on mined graphs reach 34.7–37.1%, a 46–48 point shortfall attributable to estimation alone (Table 7). The magnitude is consistent with the estimator's measured accuracy: at 29–52% coupling F1 (Table 6), roughly half the factors in a mined network are wrong or missing, and a ladder whose consecutive rungs differ by one or two relations cannot survive that. This is the paper's central empirical claim, and it is a claim about *where* the remaining work lies.

(3) The invariance failure is the substantive one, and only a negative control could reveal it. Every mined source scores $0/45$ on ORDER and $0/15$ on CONTROL (Table 7), and $0/20$ on the invariance axis under *each* of the two miners separately (Table 8). This is real movement, not absent scores: on one ORDER family the `gpt-oss` WINDOWED miner emitted 9, 16, 20, 20 and 17 edges across five rungs differing *only* in sentence order. Invariance is structural in the model — permuting sentences changes neither which claims stand in which relation nor the Precedence↔Succession compilation to NONE, so the Markov network is *literally identical* — so mining breaks the premise rather than the theorem. Notably, mined sources are *not* uniformly weak on the increase-type ladders (CONFLICT 53–59 of 112), so a corpus without CONTROL would have judged mined extraction far closer to gold than it is. That is the false positive the negative controls exist to catch.

(4) Candidate-pair policy trades recall against precision, and ladder agreement does not adjudicate it. Forward-only WINDOWED mining is structurally capped, recovering 21.9–29.5% of the 105 backward-running gold relations against 76.2–85.7% for BIDIRECTIONAL; admitting both directions triples the edges and halves coupling precision, inverting the F1 ordering (Table 6). Yet `llama`'s BIDIRECTIONAL source has the *worst* coupling F1 (29.3) and the *best* ladder agreement (37.1%), so these sources are not separated by ladder agreement and candidate policies should not be ranked on it.

(5) Given a correct graph the measure dominates every evidence-free baseline; the pipeline does not yet. All three graded readouts on gold relations sit above the best baseline's 60%, and the LCS is the only measure tested that passes both the increase and invariance axes non-vacuously (Table 8). On mined graphs that dominance does not hold: two of twelve readout–policy combinations clear 60% on increase and none clears zero on invariance (Table 8). The distinction between what the model buys and what the current pipeline delivers is the one we most want not to be blurred.

(6) The baselines fail in three distinct ways, which is why the gap is one of kind. *Cohesion metrics* are the strongest and most instructive, reaching 60% and 56% while being unable to separate a response that contradicts itself from one that does not when both repeat the same nouns — a neighbouring construct, not this one measured badly. *Contradiction-based metrics* detect conflict and then lose it in aggregation, tracking response size. *LLM judges* fail in two ways at once, and it takes both corpora to see it: on the constructed ladders G-Eval moves $0/50 \rightarrow 21/50$ on a change of judge alone, on identical prose and rubric, while every model-free baseline is bit-identical across the same two runs; on real responses the same judges saturate instead, returning three and five distinct scores across 250 items (Table 14). And three baselines reach $20/20$ invariance *vacuously*, two of them constant across all seventy items and the third moving only beneath the 10^{-6} threshold — trivially invariant and trivially unable to order anything, which is why the two axes are always reported together and never summarized into one number.

(7) On real responses the measure discriminates where the incumbents do not. Over 250 responses from five long-form factuality datasets, all four readouts under both candidate policies return between 37 and 148 distinct values, while the two LLM judges return between 3 and 16 and ROSCOE-SC in its published maximum form returns as few as two (Table 14). A measure that

assigns nearly every response the same score cannot rank responses whatever its mean or its reported correlation, so this is a precondition for usefulness rather than a margin of victory — and it is the one comparison the ladder corpus could not make, since there the judges were unstable rather than saturated.

(8) Factuality and coherence order real corpora differently, and the ablation says the separation is real. Across the five datasets, stage-one factuality rises monotonically while $\mathrm{LCS}_{\mathrm{cn}}$ does not follow it; ASKHIST is well supported by retrieved evidence and among the least internally consistent in the sample, and both backends reproduce the ordering (Table 15). This is the orthogonality claim of §1 appearing in data rather than in construction. The obvious objection — that $\mathrm{LCS}_{\mathrm{mm}}$ correlates with factuality at $\rho = +0.67$ simply because its priors *are* the factuality posteriors — is answerable rather than merely acknowledged: rescoring the same 250 items at $\pi_i \equiv \frac{1}{2}$ drops that correlation to $+0.398$ under `llama-3.3-70b` and to $+0.169$ under `gpt-oss-120b`, and leaves the per-dataset ordering intact under both (Table 11). Inheritance therefore accounts for roughly 40% of the association on one miner and nearly all of it on the other, and none of the ordering result. The same ablation carries an unwelcome finding in the other direction, reported in (9).

(9) The two-stage composition costs dynamic range, on both miners, and we would not have found that without the ablation. Removing the priors *raises* discriminative power sharply and in 7 of the 8 miner–readout cells: distinct values go from 90 to 196 of 250 for $\mathrm{LCS}_{\mathrm{mm}}$ and 74 to 163 for $\mathrm{LCS}_{\mathrm{cn}}$ under `llama-3.3-70b`, and from 148 to 224 and 134 to 212 under `gpt-oss-120b` (Tables 11 and 16); the lone exception is $\mathrm{LCS}_{\mathrm{re}}$ under `gpt-oss`. The cause is measurable: the factuality posteriors are ceiling-weighted — 82.7% of claims above 0.99 for `llama`, 72.1% for `gpt-oss` — and a saturated prior pins the posterior marginal whatever the relations say, so the more saturated miner shows the larger gain. The composition of §3.6 therefore buys interpretability, in that a claim is credited only when it is both externally supported and internally upheld, and pays for it in resolution. On this corpus the evidence-free configuration is the more discriminative one, and a practitioner choosing between them should be told so. The finding also refines the design advice: the cost comes from prior *saturation* rather than from composition as such, so a better-calibrated stage one would pay less of it.

(10) The conflict separation owes nothing to the priors. With the prior slot uniform, the consistency readout still separates responses whose mined graph contains a conflict edge from those whose does not, by $+0.457$ under `llama` (Welch $t = 32.9$) and $+0.438$ under `gpt-oss` ($t = 32.2$) — indistinguishable from the two-stage arms' $+0.489$ and $+0.438$ (§5.7). The readout is responding to relational structure it mined rather than to factuality it inherited, and the ablation shows this directly rather than by correlation.

(11) A forward-only candidate policy cannot express a backward discourse relation, and the sense labels show where. Under WINDOWED both miners emit *exactly zero* backward-running relations — 0 of 22,417 and 0 of 14,695 — against 46.7% and 45.3% under BIDIRECTIONAL. The effect localizes to a sense: EFFECT-CAUSE appears 106 times forward and 0 backward under `llama`'s windowed policy, 73 forward and 303 backward under its bidirectional one (§5.7). This is the ladder corpus's backward-recall gap (finding 4) confirmed on naturally occurring prose.

(12) A property no scalar baseline offers: the score decomposes. As a mean of per-claim marginals, the count of claims dragged below their own prior localizes an incoherence to the claims responsible. A rubric scalar, however well correlated, cannot do this.

(13) Readers reproduce the increase ladders, endorse one invariance requirement and reject the other. Four annotators over 14 screens (56 ratings, $\alpha = 0.748$) confirm 8 of 10 declared increase orderings, 32 of 40 at the rating level ($p = 1.8 \times 10^{-9}$ against a uniform null), so those orderings are not an artefact of our perturbation recipe. The invariance screens split by ladder rather than pooling: CONTROL draws 6 of 8 "equal" ratings, while ORDER draws 0 of 8, with readers naming dangling anaphora in the sentence-shuffled rungs (Fisher $p = 0.007$). Since sentence order enters the model only through which pairs are related, the ORDER invariance result is a blind spot rather than a success, and finding (1)'s $45/45$ on that ladder should be read accordingly. Against the human majority the gold-graph measure leads at $9/12$ versus the best baseline's $8/12$, but the mined arms reach only $3/12$–$7/12$, which is finding (2)'s bottleneck measured against readers instead of against the corpus (§5.6).

Lessons learned. Three of these findings only became visible because the evaluation used two corpora with opposite weaknesses, and that is the methodological lesson we would carry to the next measure of this kind. A constructed corpus can fix the ordering and therefore localize a failure — it is what tells us that scoring is sound and extraction is not (findings 1–2), and only its negative controls exposed the invariance collapse that the increase-type ladders alone would have hidden (finding 3). What it cannot do is say whether a measure is useful on text nobody built for it. A natural corpus answers that and nothing else: it has no coherence ground truth, so it cannot rank measures by correctness, but it does reveal that the practical incumbent assigns almost every real response the same score (finding 7) — a failure invisible on ladders, where the judges were unstable rather than flat. Neither corpus alone would have supported both conclusions, and a paper reporting only one of them would have been comfortably wrong in a different direction each time.

Three further lessons are narrower but cost us the most time. First, a readout's dynamic range is a property of its factor tables, not of its definition: the no-priors tables collapse the measure's range by more than half before any relation is mined (§3.3), and no amount of tuning downstream recovers it. Second, an aggregation choice can dominate a modelling choice — ROSCOE-SC differs from our measure in three separable ways, and the maximum is the one that matters, saturating to two distinct values on real prose (§5.4). Third, a component added for interpretability can quietly cost accuracy elsewhere, and only an ablation shows it: filling the prior slot with factuality posteriors halves the measure's distinct-value count on real responses (Table 11), a cost we would have reported as a feature had we not rescored the corpus without it. All three are the kind of thing an ablation finds and a leaderboard hides.

Taken together: the coherence model is sound and its guarantees are structural rather than fitted; the benchmark localizes the open problem in extraction rather than in scoring, with the negative-control ladders making that localization trustworthy; and on real responses the measure discriminates where the incumbent judges do not. The bounds are equally plain. The ladder corpus is LLM-generated and small (70 items, 202 assertions per source), so differences of a few assertions are not resolvable; the factuality sample has no coherence gold standard, so it bounds usefulness rather than correctness. §5.10 bounds the rest.

5.10 LIMITATIONS

Five of the results above are strong enough to be worth bounding precisely. The invariance guarantee holds only given a correct graph and, on one of the two invariance ladders, does not match reader judgement; the pairwise factorization excludes a class of real incoherence; two constants in the design are set by hand rather than derived; and the natural-corpus results bound usefulness rather than correctness.

Sentence-level cohesion is outside the model, and readers do not treat it as outside coherence. The human study of §5.6 validates the CONTROL invariance requirement — six of eight ratings call a Precedence↔Succession swap equal — and refutes the ORDER one: not one of eight ratings called a sentence-shuffled pair equal, and readers named dangling anaphora, a referential defect that leaves the claim set and every relation untouched. Because assertion order enters Eq. 1 only through which pairs are related (Rem. 2), this is unrepresentable rather than badly estimated, exactly as the k-ary case below is. The $20/20$ invariance figure on ORDER is therefore a blind spot and not a human-referenced result, and the invariance claim should rest on CONTROL.

On the natural-corpus point: the factuality sample carries human labels at the *atom* level only, so it supports the discriminative-power comparison of §5.7 and the stage-one check, but it cannot rank measures by agreement with human coherence judgements, which no corpus we know of supplies at this response length. The uniform-prior ($\pi_i \equiv \frac{1}{2}$) condition has been run on that sample under *both* miners (Table 11), which settles the inheritance question; what it cannot settle is whether either configuration agrees with human judgement, for the reason just given. The ladder study supplies response-level human judgement at small scale — 56 ratings over 14 screens — which is enough to validate the declared orderings and not enough to rank measures: the mined arms reach $3/12$ to $7/12$ against the human majority there, below the gold graph's $9/12$ and not above the strongest baseline. The remaining gap is therefore an addition to the corpus rather than to the model: response-level coherence labels at scale, which would turn the discriminative-power comparison into a correctness one.

Proposition 2 (Jointly inconsistent, pairwise consistent)**.** *Let* $\mathbb{P}(\mathbf{a})$ *factorize as in Eq. 1 over* a_1, a_2, a_3 *with non-negative factors. If* $\mathbb{P}(a_1{=}1, a_2{=}1, a_3{=}1) = 0$ *then* $\mathbb{P}(a_i{=}1, a_j{=}1) = 0$ *for some pair* $i \neq j$*, so no member of the family represents "the three claims cannot all hold, but any two can" (proof in App. H).*

Genuinely k-ary conflicts are therefore unrepresentable rather than merely approximated badly, and this is not hypothetical: a planted arithmetic inconsistency — 356 teams forming 32 conferences of exactly 10 members, where no two claims conflict but $32 \times 10 \neq 356$ requires all three — yields *zero* mined conflict edges and is reported as coherent, unchanged under ALL_PAIRS. Transitivity is likewise not enforced. Both defects are expressible as first-order rules in the Markov logic view of App. I, so Rem. 1's closure guarantee is exactly what buys tractability and what costs us this defect. Two constants remain hand-set ($\lambda = 0.45$ and the resolver heuristic), and production inference is weighted mini-bucket at i-bound 6 against 2^n enumeration ($n \leq 20$), an approximation we have not characterized. At 52% coupling F1 at best roughly half a mined network's factors are wrong or missing, so mined-source numbers measure current extraction quality rather than the measure's ceiling. Human scoring of the invariance rungs is no longer open — §5.6 reports it, and it found one requirement validated and one refuted — but human correlation at scale and judge dependence (42 points between judges) remain so, and the corpus is LLM-generated throughout.

6 Conclusion

Coherence is a property of a joint distribution over what a response asserts, readable off a Markov random field whose factors are the relations the response itself draws. Three things follow, each fixing a design choice a rule-counting or rubric-scoring approach leaves open. The factor tables matter: those right for grounding a claim against evidence charge a premise for being a premise, so a coherence measure built on them spends its dynamic range in the wrong place (Prop. 1). The relation vocabulary can be rich at the annotation layer and must be narrow at the inferential one: twelve senses compiling onto five couplings keeps the model a closed pairwise family while leaving the labels interpretable. And not every reading of "how coherent is this?" is admissible: two of four candidate readouts are non-monotone in conflict resolution, measuring whether a conflict is active rather than what should be believed, and a fifth is maximized by asserting no relations at all (Prop. 3). The score that is admissible by construction reads every relation against the priors alone, and is monotone in every relation edit (Thm. 2); we make it the headline and keep the mean marginal to locate an incoherence.

Contributions. Beyond the model and its taxonomy, we contribute an evaluation built to answer a question the field has not been able to ask cleanly: when a coherence measure fails, is the scoring wrong or the input? **LoCoBench** answers it by construction. Its response families hold one claim set fixed and vary only the relations among them, so the same readouts run over a *gold* relation graph and over one *mined* from the text, identical in every other respect, and the difference is attributable to extraction alone. The corpus treats negative controls as first-class — families that permute prose while holding the relation set fixed, and so assert that a coherence measure must *not* move — which turns out to be the part that earns its keep. The two-stage composition (§3.6) makes coherence and factuality two readings of one decomposition rather than two pipelines, and the score decomposes into per-claim marginals, so an incoherence is located rather than merely detected.

Takeaways. The probabilistic layer is sound and the extraction layer is not, and the benchmark makes that a measurement rather than an impression: 82.7% of 202 declared constraints satisfied from gold relation graphs against 34.7–37.1% from mined ones. The sharp form of the gap is invariance — every mined source scores $0/20$ under *both* miners on ladders that permute prose without touching the relation set, where the model's own answer cannot move because the network is literally identical. Only the negative controls could have surfaced it: mined sources are *not* uniformly weak on the increase-type ladders, so a corpus without them would have judged extraction far closer to gold than it is. Against five families of evidence-free baselines, the measure on a correct graph is the only one that both orders responses ($49/50$ for the headline score against the best baseline's $30/50$) and stays exactly invariant ($20/20$), the latter by construction rather than by fitting. The incumbent fails in two opposite ways that took both corpora to see: an LLM judge swings 42 points between judge models

on identical prose, then saturates to 3–16 distinct scores on 250 real responses where our readouts return 37–148.

What cuts against us. Two results run the other way, and both came from ablations rather than the headline comparisons. Composing with factuality costs resolution: filling the prior slot with factuality posteriors roughly halves the measure's distinct-value count on real responses, on both miners, because those posteriors are ceiling-weighted and a saturated prior pins the marginal whatever the relations say. The composition buys interpretability and pays for it in dynamic range — traceable to prior *saturation* rather than to composition as such, so a better-calibrated stage one would pay less. Second, the pairwise closure that buys tractability excludes a class of real incoherence: three claims that cannot all hold while any two can are unrepresentable here rather than badly approximated (Prop. 2), and a planted arithmetic inconsistency of that shape draws zero mined conflict edges and is reported as coherent. Neither defect sits where we expected the ceiling. Relation extraction does: at 29–52% coupling F1 roughly half a mined network's factors are wrong or missing, and no readout can be better than the graph it reads. That is where the next increment belongs, and the benchmark exists so progress on it can be measured rather than asserted.

AI USE STATEMENT

Generative AI is both an object of study and a component of the artifacts released here, and we disclose both roles. The relation estimator of §3.5 is LLM-based by design: its prompts, its logprob-derived confidences and its failure modes are contributions of the paper. The evaluation corpus of §5.1 is LLM-generated throughout, including its gold relation labels, and validated by an independent LLM committee; §5.10 states the consequences for treating it as ground truth. We additionally used AI assistants for code implementation and for editorial assistance on this manuscript. All LLM-generated code was reviewed by the authors and is covered by a test suite that pins the exact numerical values reported here; every number in §5 was traced to a stored experiment record. The research questions, the taxonomy, the readout definitions, the propositions and their proofs are the authors' own. We take responsibility for the final content of this work.

ETHICS STATEMENT

The human study of §5.6 involved four annotators, who judged which of two machine-generated responses was more logically coherent and wrote a free-text reason. No personal data was collected and none is released: the study records carry only an annotator label (A1 to A4), assigned by the import tooling so that no account identifier or email address reaches a results file. The material judged was LLM-generated expository prose on neutral technical topics, with no content about identifiable people. The fixtures include a summary of a published appellate judgment, used as a worked example of ordering-dependent coherence; it is a matter of public record and no private individual is the subject of analysis. The principal ethical risk is misreading: a coherence score is not a truth score, and a response can be perfectly coherent and entirely false. That is precisely why §3.6 keeps the two axes separable, and we caution against deploying LCS alone as a factuality filter — a fluent, internally consistent fabrication is exactly the artifact it scores highest. The released fixtures and corpus deliberately contain false claims, labelled as such, and should not be redistributed without those labels.

REPRODUCIBILITY STATEMENT

The model is fully specified in the paper: Def. 1 gives the joint, §3.3 the potentials, Def. 2 the penalized sets, §3.2 the compile map, Eq. 3 the headline readout and Eq. 5 the diagnostic, with the alternatives and the auxiliary-variable constructions two of them require in App. D. Algorithms 2 and 1 give candidate selection and the end-to-end procedure. App. K reproduces all four estimation prompts verbatim, including the trailing token boundary the surrogate readout depends on. The exact values in §3.7 and App. G are computed by explicit 2^n enumeration — 2^5 for the five-claim pair, 2^{16} for the recall report — and are pinned by regression tests in the released code, so they are reproducible without LLM access. The k-ary representability figures of App. H are likewise exact by 2^3 enumeration, with the seeded random search and the published table both reproduced by a

released script. §5 names every model used and states the sampling regime and its measured variance. Code, fixtures and corpus are released.

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A LoCoBench at a glance

This section is a self-contained summary of the benchmark, for a reader who wants the whole of it in one page; §4 gives the full treatment, with the tables and thresholds this section only names.

The problem it solves. Evaluating a coherence measure requires knowing which of two responses is more coherent — exactly what no natural corpus supplies, since human judgements are expensive, disagree, and do not say *why* one text is worse. Our own study puts numbers on the first two: a median of 15.5 minutes per screen, and $\alpha = 0.748$ with two of fourteen screens drawing no majority at all (§5.6). Perturbing a document and assuming the perturbation hurt is weaker than it looks: the edit lands on raw text and its effect on the thing being measured goes unverified. LoCoBench inverts that. Perturbations apply to a *declared relation graph* rather than to prose, so each edit's intended effect is known per item before any text exists, and the ordering is a property of the construction rather than an assumption about it.

The unit of evaluation. The unit is not a response but a *family*: five responses (*rungs*) over one fixed claim set and one fixed topic, differing along exactly one structural dimension (Def. 3). Holding the claim set fixed is what makes an ordering attributable to coherence rather than to content: every rung asserts the same claims, so every per-claim factuality verdict is identical across the family and cannot explain any score difference. Four family types divide the work (Table 1). CONFLICT varies conflict structure and CHAIN breaks and repairs an entailment chain; both assert a strictly *increasing* score, and the pair matters because a measure built on contradiction detection alone passes the first and fails the second, where the defect is a *missing* relation rather than a present conflict. ORDER and CONTROL apply only meaning-preserving edits and so assert *invariance* — the negative controls, without which no corpus can tell a coherence measure from an edit-count proxy. Constraints come in three classes: a strict increase between consecutive rungs, a *predicted* decrease or flatness, and endpoint separation. A missing score fails rather than passes.

How a family is built. Generation is a five-stage prompted pipeline (Fig. 3): a topic becomes a question (P1), the question becomes 26 atomic claims (P2), a subset of them a relation plan (P3), the plan a base response (P4), and six perturbation operators realized by ten calls (Table 2) turn that base into the four remaining rungs (P5). The surplus at P2 is deliberate — selecting from more claims than are needed lets the plan choose claims supporting the required structure rather than forcing relations onto whatever claims exist. Every prompt is fixed, versioned, and has its instruction count and placeholder contract asserted by a test against the prompt string (Table 3), so a defect is attributable to one stage and repairable there.

How a family earns admission. Three validators gate it, run by a committee that *excludes* the generating model (Table 4). **V1** is the load-bearing one: an independent model is shown the prose and asked to recover the relation graph, and the family is admitted only if enough planned relations come back with the right coupling and sense. This is what makes the gold graph a claim about the text rather than about the generator's intentions: a plan the prose does not realize fails here. **V3** audits fluency, formality and organization and rejects any leakage of the construction into the text; **V4** checks that every planned claim is present. Families are admitted or rejected whole, so no rung is scored against a ladder whose siblings were discarded.

What shipped. Fourteen families, seventy items, 1,115 claims at 15–16 and 609–698 words per item, over fourteen topics spanning four domains (four Natural Sciences, four Engineering & Technology, three Humanities, three Social Sciences), so no topic or domain carries a result. The families declare 670 gold relations, 603 of them edge-producing — the other 67 are ordering-only senses compiling to NONE — and 342 declared *non*-relations, which matter because a measure must be tested on pairs it should leave alone as well as on pairs it should link. Of the 670, 408 (0.61) are marked valid; and 105 of the 603 run *backward*, their target preceding their source in assertion order, which is what makes candidate-pair policy a substantive choice rather than a tuning detail. Generation was by `claude-opus-5`, admission by a cross-vendor committee of `gpt-5.6-terra`, `gemini-3.1-pro` and `claude-sonnet-5`. Together the families yield 202 ordering constraints per scored source (Table 5).

What it can and cannot show. Fixing the claim set isolates coherence from factuality, and acting on a declared graph lets the benchmark say *which* defect a measure misses rather than only that it scored poorly — which is how §5.3 separates a scoring failure from an extraction failure. Two limits bound it: the corpus is LLM-generated throughout, so an idiosyncrasy shared by the generator and all three validators would be invisible here, and at seventy items its 202 assertions separate sources differing by tens of points but not by a few. Both are reasons to read LoCoBench as a diagnostic instrument rather than a leaderboard.

A.1 One family end to end: `f001`

The preceding summary describes the construction in general terms. This subsection instantiates all of it on a single shipped family, `f001` — a CONFLICT ladder on Anthropology (Social Sciences), generated by `claude-opus-5` — so that every object named above can be seen at once: the five rungs, the relation graph each carries, the constraints the family declares, and the scores those constraints are checked against. Every number below is read from the released corpus (`families.json`, `items.jsonl`) and from the GOLD arm of the ladder sweep; none is illustrative.

The family's framing question is: *how do anthropologists adjudicate between rival interpretations when skeletal remains at a burial site show signs of trauma, and what evidence establishes whether the injuries were inflicted by human violence or by post-mortem processes?* All five rungs assert the same 16 claims about that question, in the same order where order is not the thing being varied.

The five rungs. Table 19 gives the ladder. Each non-base rung is a sequence of operator calls applied to the base plan ρ_1 — *not* to the rung below it, so a defect in one rung cannot propagate up the ladder. The relation counts show the mechanism directly: the number of CONTRADICTION edges falls $4 \to 3 \to 3 \to 2 \to 2$ as conflicts are repaired, the count of resolved concessions rises $1 \to 2$ at ρ_2 where the concession is discharged, and ρ_4 additionally drops the lone EXCLUSIVE edge. Because the claim set never changes, no per-claim factuality verdict differs across the five rows, and so none of them can explain any score difference.

Table 19: The five rungs of `f001`. *Calls* are the perturbation operators applied to the base plan; *parent* is the rung the perturbation is applied to, which is the base for all four. *Rel.* counts all declared relations and *edge* only the edge-producing ones (the difference is the ordering-only sense, which compiles to NONE). *Cfl.* counts CONTRADICTION edges and *Res.* resolved concessions — the two quantities the ladder is built to vary.

	Rung	Calls applied to the base	Par.	Rel.	Edge	Cfl.	Res.
ρ_0	worse	spurious_relation	ρ_1	11	10	4	1
ρ_1	base	— (the base response)	—	10	9	3	1
ρ_2	concession_resolved	add_resolution	ρ_1	10	9	3	2
ρ_3	fix_one_conflict	add_resolution, drop_relation	ρ_1	9	8	2	2
ρ_4	coherent	add_resolution, drop_relation $\times 2$	ρ_1	8	7	2	2

The readouts. Table 20 gives all four readouts on all five rungs, from the GOLD relation source. Three of them are graded; re is computed and stored but no constraint is stated over it (§D), which is why the multiplier behind the corpus's 202 assertions is three rather than four. Read down the mm column and the ladder is visible as intended, $0.750 \to 0.823$ end to end; read down cn and it is not monotone, which is the point of the next paragraph rather than a defect.

The constraints, and why there are fourteen. The declaration grid is three graded readouts $\times$ four adjacent rung pairs, plus one endpoint assertion per readout: $12 + 3 = 15$ cells. A CONFLICT family declares only 14, because one cell is deliberately left undeclared — cn on $\rho_2 \to \rho_3$, where the readout is *unconstrained*: the rung both discharges a concession and drops a conflict, and the two move the conflict term in opposite directions by amounts the construction does not fix. Asserting either direction there would be asserting something the design does not predict, so the cell appears in neither C1 nor C2. Over the eight CONFLICT families this single omission is exactly what separates the corpus's 202 assertions from the 210 a uniform 15 per family would give.

Table 20: The readouts on f001, GOLD source. The headline bl rises on every pair of rungs, including $\rho_2 \to \rho_3$ where mm does not. mm rises across the ladder otherwise; the cn dip at ρ_2 is predicted, not incidental (Prop. 4). re is stored but ungraded. The constraints of Table 21 are declared for mm, cn and lp, and are checked against those columns.

	Rung	bl	mm	cn	lp	re
ρ_0	worse	0.437648	0.750400	0.499618	0.342596	0.150678
ρ_1	base	0.465949	0.760539	0.649468	0.374712	0.221853
ρ_2	concession_resolved	0.495681	0.789580	0.494457	0.374268	0.252912
ρ_3	fix_one_conflict	0.497086	0.789239	0.500444	0.425367	0.261813
ρ_4	coherent	0.541393	0.822792	0.523759	0.488651	0.320921

The $1 \to 2$ pair is where the three readouts are asked for three *different* things, and it is the clearest illustration of why the constraint classes are per readout rather than per rung. The edit is add_resolution: a concession is discharged. Then mm must *rise* (C1), because resolving a conflict raises belief in the claims it suppressed; cn must *fall* (C2), because it counts conflict activity and the discharge raises the both-true cell mass, which is Prop. 4 made falsifiable; and lp must be *exactly flat* (C2), because the edit is weight-only and the normalized log-partition is invariant to it. A measure that is monotone across all three has not implemented the concession discount, and the C2 class is what detects that.

Table 21: All 14 declared constraints of f001, checked against Table 20. 12 hold. A constraint is one assertion about one readout on one pair of rungs; Δ is the observed difference, classified against expected with a tie tolerance of 10^{-6}. The undeclared fifteenth cell (cn on $2 \to 3$) is absent by construction, not omitted. Both failures are discussed below; note that both are smaller in magnitude than every passing assertion.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mm	$0 \to 1$	increase	increase	$+0.010139$	✓
C1	cn	$0 \to 1$	increase	increase	$+0.149850$	✓
C1	lp	$0 \to 1$	increase	increase	$+0.032116$	✓
C1	mm	$1 \to 2$	increase	increase	$+0.029040$	✓
C2	cn	$1 \to 2$	decrease	decrease	-0.155011	✓
C2	lp	$1 \to 2$	invariant	decrease	-0.000444	×
C1	mm	$2 \to 3$	increase	decrease	-0.000340	×
C1	lp	$2 \to 3$	increase	increase	$+0.051099$	✓
cn *on* $2 \to 3$*: undeclared (unconstrained)*						
C1	mm	$3 \to 4$	increase	increase	$+0.033553$	✓
C1	cn	$3 \to 4$	increase	increase	$+0.023315$	✓
C1	lp	$3 \to 4$	increase	increase	$+0.063284$	✓
C3	mm	$0 \to 4$	increase	increase	$+0.072392$	✓
C3	cn	$0 \to 4$	increase	increase	$+0.024142$	✓
C3	lp	$0 \to 4$	increase	increase	$+0.146055$	✓

What the two failures show. Both are at the tie floor, but they have different causes. mm on $2 \to 3$ misses by -3.4×10^{-4}: the two rungs differ by a single dropped CONTRADICTION edge of weak strength $p = 0.47$ (the resolution is already present at ρ_2; ρ_3 is derived from ρ_1), and since a sub-$\frac{1}{2}$ contradiction raises belief in its own both-true world, dropping it lowers the belief readout slightly, so the declared strict increase is not delivered. lp on $1 \to 2$ misses by -4.4×10^{-4} against a prediction of *exact* flatness. The prediction is sound — the edit is weight-only — but it is checked at a tie tolerance of 10^{-6} against values from weighted mini-bucket at i-bound 6, whose own approximation error is larger than that. The failure is therefore attributable to inference precision rather than to the model: the ORDER and CONTROL families pass their flatness assertions $45/45$ and $15/15$ only because their rungs compile to the *same* network, making those ties exact. We report the number as it is scored rather than widening the tolerance to absorb it, and note the consequence: f001's $12/14$ is

slightly above the corpus-wide CONFLICT rate of $89/112$, and one of its two failures is a statement about Liu & Ihler (2011) at i-bound 6, not about the coherence model.

The headline readout makes the first failure concrete. On the same five rungs bl rises at every step (Table 20): $+0.028$, $+0.030$, $+0.0014$ and $+0.044$, and $+0.104$ end to end. At $\rho_2 \to \rho_3$, where mm falls by 3.4×10^{-4}, bl rises by 1.4×10^{-3}, three orders of magnitude above the tie tolerance. The single edit there, deleting the weak Contrast contradiction $a_3 \Rightarrow \neg a_2$, is a conflict deletion, which Thm. 2 guarantees cannot lower bl at any weight; the belief readout has no such guarantee and moves the other way.

The same family under mined relations. Holding everything but the relation source fixed (§5.1) and re-running `f001` from graphs mined out of its own response text gives $4/14$, $8/14$, $7/14$ and $7/14$ for the four miner–policy arms of Table 7, against $12/14$ from gold. The claims, the priors, the readouts and every scorer setting are byte-identical across those five numbers, so the gap is caused by the relation graph and nothing else — which is the paper's central empirical claim (§5.3), visible here on one family.

A.2 THE TWO BASELINE AXES: WHERE 50 AND 20 COME FROM

Table 8 scores baselines against 50 increase assertions and 20 invariance assertions, not against the 202 constraints of Table 7. The two counts are computed from the same `families.json`, and this subsection derives both, since a reader who assumes one is a subset-by-filtering of the other will find the percentages irreconcilable — $37/50$ is 74% where $167/202$ is 82.7%, and neither is an error.

Why a baseline cannot be scored on the 202. A constraint in the corpus sense is an assertion about *one readout* on one pair of rungs, and the three graded readouts are what multiply four adjacent pairs and one endpoint pair into the corpus's per-family 14 or 15 (§A.1). A baseline returns a single number per response. It has no mm as against cn as against lp, so the multiplier has nothing to multiply, and only the *readout-independent* content of a constraint transfers to it: the rung pair, and the direction asserted over it. Both axes below are therefore *deduplicated across readouts* — a rung pair is one assertion rather than one per readout — which is the single operation separating 70 from 202.

The increase axis (50). Take the ten increase-type families, eight CONFLICT and two CHAIN. From each, take the C1 consecutive pairs and the C3 endpoint separation, deduplicated. Every one of the ten yields exactly the same five assertions:

$$\text{C1}: 0 \to 1, \quad \text{C1}: 1 \to 2, \quad \text{C1}: 2 \to 3, \quad \text{C1}: 3 \to 4, \quad \text{C3}: 0 \to 4,$$

the four adjacent pairs and the endpoint, for $10 \times 5 = 50$. An assertion is satisfied when the later rung scores strictly higher, by more than 10^{-6}.

Two consequences of the deduplication are worth stating, because they are what make the count uniform rather than ragged. First, **C2 is excluded by construction**, not by an arbitrary filter: it predicts the internal behaviour of a *particular* readout — the concession dip of Prop. 4 — and a single-score baseline makes no claim a C2 constraint could test. Were it included, every baseline would be charged with failing a prediction never made about it. Second, and less obviously, **the undeclared cell that makes a CONFLICT family 14 rather than 15 disappears here**. That family's C1 block holds nine entries over four distinct pairs ($0 \to 1$ and $3 \to 4$ with three readouts each, $2 \to 3$ with two, $1 \to 2$ with one); cn on $2 \to 3$ is unconstrained, but lp still declares an increase on that pair, so the pair survives deduplication intact. A CONFLICT family declaring 14 and a CHAIN family declaring 15 therefore contribute 5 each, and the asymmetry visible in the 202 is invisible in the 50.

The invariance axis (20). Take the four invariance-type families, three ORDER and one CONTROL, and take *every* rung pair each declares: the four adjacent pairs from C2 plus the C3 invariant endpoint, again deduplicated, for $4 \times 5 = 20$. Here C2 is *included*, which is not an inconsistency with the paragraph above: these families set flat = READOUTS, so every cell they declare is an invariance claim rather than a readout-specific prediction, and the readout-independent content of such a claim is exactly "this pair must not move". A pair is satisfied when the two scores differ by at most 10^{-6}. The pairs are read from each family's own constraints rather than from a hardcoded rung range, so a family with a different rung count would still be scored correctly.

One worked axis, on the family of §A.1. On `f001` the increase axis reduces the 14 declared constraints to five, scored here on mm: $0 \to 1$ $(+0.010139)$, $1 \to 2$ $(+0.029040)$, $2 \to 3$ (-0.000340), $3 \to 4$ $(+0.033553)$ and the endpoint $0 \to 4$ $(+0.072392)$ — four of five, the failure being the same near-cancelling $2 \to 3$ pair discussed in §A.1. Summing that computation over the ten increase families gives $4 + 3 + 5 + 4 + 3 + 3 + 3 + 3 + 4 + 5 = 37$ of 50, which is the $\mathrm{LCS}_{\mathrm{mm}}$ row of Table 8. The corresponding per-readout totals are 37, 36 and 34 of 50 for mm, lp and cn, and $20/20$ on invariance for all three; these are asserted as a reproduction check in the reporting script, which refuses to emit any mined number if the gold rows drift from them, on the grounds that an axis definition that no longer reproduces the published gold figures cannot be trusted to score anything else.

What the two counts are and are not. Table 22 sets the two schemes side by side. The 70 assertions are a strict subset of the 202 in the sense that every rung pair scored on an axis is also scored in the corpus count, but they are *not* a subset arrived at by dropping rows: the readout dimension is collapsed, and on the increase axis an entire constraint class is removed. The percentages are therefore not comparable between Table 7 and Table 8, and we report them in separate tables with separate denominators rather than in one normalized column for exactly that reason.

Table 22: The two counting schemes. Both read the same `families.json`; they differ in whether the readout dimension is retained. The right-hand column is what Table 8 uses and the left-hand what Table 7 uses; a percentage from one is not comparable with a percentage from the other. The final row is the one behavioural difference beyond the counting: in the corpus scheme an unscored rung is a *failure*, while on the axes it is dropped from the denominator — which matters only if a measure fails to return a score, and every measure reported here returned all seventy, so both denominators are full.

	Corpus constraints	Baseline axes
Count	202 per relation source	50 increase + 20 invariance
Unit	readout × rung pair	rung pair
Readouts	three, asserted separately	collapsed (single score)
Families	all 14, reported together	10 and 4, reported separately
C1 / C3	included	included
C2	included	increase: excluded; invariance: included
Tie tolerance	10^{-6}	10^{-6}
Used by	Table 7	Table 8
Unscored rung	counts as a failure	dropped from the denominator

Why two axes rather than one number. The split is the point rather than a presentational choice. The increase axis asks whether a measure can tell a more coherent response from a less coherent one; the invariance axis asks whether it stays put when nothing about coherence has changed. A measure is useful only if it does both, and the two are separately cheap to pass in ways that cancel: a constant function scores $20/20$ on invariance and $0/50$ on increase, while a measure tracking edit count can climb the increase ladder and move on every permutation. Averaging them into one figure would let a vacuous $20/20$ flatter exactly the rows that deserve it least — three of the baselines reach perfect invariance without measuring anything (§5.5) — so the two axes are reported side by side and never combined.

B Stage one: probabilities for individual claims

The coherence model consumes claim-level probabilities and reuses the inference machinery that produces them. That stage is FactReasoner (Marinescu et al., 2025); the account here is a summary sufficient to make the composition of §3.6 precise, and it also fixes the label-span estimator whose construction the relation miner of §3.5 reuses.

B.1 Decomposition, retrieval, and relation extraction

A response y is decomposed by an LLM *atomizer* into atomic claims $\mathcal{A} = \{a_1, \ldots, a_n\}$, each a single independently verifiable proposition. A *reviser* then decontextualizes each one, resolving

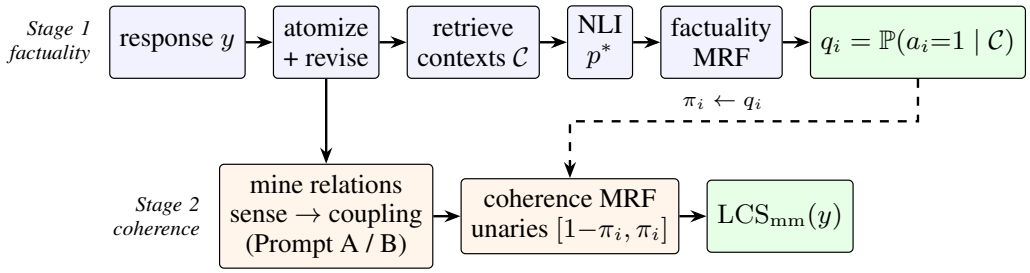


Figure 4: The two-stage pipeline. Stage 1 (§B) is FactReasoner (Marinescu et al., 2025): it grounds each atom against retrieved evidence and returns a posterior marginal q_i. Stage 2 (§3) builds a second MRF over the atoms *alone* and adds one pairwise factor per mined atom–atom relation; the mined relation already carries its Level-1 coupling, so no separate compile step intervenes. The dashed arrow is the only coupling between the stages, and carries the composition $\pi_i \leftarrow q_i$ into the $\phi_i = [1-\pi_i, \pi_i]$ of Def. 1. Atomization is shared, so the response is decomposed once and both stages score the same claim set; setting $\pi_i \equiv \frac{1}{2}$ instead recovers a pure coherence score that consults no external evidence (Rem. 3).

pronouns and anaphora so that the claim is checkable outside its original sentence; near-duplicates are collapsed. For each claim a retriever returns supporting passages, which are deduplicated into an evidence set $\mathcal{C} = \{c_1, \ldots, c_m\}$.

An LLM-based NLI extractor then labels admitted pairs. For a claim–evidence pair the label is drawn from $\{\text{entailment}, \text{contradiction}, \text{neutral}\}$; for an evidence–evidence pair both directions are scored and reconciled, with entailment in both directions relabelled *equivalence*. Neutral verdicts are dropped rather than encoded: a claim about which the evidence is silent contributes no factor and is left at its prior, which is what makes the factor set of §3 contain only the relations someone actually asserted.

B.2 Reading a probability off the generation

Each retained relation carries a probability p^*. Rather than asking the model to verbalize a confidence, we read it from the distribution over the tokens it actually emitted. The extractor is prompted to end with a bracketed label; let T be the set of generated tokens overlapping the span of that label in the decoded text. Then

$$p^* \;=\; \exp\Big(\tfrac{1}{|T|}\textstyle\sum_{t\in T} \log p_t\Big), \tag{4}$$

the per-token geometric-mean probability of the label the model produced. Three details of this construction are load-bearing, and we record them because each was a source of silent error.

Span alignment. The probability is measured over exactly the span the label *parser* reports, reconstructing the decoded string from the token list and tracking each token's character offsets. Measuring the same span the label came from makes the label and its probability structurally unable to disagree. Requiring standalone bracket-delimiter tokens instead does not work: tokenizers routinely fuse `[`, `]` and `"` into subwords with adjacent characters, so a delimiter-based rule silently fails on some backends and not others.

Token-array conventions. An earlier version of this code stripped the final entry of the logprob array as an end-of-sequence token. OpenAI-compatible and vLLM logprob arrays contain only emitted content tokens — the stop is signalled out of band — so dropping the last entry deleted a real content token, and for a bracketed label that token is the one closing the label whose confidence is being measured.

Degenerate cases. When the logprob array is empty, no label span is found, or no token aligns to it, the estimator returns 0.5 rather than 0.0. Returning zero would be read downstream as a *degenerate, impossible* relation for a label the model in fact generated, which is a stronger claim than "we could not measure the confidence".

Backends without logprobs. Some serving stacks expose no token logprobs at all. There p^* instead comes from SIMBA-UQ (Bhattacharjya et al., 2025), which samples the extractor repeatedly across

a temperature schedule, builds a pairwise similarity matrix over the samples, and aggregates each sample's similarity row into a self-consistency confidence. The winning sample supplies the label and its aggregated confidence supplies p^*, so the downstream model is unchanged.

B.3 THE FACTUALITY NETWORK AND ITS VARIANTS

The labelled graph becomes a pairwise MRF with one binary variable per claim and per evidence passage. Unary factors are $\phi_X(x) = [1 - \pi_X,\ \pi_X]$ with π_X the prior on X; each retained relation contributes one pairwise factor from Table 26 below, oriented so the evidence passage is the premise. Marginals $\mathbb{P}(A_i{=}1)$ are computed by weighted mini-bucket elimination (Liu & Ihler, 2011) with i-bound 6, and a claim is reported supported when $\mathbb{P}(A_i{=}1) > \frac{1}{2}$.

Three graph variants differ in which factors are admitted: FR1 connects each claim only to the passages retrieved for it; FR2 deduplicates passages and scores every claim against all of them; FR3 adds evidence–evidence factors. We use FR2. Because scoring every claim against every passage is $O(nm)$ LLM calls, a candidate-pair policy can restrict the pairs actually scored; the calibration of that policy is instructive for the analogous choice in §3.5. Replaying policies against recorded verdicts on a 20-claim narrative with 60 passages, a provenance-restricted policy with an embedding similarity gate holds recall at 1.000 through a threshold of 0.20, slipping to 0.985 at 0.30 and 0.833 at 0.40. Substituting token-Jaccard similarity for sentence embeddings, however, loses 22 of 72 non-neutral relations and moves the factuality score by 0.05 — and *no* threshold recovers full recall, even 0.05. Cheap surrogates for semantic similarity are not safe here, and the saving is workload-dependent rather than constant: roughly $5\times$ on unrelated subtopics but $1.2\times$ on a narrative where every claim shares characters.

Conventions. We use the with-priors factor tables (Table 26) throughout, with $\pi_C = 0.9$ for evidence and $\pi_A = 0.5$ for claims. The published factuality model uses the no-priors tables and a higher evidence prior. The difference is immaterial to the coherence results below, which set claim priors explicitly, but the convention must be stated because the two table families are not interchangeable — that is the subject of §3.3.

C THE RELATION TAXONOMY IN FULL

§3.2 states the two-level taxonomy compactly. This appendix grounds every entry in a realistic response excerpt, gives the compile map with its glosses and flags, and works one coupling through by hand.

C.1 ONE EXCERPT PER INFERENTIAL COUPLING

Table 23 gives a response excerpt for each coupling, set against the world that coupling denies; since a coupling *is* its penalized set, reading the two together is the quickest way to see what each one commits the response to. The first three couplings are exactly the NLI edge types of §B, which is what lets a mined relation reuse the existing graph machinery unchanged. The last two are additions required by argumentative prose, and the distinction between them and CONTRADICTION is the one the taxonomy exists to draw.

Why exhaustiveness needs its own coupling. CONTRADICTION forbids only the both-true world: two claims that cannot both hold, but might both fail. EXCLUSIVE additionally forbids the both-false world, which is what *exhaustive* alternatives assert. "No one was harmed" and "three people died" can neither both hold nor both fail — either someone was harmed or no one was — so a model that treats them as mere opposition has mis-described the situation in a way that matters for scoring: it leaves available a world in which *neither* claim is true, and so under-penalizes. Conversely CO_NECESSITY is the dual, penalizing only the both-false world, and is what a disjunction asserts ("at least one of these two findings must hold"). Adjudicated disputes — incident investigations, litigation, competing hypotheses — are precisely the domains where exhaustive pairs occur, because the function of a finding is to close one. The CONTRADICTION, EXCLUSIVE and CO_NECESSITY rows of Table 23 state one two-part weld failure three times over, so that the only thing varying across them is which same-value worlds the response leaves standing.

Table 23: One realistic response excerpt per Level-1 coupling of Def. 2. As in Table 25, each excerpt is prose containing both claims — A is the source, underlined, B **the target, bold**, the *italic* words drawing the link — and $V(\tau)$ is repeated from Def. 2 so the excerpt can be read against the world it rules out. The *Why* column names the world the response denies, which is what distinguishes each coupling from the one above it; the three conflict-adjacent couplings CONTRADICTION, EXCLUSIVE and CO_NECESSITY are illustrated on one two-part weld failure so that only exhaustiveness varies. NONE produces no factor, so its excerpt is one where the text relates the claims without constraining their truth.

Coupling	Reading	$V(\tau)$	Response excerpt, and the world it denies
ENTAILMENT	$a_s \Rightarrow a_t$	$\{(1,0)\}$	"The alloy is chemically identical to the certified reference, *and therefore* **it meets the airframe specification**." *Why:* the response rules out the world where the source holds and the target fails — $(1,0)$ — and says nothing about the case where the alloy differs, which is why the a_s=0 row returns the target to its prior.
CONTRADICTION	$a_s \Rightarrow \neg a_t$	$\{(1,1)\}$	"The weld passed the tensile test, *yet* **the same joint was porous along its full length**." *Why:* only the both-true world $(1,1)$ is denied. Both can still fail together — if the wrong joint was tested, neither claim holds — so $(0,0)$ must stay available.
EXCLUSIVE	exactly one holds	$\{(0,0),(1,1)\}$	"The joint was destructively examined, so *exactly one of* the weld met the tensile minimum *and* **the weld fell short of it** *must be the finding — there is no third outcome*." *Why:* both same-value worlds are denied. The added $(0,0)$ is the whole difference from CONTRADICTION: once the joint has actually been examined the pair cannot both fail, so leaving that world open — as CONTRADICTION does — would under-penalize the conflict.
CO_NECESSITY	at least one holds	$\{(0,0)\}$	"The fracture began at the joint, so *at least one of* the weld *and* **the fastener** *must have given way* — conceivably both." *Why:* the dual of EXCLUSIVE. Only the both-false world is denied, so asserting the disjunction is evidence *for* both endpoints (Example 1).
EQUIVALENCE	$a_s \Leftrightarrow a_t$	$\{(0,1),(1,0)\}$	"The shipment was certified to specification — *that is to say*, **every contractual requirement had been met**." *Why:* the two disagreeing worlds are denied, in both directions; unlike ENTAILMENT there is no vacuous row, since a false source now also forces a false target.
NONE	no coupling	$\varnothing$	"The replacement units were installed in March, *some three months before* **the first in-flight failures were reported**." *Why:* no world is denied. The response genuinely relates the claims, but ordering them in time constrains neither one's truth, so the pair contributes **no factor** and both claims stay at their priors.

Example 1 (A CO_NECESSITY factor in isolation)**.** *Take a single* CO_NECESSITY *relation with* $p = 0.9$ *over two claims with priors* $\pi = 0.5$*, and no other factors. From Table 26 the pairwise table is* $[1-p,\ \pi_s,\ \pi_s,\ p] = [0.1, 0.5, 0.5, 0.9]$*, and multiplying in the unaries and normalizing gives world masses*

$$\mathbb{P}(0,0) = 0.05, \quad \mathbb{P}(0,1) = 0.25, \quad \mathbb{P}(1,0) = 0.25, \quad \mathbb{P}(1,1) = 0.45.$$

Table 24: The compile map $\mathcal{C}$ of §3.2, with glosses. *Ordering-only* senses record assertion order but induce **no factor**, which is what makes an order-only edit provably score-invariant (Rem. 2). Concession is a conflict the text may itself resolve, handled by Eq. 2. The nine senses marked $\star$ form the admissible vocabulary of §3.5.

Level-2 sense	$\mathcal{C}$ coupling	prior	dir.	flag	gloss
Cause-Effect*	ENTAILMENT	$\perp$	yes		source brings target about
Effect-Cause*	ENTAILMENT	$\perp$	yes		source is the effect, target the cause
Evidence*	ENTAILMENT	$\perp$	yes		source is evidence for target
Condition	ENTAILMENT	$\perp$	yes		source conditions target
Instantiation	ENTAILMENT	$\perp$	yes		target instantiates source
Restatement*	EQUIVALENCE	0.90	no		definitional identity
Contrast*	CONTRADICTION	$\perp$	no		opposed, *not* exhaustive
Concession*	CONTRADICTION	$\perp$	yes	conc.	a tension the text resolves
Alternative*	EXCLUSIVE	$\perp$	no		exhaustive: exactly one holds
Disjunction*	CO_NECESSITY	$\perp$	no		at least one holds
Precedence*	NONE	—	yes	ord.	temporal order only
Succession	NONE	—	yes	ord.	temporal order only

The both-false world is suppressed from the 0.25 *it would receive under independent fair priors to* 0.05*, while the three worlds satisfying "at least one holds" share the recovered mass in proportion to how many endpoints they assert. Each marginal rises from* 0.5 *to* 0.7*: asserting that at least one of two claims holds is evidence for both, which is the intended reading.*

C.2 THE COMPILE MAP, WITH GLOSSES

Couplings are the right vocabulary for a factor table and the wrong one for annotation. Asked whether two claims stand in ENTAILMENT, an annotator — human or model — has been handed a question about logic, when the fact actually observable in the text is a question about discourse: does this sentence give a cause, offer evidence, concede a tension, or lay out alternatives? We therefore mine an interpretable layer and compile it down.

C.3 ONE EXCERPT PER DISCOURSE SENSE

Table 25 grounds each sense in a response excerpt that contains both claims and the wording that licenses the link, since what an annotator must actually recognize is a form of words rather than a truth-functional pattern. Several entries encode judgements worth defending. Restatement carries a fixed strength prior of 0.90 because a marked paraphrase ("in other words", "equivalently") is a claim about identity of content, and a model asked to estimate its strength has little to add; the estimator's value is used when available, and the prior seeds it otherwise. Five senses collapse onto ENTAILMENT, which is a deliberate loss: the model does not distinguish causation from evidence, because a pairwise potential over truth values cannot. The grouped ENTAILMENT block of Table 25 shows exactly what compilation discards — five distinct discourse facts, one factor table. What survives compilation is the inferential force, and the sense is retained alongside it for interpretability and for error analysis.

The Contrast/Alternative boundary is exactly the exhaustiveness question of §3.2, and the Contrast, Alternative and Disjunction rows of Table 25 are that question in concrete form: one two-part weld failure written up three ways — as mere opposition, as an exhaustive choice, and as a disjunction — differing in nothing but which same-value worlds the response leaves available. The two ordering-only senses are the taxonomy's way of representing a relation the text genuinely draws while asserting that it has no truth-functional content: a narrative that says one event preceded another has related the two claims, and has said nothing about whether either is true.

Table 25: One realistic response excerpt per Level-2 sense of Table 24, in the same row order and grouped by the coupling each compiles to. Each excerpt is prose of the kind the estimator actually sees and *contains both claims*: A is underlined, B **is bold**, and the *italic* words are the ones that draw the link — the span §3.5 obliges the model to quote, which is why it must be findable in the response and not in either claim. Senses are marked ⋆ when admissible; the three unmarked ones are included because their breadth is the argument for excluding them (§3.5). The Contrast, Alternative and Disjunction rows describe one two-part weld failure three ways, separated only by exhaustiveness (§3.2); they reuse the scenario of Table 23 deliberately, so that a discourse sense and the coupling it compiles to can be compared on the same prose.

Level-2 sense	$\mathcal{C}$	Response excerpt, and why this sense rather than a neighbour
Cause-Effect⋆	ENTAILMENT	"The company shipped a batch it knew to be out of tolerance, and *as a direct result* **its share price fell** 15% **that quarter**." *Why:* the response itself presents A as bringing B about; the marker names the direction.
Effect-Cause⋆	ENTAILMENT	"The entire fleet was grounded on 14 March. The order came *because* **inspectors had found a cracked wing spar on two airframes**." *Why:* the same support runs from the stated effect to its cause. Direction of *explanation* is reversed; direction of inferential support is not.
Evidence⋆	ENTAILMENT	"The maintenance log records no inspection between January and June, *which shows that* **the mandatory 500-hour check was skipped**." *Why:* A is offered in support of B without any claim that it caused it — a record is evidence, not a mechanism.
Condition	ENTAILMENT	"**A part is cleared for airframe use** *only if* its alloy meets the reference specification." *Why:* B is asserted contingent on A. Note how weakly the cue constrains: nearly any dependence can be reread as a condition, which is why the sense is inadmissible.
Instantiation	ENTAILMENT	"Several suppliers failed the 2023 audit — *for instance*, **the Tucson plant failed on two counts**." *Why:* B is one case falling under A's generalization. Like Condition, broad enough to fit almost any pair that shares a topic.
Restatement⋆	EQUIVALENCE	"The shipment was certified to specification — *in other words*, **every requirement in the contract had been met**." *Why:* a *marked* paraphrase asserts identity of content, which is why the strength prior is fixed at 0.90 rather than estimated.
Contrast⋆	CONTRADICTION	"The weld passed the tensile test, *but* **the same joint was porous along its full length**." *Why:* opposed and *not* exhaustive. A third possibility — the wrong joint was tested — leaves both false, so the both-false world must stay available.
Concession⋆	CONTRADICTION	"*Although* the airline argued throughout that pilot error was to blame, **the final report identified a metallurgical defect**; the tribunal *ultimately held* for the latter." *Why:* a tension the text itself settles. The resolving holding triggers the discount of Eq. 2 rather than counting as a live conflict.
Alternative⋆	EXCLUSIVE	"The two accounts cannot be reconciled: *either* no one aboard was harmed, *or* **three passengers died**, and *exactly one of them is true*." *Why:* exhaustive. Unlike Contrast the pair can neither both hold nor both fail, so the both-false world is penalized too.
Disjunction⋆	CO_NECESSITY	"Metallurgy confirms the fracture began at the joint, so *at least one of* the weld *and* **the fastener** *must have failed* — possibly both." *Why:* at least one holds and both may. The dual of Alternative: only the both-false world is penalized.
Precedence⋆	NONE	"The replacement units were installed in March, *some three months before* **the first in-flight failures were reported in June**." *Why:* the text genuinely relates the two claims, and says nothing about either one's truth — so the sense induces **no factor**.
Succession	NONE	"**The first in-flight failures were reported in June**, *fully three months after* the replacement units had been installed." *Why:* the mirror of Precedence — the same two claims, the same absence of truth-functional content. Offering both senses invites an arbitrary choice on a relation that produces no factor either way.

Table 26: Pairwise potentials $\psi_{\tau,p}(a_s, a_t)$, listed row-major over the assignments $(0,0),(0,1),(1,0),(1,1)$; π_s is the factor family's fixed source-prior parameter, set by the edge's link type ($\frac{1}{2}$ for claim–claim edges) and *not* the source claim's own prior; bold marks the penalized set $V(\tau)$ of Def. 2. The *no-priors* family is correct for claim–evidence factuality, where the source passage's own truth is at stake; the *with-priors* family is what coherence requires (Prop. 1) and what Def. 1 uses. The two symmetric couplings EQUIVALENCE and EXCLUSIVE coincide in both families, having no source-prior term to correct.

Coupling τ	No priors	With priors	Symmetry
ENTAILMENT	$[p,\ p,\ \mathbf{1-p},\ p]$	$[1-\pi_s,\ \pi_s,\ \mathbf{1-p},\ p]$	asymmetric
CONTRADICTION	$[p,\ p,\ p,\ \mathbf{1-p}]$	$[1-\pi_s,\ \pi_s,\ p,\ \mathbf{1-p}]$	asymmetric
EQUIVALENCE	$[p,\ \mathbf{1-p},\ \mathbf{1-p},\ p]$		symmetric
EXCLUSIVE	$[\mathbf{1-p},\ p,\ p,\ \mathbf{1-p}]$		symmetric
CO_NECESSITY	$[\mathbf{1-p},\ p,\ p,\ p]$	$[\mathbf{1-p},\ \pi_s,\ \pi_s,\ p]$	symmetric

D POTENTIALS AND THE FOUR COHERENCE READOUTS

D.1 BOTH FAMILIES OF PAIRWISE POTENTIALS

§3.3 states Prop. 1 and the collapse it prevents. The table above gives both families side by side: the *no-priors* family is correct for claim–evidence factuality, where the source passage's own truth is at stake, and the *with-priors* family is what coherence requires. The two symmetric couplings EQUIVALENCE and EXCLUSIVE coincide in both families, having no source-prior term to correct.

D.2 THE FOUR READOUTS

Definition 1 specifies a distribution; a coherence score is a functional of it, and several are defensible. We state five criteria, define four candidates with their exact constructions, and then decide between them by exact inference rather than by preference. Two more, introduced after the comparison, come with proofs of R3: (e) in conflict edits (§D.8) and (f) in support edits as well (§D.9).

Definition 4 (Criteria)**.** *A coherence readout should be:* ***(R1)*** *a genuine functional of the joint $\mathbb{P}(\mathbf{a})$, not of the relation list alone;* ***(R2)*** *valued in $[0,1]$ and interpretable;* ***(R3)*** monotone *— resolving or removing a conflict should not decrease it;* ***(R4)*** *sensitive to conflict without hand-tuned constants;* ***(R5)*** *computable from standard inference queries.*

R1 deserves comment, since it is the criterion that does the most work. A readout that depends only on $\mathcal{R}$ — how many relations there are, what types, what weights — is not measuring the response; it is measuring its own annotation. Proposition 3 below shows this is not a hypothetical concern. R3 is harder than it looks: none of the four candidates satisfies it on every network (Prop. 6, Rem. 4), so §D.7 tests it on a coherence-ordered family and Prop. 5 states the part that can be proved for them. §D.8 and §D.9 then give readouts for which it is a theorem.

D.3 (A) MEAN POSTERIOR MARGINAL

$$\mathrm{LCS}_{\mathrm{mm}}(y) \;=\; \frac{1}{n}\sum_{i=1}^{n}\mathbb{P}(a_i = 1)\,. \tag{5}$$

The marginals come directly from a single MAR query. The interpretation is the expected fraction of the response's claims that the argument, taken as a whole, upholds. A conflict suppresses the marginal of its target (Prop. 5), while its source and the rest of the graph may move either way; a satisfied entailment chain propagates support along itself; a claim no relation touches stays at its prior and contributes neutrally.

Alongside the scalar we report three diagnostics that come free with the same query: the per-claim marginals themselves, which localize an incoherence to specific claims; the count of claims dragged *below their own prior*, which is the sharpest single indicator of an active conflict; and the mean normalized binary entropy $\mathcal{E} = \frac{1}{n}\sum_i H_2(q_i)$, lower being more decisive.

D.4 (B) CONSISTENCY: A TWO-TERM PROBABILITY

The obvious alternative to averaging marginals is to ask for the probability that the response is consistent. Making that precise turns out to require two terms:

$$\mathrm{LCS}_{\mathrm{cn}} = \tfrac{1}{2}\big(\mathcal{K} + \mathcal{S}\big), \quad \mathcal{K} = \mathbb{P}\left(\bigwedge_{r:\, \tau_r = \text{CONTRADICTION}} \neg\big(a_{s_r} \wedge a_{t_r}\big) \right), \quad \mathcal{S} = \frac{\sum_{r \in \mathcal{R}_\mathcal{S}} p_r\, U_r}{\sum_{r \in \mathcal{R}_\mathcal{S}} p_r}, \tag{6}$$

where $\mathcal{R}_\mathcal{S}$ ranges over $\{\text{ENTAILMENT}, \text{EQUIVALENCE}, \text{EXCLUSIVE}\}$ and U_r is the probability that relation r is *actively upheld*:

$$U_r = \mathbb{P}(a_{s_r}{=}1 \wedge a_{t_r}{=}1) \ \text{ for ENTAILMENT, EQUIVALENCE}, \qquad U_r = \mathbb{P}(a_{s_r} \neq a_{t_r}) \ \text{ for EXCLUSIVE}.$$

The two coupling sets are deliberately different. The conflict event $\mathcal{K}$ keys on CONTRADICTION alone, even though $\mathcal{T}_{\mathrm{conf}} = \{\text{CONTRADICTION}, \text{EXCLUSIVE}\}$. The reason is that an EXCLUSIVE spreads its incoherence over *both* same-value worlds, so reading only its both-true half would describe half the coupling and silently report the other half as satisfactory. EXCLUSIVE therefore earns credit in $\mathcal{S}$ instead, where its exactly-one world is simultaneously where the exclusion is honoured and where it is informative. CO_NECESSITY enters neither term: its defect is the both-false world, which (a)'s marginals already see directly.

Computing $\mathcal{K}$ exactly. $\mathcal{K}$ is an event probability over a conjunction of $|\{r : \tau_r = \text{CONTRADICTION}\}|$ pairwise events, which a naive encoding would represent as a factor of exponential arity. Instead we augment the network with deterministic auxiliary variables: one u_r per conflict edge with a ternary factor enforcing $u_r = a_{s_r} \wedge a_{t_r}$, then a chained running disjunction $U_k = U_{k-1} \vee u_k$ with ternary factors, and finally $\mathcal{K} = \mathbb{P}\big(U_{|\cdot|}{=}0\big)$. Every auxiliary factor is deterministic and at most ternary, so no $2^{|\mathcal{R}|}$ table is ever built and the augmented network is read with one MAR.

Computing $\mathcal{S}$ exactly. $\mathcal{S}$ is not an event probability but a weighted sum of *pairwise joints* — a linear functional of the joint — so it cannot be read off the same accumulator chain. It needs one deterministic indicator variable per supported edge and its own MAR. Because the indicators are mutually independent given the base network, they may be batched across several queries without approximation, which matters only for inference backends that enumerate.

Why "actively upheld" and not "satisfied". The definition of U_r credits a relation only in the informative world, not in every world where it is technically satisfied. An entailment $a_s \Rightarrow a_t$ is satisfied vacuously whenever $a_s = 0$, and crediting that is fatal:

Proposition 3 (The vacuous-truth trap)**.** *Let $\mathcal{K}' = \mathbb{P}($every relation in $\mathcal{R}$ is satisfied$)$, the apparently natural extension of $\mathcal{K}$ to all couplings. Then a response with no mined relations satisfies $\mathcal{K}' = 1$, the maximum attainable value.*

Proof. $\mathcal{K}'$ is the probability of an intersection of $|\mathcal{R}|$ events; for $\mathcal{R} = \varnothing$ the intersection is the whole space. □

The proposition is immediate, but its consequence is not, and it is the reason $\mathrm{LCS}_{\mathrm{cn}}$ has the form it does. Measured on the family of §G.2, a disconnected bag of claims scores $\mathcal{K}' = 1.000$ while a response whose causal spine is fully satisfied scores 0.221 — an inversion, not a tie. Nor is the defect repaired by count normalization: the geometric mean over relations gives 0.883 for the base response against 0.882 for the coherent one, non-monotone *and* still 1.000 on the bag; the expected fraction of satisfied relations gives 0.857 against 0.852, likewise inverted; and normalizing against a prior ceiling leaves $[0, 1]$ altogether (8.82 against 5.46) while still running backwards. The common cause is that "are the asserted relations honoured?" is a property of $\mathcal{R}$ rather than of y — a violation of R1 — and it is maximized by asserting nothing. Crediting only the informative world removes the free lunch: a bag of claims upholds nothing and scores $\mathcal{S} = 0$.

The two terms are averaged rather than multiplied so that neither can annihilate the other. A live contradiction should not erase a well-supported spine, and a relation-free response should not inherit a perfect score from an empty conflict set.

D.5 (C) A REIFIED COHERENCE NODE

The most direct reading of "is this response coherent?" as a single probability is to introduce a variable for it. Add a binary R with Bernoulli prior ρ and, for each relation, a ternary *vote* factor

$$h_r(R, a_s, a_t) = \begin{cases} 1 & R = 0 \quad \text{(the branch is flat)} \\ 1 & R = 1,\ (a_s, a_t) \notin V(\tau_r) \\ 1 - p_r & R = 1,\ (a_s, a_t) \in V(\tau_r), \end{cases} \qquad \mathrm{LCS}_{\mathrm{re}} = \mathbb{P}(R = 1)\,. \tag{7}$$

In the $R = 1$ branch each violated relation charges $1 - p_r$, so R behaves as a noisy-AND over relation satisfaction; in the $R = 0$ branch nothing is charged, so $R = 0$ carries no commitment. With no relations R is decoupled and $\mathrm{LCS}_{\mathrm{re}} = \rho$, correctly signalling that nothing was measured. Note that the violated set is $V(\tau)$ from Def. 2, so ENTAILMENT is treated as satisfied at $a_s = 0$ — vacuously — which is where (c) inherits a weaker form of Prop. 3's problem. Appendix J works a three-claim subnet through by hand, giving $\mathbb{P}(R{=}1) = 0.453$.

D.6 (D) NORMALIZED LOG-PARTITION

The partition function measures how much probability mass the constraints leave standing, which is a global coherence signal requiring no marginals at all. It is unnormalized, so we grade it between two references built from the same edge skeleton:

$$\mathrm{LCS}_{\mathrm{lp}} \;=\; \frac{\log Z - \log Z_{\min}}{\log Z_{\max} - \log Z_{\min}}. \tag{8}$$

For the ceiling, $Z_{\max}$ is the partition function of the same network with every conflict edge removed: deleting constraint factors only adds mass, so $\log Z \le \log Z_{\max}$ always. For the floor we take the *MAP world mass* of the base network, $Z_{\min} = \max_{\mathbf{a}} \prod_i \phi_i \prod_r \psi_r$.

Lemma 1 (The MAP world mass is a valid floor)**.** $\log Z_{\min} \le \log Z \le \log Z_{\max}$ *for every network, so Eq. 8 lies in* $[0, 1]$ *whenever* $Z_{\max} > Z_{\min}$.

Proof. $Z = \sum_{\mathbf{a}} \mathrm{mass}(\mathbf{a})$ is a sum of non-negative terms, so its single largest term $Z_{\min}$ is a lower bound. Every entry of every table in Table 26 lies in $[0, 1]$, so deleting a conflict factor multiplies no world's mass by less than one, and $Z \le Z_{\max}$. □

Remark 4 (The floor is not monotone in conflict strength)**.** *It is tempting to add that* $Z_{\min}$ *is non-increasing in conflict strength, on the grounds that raising* p_r *only down-weights the penalized worlds* $V(\tau_r)$*. That is false for the with-priors tables. Raising* p_r *does lower the* $(1, 1)$ *cell* $1 - p_r$ *of a* CONTRADICTION *table* $[1 - \pi_s,\ \pi_s,\ p_r,\ 1 - p_r]$*, but it* raises *the* $(1, 0)$ *cell* p_r*, so the source-true, target-false world gains mass and the maximum over worlds can grow. For a lone* CONTRADICTION *edge at uniform priors,* $Z_{\min} = \max(\frac{1}{8}, \frac{1}{8}, \frac{p}{4}, \frac{1-p}{4}) = p/4$ *for* $p \ge \frac{1}{2}$*, which* increases *from* 0.150 *to* 0.200 *to* 0.2375 *as* p *goes* $0.6 \to 0.8 \to 0.95$*. The same holds for* EXCLUSIVE*, whose* $(0, 1)$ *and* $(1, 0)$ *cells are* p_r*. Only the first half of the original statement, the bound, is needed for Eq. 8 to be well defined. Because the floor moves with the edge set, though,* $\mathrm{LCS}_{\mathrm{lp}}$ *inherits no monotonicity from it: on four claims at uniform priors with relations* a_3 EXCLUSIVE a_1 (0.8)*,* $a_4 \Rightarrow \neg a_2$ (0.9)*,* $a_4 \Rightarrow \neg a_3$ (0.8) *and* $a_1 \Rightarrow \neg a_4$ (0.8)*, deleting the last conflict raises* $\log Z$ *from* -2.828 *to* -2.079 *but raises* $\log Z_{\min}$ *further, from* -4.605 *to* -3.324*, so* $\mathrm{LCS}_{\mathrm{lp}}$ falls *from* 0.386 *to* 0.374 *(exact,* $\log Z_{\max} = 0$ *in both).*

Lemma 1 is not a technicality; it replaces two floors that look more natural and are wrong. Retyping every edge to CONTRADICTION, or saturating conflicts to $p = 1$, both fail to bound $\log Z$ below for the row-stochastic tables of Prop. 1: an isolated conflict removes no mass at all when its endpoints sit at the uniform prior — the setting in which both candidates were checked — and little away from it, whereas a mass-concentrating entailment backbone concentrates a great deal, so a network combining a strong backbone with several soft conflicts can have a base $\log Z$ *below* either candidate floor. We observed exactly this on real graphs, which put Eq. 8 outside $[0, 1]$ until the MAP floor replaced them. In practice we clamp the ratio to $[0, 1]$ regardless, since approximate inference computes $\log Z$ and the MAP bound with finite i-bound and small tolerance violations are possible.

Table 27: All five readouts across a coherence-ordered family of five variants of one response, by exact inference over 2^{16} worlds. The variants run from least to most coherent top to bottom in the intended ordering. Bold marks a value that moves the *wrong* way relative to its predecessor. (a), (d), (e) and (f) are monotone throughout on this family. Only (e) and (f) are monotone on every network (Thms. 1 and 2); (a) and (d) are not (Prop. 6, Rem. 4).

Variant	(a) $\mathrm{LCS}_{\mathrm{mm}}$	(b) $\mathrm{LCS}_{\mathrm{cn}}$	(c) $\mathrm{LCS}_{\mathrm{re}}$	$\log Z$	(d) $\mathrm{LCS}_{\mathrm{lp}}$	(e) $\mathrm{LCS}_{\mathrm{es}}$	(f) $\mathrm{LCS}_{\mathrm{bl}}$
worse	0.586	**0.778**	0.116	−10.54	0.68	0.181	0.5286
base	0.607	0.781	0.148	−9.64	0.78	0.321	0.5418
concession	0.612	**0.772**	**0.134**	−9.64	0.80	0.388	0.5477
fixcasualty	0.613	**0.764**	0.156	−8.95	0.89	0.600	0.5612
coherent	0.620	**0.752**	0.181	−8.25	1.00	1.000	0.5768
monotone?	**yes**	no	no	**yes**	**yes**	**yes**	**yes**

Table 28: The five readouts against the criteria of Def. 4, with the inference queries each requires. Under R3, ✓ means monotone on every network (Thm. 1); (✓) means monotone on the family of Table 27 but not on every network (Prop. 6, Rem. 4); × means non-monotone on that family itself. ✓$^+$: monotone on every network in support edits as well as conflict edits (Thm. 2).

Readout	Queries	R1	R2	R3	R4	R5	Note
(a) mean marginal	MAR	✓	✓	(✓)	✓	✓	diagnostic
(b) consistency	MAR×2	✓	✓	×	✓	✓	concession dip
(c) reified	MAR	✓	✓	×	×	✓	free ρ; dip
(d) norm. $\log Z$	PR, MAP	✓	✓	(✓)	✓	✓	needs 2 references
(e) conflict escape	PR×2	✓	✓	✓	✓	✓	ignores support
(f) balanced	PR	✓	✓	✓$^+$	✓	✓	**selected**; paddable

D.7 Deciding between the four

Table 27 settles the question empirically and Table 28 summarizes. Readouts (b) and (c) both fail R3, and they fail it at the *same* variant — the one where a conceded tension is resolved — for a shared reason that is worth stating precisely, because it looks like an implementation bug and is not.

Proposition 4 (Belief versus event activity)**.** *Softening a resolved conflict via Eq. 2* increases *the probability of its both-true world. For a single conflict edge with uniform priors, discounting p from* 0.80 *to* 0.55 *moves* $\mathbb{P}(a_s{=}1, a_t{=}1)$ *from* 0.10 *to* 0.225.

Proof. Under both conflict couplings the $(1,1)$ cell of Table 26 is $1-p$, which increases as p falls. Take a lone CONTRADICTION edge with $\pi_s = \pi_t = \frac{1}{2}$, so the with-priors table is $[\frac{1}{2}, \frac{1}{2}, p, 1-p]$ and each world carries an equal unary factor of $\frac{1}{4}$. The unnormalized masses are then proportional to the table itself, giving $\mathbb{P}(1,1) = (1-p)/\big(\frac{1}{2} + \frac{1}{2} + p + (1-p)\big) = (1-p)/2$, which is monotone decreasing in p. Evaluating at $p = 0.80$ and $p = 0.55$ gives 0.100 and 0.225. □

The consequence sorts the four readouts into two kinds. A readout that asks *is the conflict active* — (b) counts exactly that event, (c) charges $1 - p_r$ for it — must register a softened conflict as *more* often active, and therefore dips. A readout that asks *how strongly should each claim be believed* — (a) through marginals, (d) through mass — sees less suppression of the conflict's target, which provably rises (Prop. 5), and on this family the readout rises with it; Prop. 6 shows that the readout's rise is not guaranteed elsewhere. Since resolving a concession is by construction an increase in coherence, a coherence score must track belief rather than event activity. That is the substantive finding of this section, and it is not specific to our parameterization: any readout defined as the probability of a conflict-free configuration *under the posterior* will invert under any discount-based treatment of resolved concessions, because the posterior already contains the conflict being measured. The qualifier matters. Measured instead under a distribution from which the conflicts have been removed, the same event does not invert, and §D.8 builds a readout on exactly that observation.

A second failure mode of (b). Table 27 shows (b) failing worse than a single dip, and the reason is instructive about $\mathcal{K}$ and $\mathcal{S}$ jointly. In this family both conflicts are EXCLUSIVE rather than CONTRADICTION, so $\mathcal{K} \equiv 1.000$ throughout by the coupling-set decision above, and the score is carried entirely by $\mathcal{S}$. But removing an exclusion removes *support* credit as well as conflict, so the score runs backwards from 0.778 to 0.752 across the whole ladder. On three-coupling variants of the same response, where the conflicts are CONTRADICTION and $\mathcal{K}$ is live, (b) instead runs $0.576 \to 0.654 \to \mathbf{0.582} \to 0.672 \to 0.752$: correctly ordered apart from the concession dip. Either way it is non-monotone, and the lesson is that $\mathcal{K}$ and $\mathcal{S}$ should be read separately as diagnostics rather than averaged into a headline.

What R3 does and does not guarantee. Table 27 is evidence on one family, not a proof, and the property it exhibits does not extend to every network. What does extend is local to the edited relation:

Proposition 5 (A contradiction suppresses its target)**.** *Let r be a* CONTRADICTION *relation from s to t, and fix the priors and every other factor. Then $\mathbb{P}(a_t{=}1)$ is non-increasing in p_r, for any priors and any other relations. If in addition the table's source-prior entry is $\pi_s = \frac{1}{2}$, deleting r is equivalent to setting $p_r = \frac{1}{2}$, so deleting a contradiction with $p_r \geq \frac{1}{2}$ never lowers its target's marginal.*

Proof. Write $p = p_r$. The idea is that p only moves mass among worlds where the source is true, from worlds where the target is true to worlds where it is false.

Step 1: reduce the network to four numbers. Group the worlds by the values (a_s, a_t) of the two endpoints, and let $M_{ij} \geq 0$ be the total mass that every factor *other than* ψ_r gives to the worlds with $(a_s, a_t) = (i, j)$. None of the M_{ij} depends on p. The factor $\psi_r = [1 - \pi_s,\ \pi_s,\ p,\ 1 - p]$ then multiplies each group by one constant:

(a_s, a_t)	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
mass with ψ_r	$(1-\pi_s)M_{00}$	$\pi_s M_{01}$	$p\,M_{10}$	$(1-p)M_{11}$

Step 2: split by the target. The mass of the target-true worlds, $(0,1)$ and $(1,1)$, and the mass of the target-false worlds, $(0,0)$ and $(1,0)$, are

$$T(p) = \pi_s M_{01} + (1-p)\,M_{11}, \qquad F(p) = (1-\pi_s)M_{00} + p\,M_{10}.$$

Since $M_{11}, M_{10} \geq 0$, T is non-increasing and F non-decreasing in p.

Step 3: conclude. The target's marginal is the target-true share of the mass,

$$\mathbb{P}(a_t{=}1) = \frac{T(p)}{T(p) + F(p)} = \frac{1}{1 + F(p)/T(p)},$$

and F/T is non-decreasing in p, so $\mathbb{P}(a_t{=}1)$ is non-increasing. If $T(p) = 0$ the marginal is 0, its least possible value, so the claim holds there too.

Deletion. At $\pi_s = p = \frac{1}{2}$ every entry of ψ_r is $\frac{1}{2}$. A constant factor cancels on normalization, so this network is exactly the one without r. Deleting r is therefore the same as lowering p_r to $\frac{1}{2}$, and by Step 3 that cannot lower $\mathbb{P}(a_t{=}1)$ when $p_r \geq \frac{1}{2}$. □

Softening a resolved concession by Eq. 2 therefore always raises belief in the conflict's target, which is the sense in which belief-tracking readouts respond correctly to it. The guarantee is about the target alone. The source's marginal can move either way, the deletion corollary can fail when $\pi_s \neq \frac{1}{2}$ (a lone CONTRADICTION at $p = \frac{1}{2}$ with $\pi_s = 0.75$, $\pi_t = 0.45$ holds the target at 0.513, and deleting it returns the target to 0.45), and the other claims can move against the edited pair:

Proposition 6 (Removing a contradiction can lower $\mathrm{LCS}_{\mathrm{mm}}$)**.** *There are networks in which deleting a* CONTRADICTION *relation, or softening it towards $p = \frac{1}{2}$, strictly decreases* $\mathrm{LCS}_{\mathrm{mm}}$.

Proof. We build a small network in which the score can be computed by hand. All priors are $\frac{1}{2}$. A *hub* claim a_2 contradicts k *spoke* claims $a_3, \ldots, a_{k+2}$, each relation $a_2 \Rightarrow \neg a_j$ at weight p. A *blocker* claim a_1 contradicts the hub, $a_1 \Rightarrow \neg a_2$, at weight b. We show that $\mathrm{LCS}_{\mathrm{mm}}$ strictly increases with b whenever $k(2p-1) > 2$. Softening the blocker then lowers the score, and so does deleting it.

Step 1: read the network as a Bayesian network. With $\pi_s = \frac{1}{2}$ each CONTRADICTION table is $[\frac{1}{2}, \frac{1}{2}, p, 1-p]$, whose rows are the conditionals $\mathbb{P}(a_t \mid a_s)$ (Prop. 1). Multiplying in the target's prior $\frac{1}{2}$ gives $\frac{1}{2}\,\mathbb{P}(a_t \mid a_s)$. Every claim except a_1 is the target of exactly one relation, so up to a constant the joint is

$$\mathbb{P}(a_1)\ \mathbb{P}(a_2 \mid a_1)\ \prod_{j=3}^{k+2} \mathbb{P}(a_j \mid a_2)\,, \qquad \mathbb{P}(a_1{=}1) = \tfrac{1}{2}.$$

The marginals can therefore be read off from the top down: first a_1, then the hub, then the spokes.

Step 2: compute the marginals. The blocker stays at $\frac{1}{2}$. The hub is true with probability $\frac{1}{2}$ when the blocker is false and $1-b$ when it is true, so

$$c \;:=\; \mathbb{P}(a_2{=}1) \;=\; \tfrac{1}{2}\cdot\tfrac{1}{2} + \tfrac{1}{2}\,(1-b) \;=\; \frac{3-2b}{4}.$$

Each spoke is true with probability $\frac{1}{2}$ when the hub is false and $1-p$ when it is true, so

$$\mathbb{P}(a_j{=}1) \;=\; \tfrac{1}{2}\,(1-c) + (1-p)\,c \;=\; \tfrac{1}{2} - c\,\big(p - \tfrac{1}{2}\big).$$

Step 3: add them up. Over the $k+2$ claims,

$$(k+2)\,\mathrm{LCS}_{\mathrm{mm}} \;=\; \tfrac{1}{2} + c + k\Big(\tfrac{1}{2} - c\,\big(p-\tfrac{1}{2}\big)\Big) \;=\; \frac{k+1}{2} + c\,\Big(1 - k\,\big(p-\tfrac{1}{2}\big)\Big).$$

Raising belief in the hub by c adds c to the hub's marginal but takes $c\,(p-\frac{1}{2})$ from each of the k spokes. When $k(2p-1) > 2$ the bracket is negative, so $\mathrm{LCS}_{\mathrm{mm}}$ strictly decreases in c. Since $c = (3-2b)/4$ strictly decreases in b, $\mathrm{LCS}_{\mathrm{mm}}$ strictly increases in b. Deleting the blocker gives the same network as $b = \frac{1}{2}$, because its table is then the constant $\frac{1}{2}$. So for any $b > \frac{1}{2}$ deletion lowers $\mathrm{LCS}_{\mathrm{mm}}$ too.

Step 4: a concrete instance. Take $k = 3$ and $p = b = 0.95$, so $k(2p-1) = 2.7 > 2$ and the bracket is $1 - 3 \times 0.45 = -0.35$. With the blocker, $c = \frac{11}{40}$, each spoke is at $\frac{301}{800}$, and $\mathrm{LCS}_{\mathrm{mm}} = \big(2 - 0.35 \times \frac{11}{40}\big)/5 = 1523/4000 = 0.381$. Without it, $c = \frac{1}{2}$, each spoke is at $\frac{11}{40}$, and $\mathrm{LCS}_{\mathrm{mm}} = (2 - 0.35 \times \frac{1}{2})/5 = 73/200 = 0.365$. Softening instead of deleting gives 0.381, 0.376, 0.369 and 0.365 at $b = 0.95$, 0.8, 0.6 and 0.5. Exact enumeration over the 2^5 worlds gives the same values. □

Step 3 isolates the mechanism. The deleted conflict was holding down a hub that contradicts three other claims. Once released, the hub suppresses all three: one marginal rises by 0.225 and three fall by 0.101 each. Proposition 5 holds, since a_2 is the target and rises, yet the mean falls. The failure is not an artefact of the with-priors tables either. With a penalty-only table $[1, 1, 1, 1-p]$ and five spokes at $p = 0.99$, the same deletion lowers the mean from 0.376 to 0.325. Any readout that averages per-claim belief is exposed to it, and Remark 4 gives the corresponding counterexample for (d). For (a) and (d), R3 is therefore reported in Table 28 as satisfied on the test family rather than in general, and the ladder constraints of §4 remain empirical tests of it rather than consequences of the model.

D.8 (E) CONFLICT ESCAPE: A READOUT THAT IS PROVABLY MONOTONE

The two failures above have one source. Both (a) and (d) read the conflicts through the posterior, where a conflict's effect on everything else is part of what is measured, so deleting one conflict can strengthen another (Prop. 6). The remedy is to evaluate the conflicts against a distribution that does not contain them. Let $\mathcal{R}_{\mathrm{c}} = \{r : \tau_r \in \mathcal{T}_{\mathrm{conf}}\}$ be the conflict relations, and let P_0 be the MRF of Def. 1 with every conflict factor removed — priors and support relations only, the network whose partition function is the ceiling $Z_{\max}$ of Eq. 8. Give each conflict a *penalty* $w_r = p_r$, clipped below 1, and define

$$\mathrm{LCS}_{\mathrm{es}}(y) \;=\; \mathbb{E}_{\mathbf{a}\sim P_0}\Big[\prod_{r\in\mathcal{R}_{\mathrm{c}}} \big(1 - w_r \mathbb{1}[(a_{s_r}, a_{t_r}) \in V(\tau_r)]\big)\Big] \;=\; \frac{Z_{\mathrm{pen}}}{Z_{\max}}, \tag{9}$$

where Z_{pen} is the partition function of P_0 multiplied by one *penalty-only* factor per conflict, equal to $1 - w_r$ on $V(\tau_r)$ and to 1 elsewhere. The reading is a probability: draw a world from what

the response's priors and support relations make plausible, let each conflict independently catch its violation with probability w_r, and ask whether the world escapes every conflict. Any penalty map non-decreasing in p_r with values in $[0, 1)$ would serve the theorem below; the linear map is the one that leaves no conflict invisible, and the ladders (§D.8, Table 29) show why that matters: a map that is zero below some weight makes resolving a conflict of that weight a no-op.

Theorem 1 ($\mathrm{LCS}_{\mathrm{es}}$ is monotone in conflict resolution)**.** *For any priors and any support relations, deleting a conflict relation or lowering its weight p_r never decreases* $\mathrm{LCS}_{\mathrm{es}}$. *Moreover* $\mathrm{LCS}_{\mathrm{es}} \in [0, 1]$, *with* $\mathrm{LCS}_{\mathrm{es}} = 1$ *when no conflict has* $p_r > \frac{1}{2}$.

Proof. Write $g(\mathbf{a}) = \prod_{r \in \mathcal{R}_{\mathrm{c}}} \big(1 - w_r \mathbb{1}[(a_{s_r}, a_{t_r}) \in V(\tau_r)]\big)$, so that $\mathrm{LCS}_{\mathrm{es}} = \mathbb{E}_{P_0}[g]$. Three facts suffice.

(i) P_0 does not move. By construction P_0 contains no conflict factor, so neither deleting a conflict nor changing its weight changes P_0.

(ii) Each factor of g lies in $[0, 1]$ and only grows when the conflict is relaxed. The factor for r is $1 - w_r$ in the worlds that violate r and 1 in the others. Lowering p_r lowers w_r, which raises $1 - w_r$ and leaves the other worlds at 1. Deleting r replaces the factor by the constant 1, which is the same as setting $w_r = 0$.

(iii) So g grows world by world, and so does its mean. By (ii), relaxing a conflict multiplies $g(\mathbf{a})$ in every world by a number at least one (or leaves it unchanged). Since $g \geq 0$, $g(\mathbf{a})$ cannot decrease in any world, and by (i) the weights $P_0(\mathbf{a})$ that average it are unchanged. An average of non-decreasing terms with fixed non-negative weights cannot decrease.

Finally, g is a product of numbers in $[0, 1]$, so its mean is in $[0, 1]$; and when every $w_r = 0$, $g \equiv 1$. □

The proof is short because the construction removes the mechanism behind Prop. 6: in (a) a conflict changes the very distribution against which the other conflicts are scored, so removing one can strengthen another, whereas in (e) the scoring distribution P_0 is fixed and conflicts interact only by multiplying their penalties. Two consequences are worth recording. On the star of Prop. 6, deleting the blocker raises $\mathrm{LCS}_{\mathrm{es}}$ from $2754481/5120000 = 0.538$ to $73261/128000 = 0.572$, where it lowered $\mathrm{LCS}_{\mathrm{mm}}$. And softening a lone contradiction at uniform priors from $p = 0.80$ to 0.55 raises it from $1 - p/4 = 0.800$ to 0.863, where (b) and (c) dip (Prop. 4): the both-true world is now weighed under P_0, which the softening does not touch, rather than under a posterior that the softening enlarges.

Both ingredients of the construction are needed. Keeping the with-priors conflict tables and taking $Z/Z_{\max}$ instead fails R3 under softening, because the $(1, 0)$ cell p of the CONTRADICTION table rises with p: a lone CONTRADICTION with $\pi_s = \frac{1}{2}$, $\pi_t = 0.2$ gives $Z/Z_{\max} = 0.59$ at $p = 0.8$ and 0.50 at $p = 0.5$. Penalty-only tables alone are not enough either, as long as the conflicts are read through the posterior: Prop. 6 survives them.

What (e) gives up is equally definite. It is a statement about conflicts only; support relations enter through P_0 but earn no credit of their own, so a disconnected bag of claims and a well-supported spine both score 1 if neither has a conflict. That is what R3 demands of a conflict-free response, and it is the opposite failure from Prop. 3, which concerned support. A support term computed under P_0 — such as $\mathcal{S}$ of Eq. 6 with P_0 in place of the posterior — is independent of the conflicts and may be averaged in without losing Thm. 1. It also does not decompose into per-claim contributions; its natural diagnostic is per conflict, the gain $\mathrm{LCS}_{\mathrm{es}}(\mathcal{R} \setminus \{r\}) - \mathrm{LCS}_{\mathrm{es}}(\mathcal{R}) \geq 0$ from deleting r. On the family of Table 27 it is strictly monotone and separates the rungs more widely than any other readout, $0.181 < 0.321 < 0.388 < 0.600 < 1.000$; in particular it is the only readout to move appreciably at the concession rung, where $\log Z$ is flat and (d) moves only through its floor.

(e) on the ladders. Readout (e) postdates the ladder runs, but every cached run stores the relation graph and priors it was scored on, so (e) can be added without re-mining: we rebuild each item's network from its record, rescore it, and require the rebuilt (a) and (d) to match the cached values before any (e) value is used (exactly on the seventy-item corpus, scored by enumeration; to 2×10^{-5} on the hundred-family release, scored by Merlin, whose printed $\log Z$ carries five to six decimals). The ladder constraints are declared per readout and none are declared for (e), so Table 29 uses the two readout-independent axes of Table 8: *Increase*, the C1 and C3 pairs of the CONFLICT and CHAIN ladders, and *Invariance*, every declared pair of the ORDER and CONTROL ladders. Since Thm. 1

Table 29: Readout (e) on the ladders, against (a), (d) and (b), on the two readout-independent axes of Table 8. *Increase*: C1 and C3 pairs of CONFLICT and CHAIN ladders, satisfied by a strict rise. *Invariance*: all declared pairs of ORDER and CONTROL ladders, satisfied when the score moves by at most 10^{-6}. (e) uses the linear map $w_r = p_r$ of Eq. 9; ($\mathrm{e_{th}}$) is the threshold ablation $w_r = \max(0, 2p_r - 1)$. Mined rows give the range over the four miner–policy sources. Generated by `scripts/rescore_ladder_conflict_escape.py`.

		Seventy items		Hundred-family release	
Graph	Readout	Increase	Invariance	Increase	Invariance
GOLD	(a) $\mathrm{LCS_{mm}}$	$37/50$	$20/20$	$\mathbf{273}/335$	$165/165$
	(d) $\mathrm{LCS_{lp}}$	$36/50$	$20/20$	$233/335$	$165/165$
	(b) $\mathrm{LCS_{cn}}$	$34/50$	$20/20$	$242/335$	$165/165$
	(e) $\mathrm{LCS_{es}}$	$\mathbf{44}/50$	$20/20$	$271/335$	$165/165$
	($\mathrm{e_{th}}$)	$34/50$	$20/20$	$205/335$	$165/165$
mined	(a) $\mathrm{LCS_{mm}}$	20–28	$0/20$	158–188	0–5
	(d) $\mathrm{LCS_{lp}}$	25–35	$0/20$	162–175	0–4
	(b) $\mathrm{LCS_{cn}}$	23–34	$0/20$	149–170	0–6
	(e) $\mathrm{LCS_{es}}$	22–27	0–3	117–177	6–62
	($\mathrm{e_{th}}$)	19–23	0–3	117–158	19–62

leaves the penalty map free, we also report an ablation, the *threshold* map $w_r = \max(0, 2p_r - 1)$, which charges nothing below $p_r = \frac{1}{2}$; (e) as defined in Eq. 9 uses the linear map.

The theorem's prediction holds on every gold pair to which it applies. On a gold CONFLICT ladder consecutive rungs differ only in conflict relations, so Thm. 1 predicts that (e) never falls; it rises on all 40 such pairs of the seventy-item corpus and all 230 of the release. (a) falls on 11 of the seventy-item pairs. The pairs on which (e) does fall are all CHAIN pairs whose edit adds or removes a *support* relation (3 of 50 seventy-item pairs, 23 of 335 release pairs), which changes P_0 and so lies outside the theorem.

Why the map is linear. An earlier version of (e) used the threshold map, and under it gold (e) ties on ten seventy-item pairs, all from a single interaction: the gold strength band *weak* spans $[0.35, 0.59]$ and enters at its midpoint 0.47, just below the threshold's zero at $p = \frac{1}{2}$, so every weak conflict carries no penalty and resolving it cannot move the score. Under the with-priors table such a relation acts as support, which is why (a) instead *falls* on those pairs. The linear map charges weak conflicts and resolves every tie, giving $44/50$, the best gold Increase count of any of (a)–(e) on that corpus. On the release, which played no part in choosing the map, (e) gives $271/335$ against $273/335$ for (a), so the advantage over (a) does not replicate, although the threshold map's deficit ($205/335$) does. The cause is the map, not the construction: the same threshold applied to the penalties of readout (f) below lowers it from $49/50$ to $39/50$ and from $335/335$ to $275/335$.

On mined graphs (e) is not more robust. Its Increase counts lie at or below the other readouts' on both corpora, consistent with §5.2: a guarantee about the score cannot repair the graph it is computed on, and every mined rung pair changes support and conflict relations together, so the theorem never applies there. The higher Invariance counts on the release should *not* be read as robustness to reordering. They come from relations that (e) cannot see — mined conflicts at weight 10^{-6}, which carry no penalty under either map, so rungs that differ only in such relations score identically — and from scores saturating near 0 or 1, where rung differences fall below the 10^{-6} tie tolerance.

D.9 (F) BALANCED: MONOTONE IN SUPPORT AS WELL AS CONFLICT

Readout (e) is monotone only in conflict edits, and the CHAIN pairs on which it falls show why: its base distribution P_0 contains the support relations, so adding one changes the distribution the conflicts are scored against. Every edit a gold ladder makes is of one of four kinds — weaken or delete a conflict, strengthen or add a support — and never the reverse; claim sets and priors are fixed within a family. A readout monotone in each relation, in the direction that edit requires, would therefore satisfy every satisfiable gold ladder assertion by construction. The construction that gives this takes (e) one step further: move the supports out of the base distribution and into the product, as

rewards. Let $\mathcal{R}_{\mathrm{s}}$ be the support relations, every edge-producing coupling that is not a conflict, and write π for the product of the priors, $\pi(\mathbf{a}) = \prod_i \pi_i^{a_i}(1-\pi_i)^{1-a_i}$. Define

$$G(\mathbf{a}) = \prod_{r\in\mathcal{R}_{\mathrm{c}}} \big(1 - p_r \mathbb{1}[(a_{s_r}, a_{t_r}) \in V(\tau_r)]\big) \prod_{r\in\mathcal{R}_{\mathrm{s}}} \big(1 + p_r \mathbb{1}[(a_{s_r}, a_{t_r}) \in C(\tau_r)]\big),$$
$$\mathrm{LCS}_{\mathrm{bl}}(y) = \sigma\Big(\tfrac{2}{n} \log \mathbb{E}_{\mathbf{a}\sim\pi}\big[G(\mathbf{a})\big]\Big), \tag{10}$$

with σ the logistic function and $C(\tau)$ the worlds in which a support is credited: the both-true world for ENTAILMENT and EQUIVALENCE, where the relation is actively upheld as in §D.4, and the two exactly-one worlds for CO_NECESSITY, where a disjunction does work. Penalties use the linear map $w_r = p_r$ of Eq. 9 and rewards the same, $u_r = p_r$. $\mathbb{E}_\pi[G]$ is the partition function of the prior unaries times one pairwise factor per relation, so (f) costs one PR call. The factor 2 is a fixed display scale, not a fitted constant, and its value comes from the model: since $p_r \le 1$, no relation factor exceeds 2, so one support relation adds at most one bit of log-evidence. Writing $b = \log_2 \mathbb{E}_\pi[G]/n$ for the net evidence per claim in bits, $\mathrm{LCS}_{\mathrm{bl}} = 4^b/(1+4^b)$: each bit per claim multiplies the odds by 4, so one maximal support per claim reads 0.8. Any fixed scale $c > 0$ in place of 2 gives the same orderings and the same theorem; c only sets the spread.

Theorem 2 ($\mathrm{LCS}_{\mathrm{bl}}$ is monotone in every relation)**.** *For any priors, deleting or weakening a conflict relation, and adding or strengthening a support relation, never decreases* $\mathrm{LCS}_{\mathrm{bl}}$. *A response with no relations scores exactly* $\frac{1}{2}$.

Proof. The weights $\pi(\mathbf{a})$ of the expectation are the priors alone and depend on no relation. Each conflict factor lies in $[0,1]$ and does not increase with p_r; deleting the conflict replaces it by 1, which is at least as large. Each support factor is at least 1 and does not decrease with p_r; adding a support multiplies G by a factor at least 1. So each of the four edits leaves $G(\mathbf{a})$ unchanged or larger in every world, and since $G \ge 0$ and the weights are fixed, $\mathbb{E}_\pi[G]$ does not decrease. The map $x \mapsto \sigma(2x/n)$ is strictly increasing and n is unchanged by any relation edit, so the order passes to $\mathrm{LCS}_{\mathrm{bl}}$. With no relations $G \equiv 1$, the expectation is 1, and $\sigma(0) = \frac{1}{2}$. □

The difference from (e) is exactly the one that matters. In (e) the supports shape the distribution under which conflicts are counted, so a support edit can move the score either way; in (f) both kinds of relation enter only as factors of G against a base that neither touches. On the star of Prop. 6 it rises from 0.438 to 0.444 when the blocker is deleted, and on the family of Table 27 it is strictly ordered. The theorem is also checked directly: on random networks of two to seven claims with arbitrary priors and all five couplings, none of 20,000 edits in the coherence-improving direction lowered it.

On the ladders. Table 30 scores (f) exactly as Table 29 scores (e), from the cached graphs with (a) and (d) reproduced first. On gold graphs it satisfies *every satisfiable* assertion on both corpora, with no rung pair on which it falls: 49/50 Increase on the seventy-item corpus, the one miss being family `f009`'s rungs 2 and 3, whose gold graphs are byte-identical so that no function of the graph can separate them; 335/335 on the hundred-family release; and 20/20, 165/165 Invariance. Held instead to the seventy assertions declared for (a), it satisfies 69 against (a)'s own 57. The remaining C2 assertions are predictions about the internals of (b) and (d) — that (b) dips at a resolved concession, that (d) is flat under a weight-only edit — and (f) is constructed not to satisfy the first, so it is not scored on them. On mined graphs (f) is the strongest Increase readout on the release (168–198 of 335, against 158–188 for (a)) but does not escape §5.2: Invariance stays near zero, because a mined graph changes when the prose is reordered and Thm. 2 says nothing about a graph that changes in both directions at once.

What it costs. Three properties come with the guarantee, and none can be removed without removing it. *Padding.* If adding a support can never lower the score then adding many must raise it: fifteen entailments at $p = 0.99$ over six claims lift a response with one contradiction from 0.476 to 0.783. This is the converse of Prop. 3, whose trap was that asserting nothing scored perfectly; here asserting much scores well. (f) measures how strongly the asserted relations cohere, not whether they were worth asserting, so it inherits the estimator's precision (§3.5) as its only defence against over-assertion. Crediting CO_NECESSITY in its exactly-one worlds rather than in all three of its satisfying worlds keeps a near-vacuous disjunction from being the cheapest relation to pad with.

Table 30: Readout (f) on the ladders, on the axes of Table 29, with (a) and (e) for comparison. Gold Increase is bounded by the satisfiable assertions: 49 of the seventy-item corpus's 50, since one pair has identical gold graphs. Mined rows give the range over the four sources. Generated by `scripts/rescore_ladder_conflict_escape.py`.

		Seventy items		Hundred-family release	
Graph	Readout	Increase	Invariance	Increase	Invariance
GOLD	(a) $\mathrm{LCS}_{\mathrm{mm}}$	$37/50$	$20/20$	$273/335$	$165/165$
	(e) $\mathrm{LCS}_{\mathrm{es}}$	$44/50$	$20/20$	$271/335$	$165/165$
	(f) $\mathrm{LCS}_{\mathrm{bl}}$	$\mathbf{49}/50$	$20/20$	$\mathbf{335}/335$	$165/165$
mined	(a) $\mathrm{LCS}_{\mathrm{mm}}$	20–28	$0/20$	158–188	0–5
	(e) $\mathrm{LCS}_{\mathrm{es}}$	22–27	0–3	117–177	6–62
	(f) $\mathrm{LCS}_{\mathrm{bl}}$	22–31	0–1	168–198	0–9

Compression. With $p_r \leq 1$ each relation moves $\log \mathbb{E}_\pi[G]$ by at most $\log 2$, and the division by n that keeps the order fixed also keeps gold scores near $\frac{1}{2}$: within 0.40–0.60 at the display scale 2, against 0.45–0.55 at scale 1, with a median rung rise of 0.019 and a minimum of 5×10^{-5}, well clear of the 10^{-6} tie tolerance. A larger fixed scale would widen the range further at no cost to the theorem, but the scale is chosen by the bit-per-claim reading above, not to spread the ladders. Normalizing instead by references built from the response's own graph, or by the relation count, breaks the theorem, for the same reason as the floor of Rem. 4: the reference moves with the edit. *Belief.* The expectation is under the priors, not the posterior, so (f) does not propagate support along a chain the way the marginals do and does not localize an incoherence to a claim; its natural diagnostic is per relation, the change in (f) from deleting r.

Selection. We report (f) as the headline. It is the only readout of the six that satisfies R3 as a theorem in every kind of relation edit (Thm. 2); on gold relation graphs that guarantee shows up as every satisfiable ladder assertion satisfied, on both corpora (Table 30); it satisfies R1, R2, R4 and R5; and it needs one PR call. Its costs are stated above and are the price of the guarantee: a response padded with support relations scores higher, and the score does not decompose into per-claim contributions. We therefore report (a) alongside it as the *diagnostic* readout. (a) is the only one of the six whose value decomposes per claim, which is what makes an incoherence *locatable* rather than merely detectable; it satisfies R3 on the family of Table 27 but not on every network (Prop. 6). (e) is the conflict-only special case of (f), for a use that should ignore support. (d) is a genuine alternative to (a) through a different route, mass rather than marginals, and $\mathcal{K}$, $\mathcal{S}$ carry diagnostic information the marginals do not expose. The experiments of §5 were run before (f) was defined and report (a)–(d); (f) is scored on their cached graphs in Tables 29 and 30.

D.10 Algorithm and cost

Algorithm 1 is dominated by steps 4–9: $|P|$ Prompt-A calls plus one Prompt-B call per surviving edge, which the candidate policy keeps near-linear in n rather than quadratic. Inference is negligible by comparison.

Answering all four readouts requires seven inference calls: the base MAR and PR, the $\mathcal{K}$ accumulator MAR, the $\mathcal{S}$ indicator MAR, the reified MAR, the conflict-free ceiling PR, and the base MAP. Computed independently they would cost fourteen, since each readout re-runs the shared base pair. Readout (e) adds one PR, on the penalty network Z_{pen}; its denominator is the ceiling already computed for (d), so (a)–(e) cost eight calls. Readout (f) adds one more PR, on the prior unaries with one reward or penalty factor per relation, for nine. Production inference is weighted mini-bucket (Liu & Ihler, 2011) with i-bound 6; every *exact* value reported in this paper instead comes from explicit 2^n enumeration, which we use as a regression oracle for $n \leq 20$ and which is the source of Tables 27 and the values in §G.

Algorithm 1 COMPUTELCS(y) — the end-to-end coherence score.

Require: response y; relation estimator $\mathcal{M}$; discount λ; priors π
Ensure: headline $\mathrm{LCS}_{\mathrm{bl}} \in (0,1)$, marginals $\{q_i\}$, diagnostics

$\mathcal{A} \leftarrow$ ATOMIZEANDDECONTEXTUALIZE(y)
$P \leftarrow$ SELECTCANDIDATEPAIRS$(\mathcal{A}, y, \dots)$ ▷ Algorithm 2
$\mathcal{R} \leftarrow \varnothing$
for all $(a_i, a_j) \in P$ **do** ▷ concurrent, rate-limited
  $(\text{sense}, \tau, c) \leftarrow$ PROMPTA(a_i, a_j, y) ▷ $c = \mathbb{P}(\tau \mid \cdot)$ from Eq. 4
  if $\tau =$ NONE **then continue**
  $\sigma \leftarrow$ PROMPTB(a_i, a_j, τ, y) ▷ surrogate-token strength
  $\mathcal{R} \leftarrow \mathcal{R} \cup \{(i, j, \tau,\ \mathrm{clip}_{[0,1]}(c \cdot \sigma))\}$
for all $r \in \mathcal{R}$ with $\tau_r \in \mathcal{T}_{\mathrm{conf}}$ sensed as Concession **do**
  if a resolving holding h exists **then** $p_r \leftarrow p_r(1 - \lambda\pi_h)$
  else mark r unresolved
$\mathcal{N} \leftarrow$ network of Eq. 1 over $(\mathcal{A}, \mathcal{R}, \pi)$ ▷ with-priors tables
$\{q_i\} \leftarrow$ INFER$(\mathcal{N}, \text{MAR})$
$\ell \leftarrow \log \mathbb{E}_\pi[G]$ ▷ one PR, Eq. 10
return $\mathrm{LCS}_{\mathrm{bl}} = 1/(1 + e^{-2\ell/n})$, diagnostic $\frac{1}{n}\sum_i q_i$, $\{q_i\}$, $\big(\#\{q_i < \pi_i\},\ \mathcal{E},\ \log Z\big)$

Algorithm 2 SELECTCANDIDATEPAIRS: response-anchored candidate selection.

Require: claims $\mathcal{A} = \langle a_1, \dots, a_n \rangle$ in assertion order; response y; policy; window w; gate threshold θ_g; discourse thresholds θ_d, δ
Ensure: ordered candidate pairs P, plus a coverage record

if policy $=$ ALL_PAIRS **then return** every ordered pair (a_i, a_j), $i \neq j$
$W \leftarrow \{(a_i, a_j) : 0 < j - i \leq w\}$ ▷ forward order window
if policy $=$ BIDIRECTIONAL **then** $W \leftarrow W \cup \{(a_j, a_i) : 0 < j - i \leq w\}$
$G \leftarrow \varnothing$
if policy $=$ GATED **then**
  $G \leftarrow \{(a_i, a_j) : j - i > w,\ \mathrm{sim}(a_i, a_j) \geq \theta_g\}$
align each claim to its best-matching sentence of y
$Prom \leftarrow \{(a_i, a_j) \notin W \cup G : \text{DISCOURSEADJACENT}(i, j)\}$
$Dem \leftarrow \{(a_i, a_j) \in W : \neg\, \text{DISCOURSEADJACENT}(i, j)\}$
return $\big(\text{DEDUP}(W \cup G \cup Prom) \setminus Dem\big),\ (|W|, |G|, |Prom|, |Dem|)$

E RELATION MINING IN DETAIL

E.1 CANDIDATE-PAIR SELECTION

Each claim is aligned to its originating sentence; a pair is *discourse-adjacent* if the two claims share content tokens above a Jaccard threshold, if their sentences are within a short span, or if the target's sentence opens with a discourse connective. Out-of-window pairs that are discourse-adjacent are *promoted* into the candidate set; in-window pairs that are not are *demoted* out of it. The refinement is a correlated filter — it selects for the same surface adjacency that the sense prompt reads as evidence of relatedness — which is why we report it as a policy component and measure its effect rather than treating it as neutral preprocessing.

E.2 WHY THE EVIDENCE SPAN MUST QUOTE THE RESPONSE

The evidence-span requirement of §3.5 obliges the model to quote the words that draw the link from the *response*, not from either claim, and the parser then verifies the quotation is genuinely a substring of the response. Quoting the claims would not work: claims are decontextualized rewrites, and only about 27% of them appear in the response verbatim even after normalization, so a claim-quoting rule would reject most genuine relations.

E.3 THE TWO PROMPT GENERATIONS

Two prompt generations. The first-generation Prompt A over-asserts badly: against a corpus in which 7.2% of candidate pairs are genuinely related, it answers "related" on 72% of the pairs a windowed policy hands it. Five properties push it that way, and the revision addresses each. (i) Its few-shot balance was three related to one unrelated; class balance in the exemplars is a strong prior on the answer distribution, and this one pointed the wrong way by an order of magnitude. The revision is two related to three unrelated. (ii) Its single negative exemplar had the response *explicitly disavow* a link ("the two processes were not related this year"), which teaches that NONE requires denial, whereas real prose simply stays silent; the revision's negatives are all silence. (iii) It asked for grounding but gave the model nowhere to show it and nothing to check it — hence the evidence span. (iv) Its first reasoning step was a leading question listing eight affirmative relation types before mentioning NONE; the revision asks the discriminating question first, whether the text draws the link or the claims are merely near each other. (v) It gave NONE one line against nine for relation-bearing senses; the revision gives it first position and equal prose. The revision also names *transitivity* explicitly as a non-relation, since roughly 85% of the first generation's false positives are the model closing a chain the text laid out pairwise, and 96.6% of them touch an endpoint of a real relation. Both generations are retained permanently so that any claim about the prompt can be measured rather than asserted; §5 reports both.

E.4 COUNTING FAILURES

Failures must be counted, not absorbed. Mining issues one LLM call per candidate pair for Prompt A and one more for each pair that yields an edge, run concurrently under a rate limit. A throttled or failed call returns no parse, and a parser that reads "no parse" as "no relation" converts an infrastructure failure into a sparse graph — which then scores as a *more* coherent response, since fewer conflicts are asserted. We therefore tally call errors separately from genuine NONE answers and report them, and distinguish an explicit NONE from an unparseable coupling for the same reason.

E.5 ALIGNING CLAIMS ACROSS THE TWO STAGES

Aligning claims across stages. The two stages must agree on which claim is which, and the obvious key is the wrong one. Matching proceeds by object identity first (the shared-atomization path, where the response is decomposed once and both stages see the same objects), then by normalized text (casefolded, whitespace-collapsed, terminal punctuation stripped), and only then by identifier. Text precedes identifier deliberately: near-duplicate removal keeps the first-seen claim and leaves the surviving identifier set *sparse* — $a_0, a_1, a_3, \ldots$ — so two independent decompositions of the same response can disagree about which claim occupies which index. Matching on identifiers alone would then attach a prior to the *wrong* claim, which is strictly worse than attaching none, since it silently corrupts the score rather than degrading it. When coverage falls below one half the run is flagged degraded, and the configured policy is to warn, to raise, or to discard every prior and fall back to a clean coherence-only score rather than report a half-primed mixture.

F EVALUATION PROTOCOL AND BASELINES IN FULL

F.1 WHAT EACH FAMILY IS DIAGNOSTIC OF

A coherence measure should be compared against controls chosen to fail in informative ways. We describe five families and state what each is diagnostic of. Four are implemented and share the LCS's own decomposition — each baseline is handed the same atoms the LCS was scored on and, like the coherence MRF, consults no retrieved evidence — so the comparison is an ablation rather than a separate experiment. Metrics requiring a source document or a reference are therefore excluded by construction, and we say so where the exclusion is not obvious. §5.4 states what each column computes; the subsections after this one state how, in enough detail to reimplement.

1. *Ladder-blind controls*: claim count, relation count, response length, and number of perturbation operators applied. These cost nothing and they catch the most likely way a coherence measure is secretly wrong — by tracking edit count or graph density rather than coherence. A measure that

cannot beat them on C3, or that fails to stay flat where they do, has not earned its complexity. Steen & Markert (2022) make the same argument for coherence measures generally, and introduce intra-system correlation and bias matrices to identify measures whose apparent performance comes from system-level confounders; we adopt the controls in that spirit. A control that *succeeds* is bad news for the item, not good news for the control, and §F.9 reports one such case.

2. *Local contradiction detection*: pairwise NLI over claim pairs with no global model, in the spirit of SelfCheckGPT's NLI variant (Manakul et al., 2023). This isolates precisely what joint inference adds over local detection. The prediction is that it does creditably where the defect is a *present* contradiction and poorly where the defect is a *missing* entailment, since a broken chain has no contradictory pair to find. We run it with the same NLI extractor as §B, over the same atoms, so the only differences from the LCS are relation *typing* and *propagation*.
3. *Structured reasoning consistency*: ROSCOE's Self-Consistency (Golovneva et al., 2023), which is evidence-free like ours but differs in three separable ways — it is a *maximum*, so a second conflict changes nothing; it is *forward-only*, so a relation running backward in claim order is unreachable (§3.5 measures 105 of 603 gold relations as both directed and backward); and it is *untyped*. We therefore also report a mean-aggregated variant, and implement a symmetric one, which turn one baseline into an ablation identifying *which* of the three differences accounts for any gap. ROSCOE targets step-by-step reasoning chains, and a long-form response is not a chain, so we label it adapted.
4. *LLM judges*: G-Eval's coherence dimension (Liu et al., 2023), with the SummEval coherence rubric (Fabbri et al., 2021), and a direct "rate the logical coherence from 1 to 5" prompt as the no-rubric control. These are the practical incumbent and the comparison reviewers will want. The ladder is a demanding test for them, since consecutive rungs differ by a few words, so a judge must be sensitive at very small edit distances while remaining flat where meaning is preserved. We report each judge as a mean over five runs with its standard deviation, because the determinism of Rem. 2 is only an advantage next to a measurement of the judge's instability, and because a spread wider than the gap being discussed bounds what the comparison can conclude — though §F.6 reports that how much it bounds depends sharply on the backend.
5. *Discourse representations*: DiscoScore (Zhao et al., 2023), with an entity-grid model (Barzilay & Lapata, 2008) as the classical reference. DiscoScore is presented as reference-based and most of its variants consume a reference, which a standalone response does not have; three of them (RC, LC and ENTITYGRAPH) ignore the reference argument entirely and score a single document, and those are the three we use — the last of which *is* an entity grid, so the classical reference comes from the same definition. All three reduce to noun-lemma repetition across sentences, with no representation of entailment or contradiction, which makes them a sharp test of whether coherence is separable from *cohesion* rather than a weak competitor: §F.9 reports a pair of responses with identical nouns and identical repetition structure, one self-consistent and one self-contradictory, on which all three are exactly flat.
6. *Internal ablations*: uniform versus factuality priors (§3.6); the four readouts of §D; the four candidate-pair policies; the two Prompt-A generations; and the gold-relation ceiling. These separate the model's design choices from the estimator's accuracy.

Beyond ladder ordering we recommend Spearman correlation against human coherence ratings on the same items, reported alongside per-family ordering agreement. Ordering agreement is the sharper instrument, being a controlled contrast with a known answer; human correlation is what establishes that the construct being ordered is the one readers care about. Neither substitutes for the other.

A baseline we cite but do not run. The closest existing work to our relation-mining stage is Liu & Strube (2025), who also build a graph over sentence pairs labelled with PDTB 3.0 senses — via a discourse parser rather than an LLM — and add entity links. We do not report a head-to-head number, and the reason is a difference in output type rather than a convenience: their model emits a *three-way* label (low, medium, high), and a three-way label cannot be checked against the ordering constraints of §5.1, which ask whether consecutive rungs move in a declared direction. The comparison would also require their trained classifier and a PDTB parser, neither of which is a pretrained scorer one can apply to a new response. Three differences are worth recording regardless, because they are the axes on which the two approaches diverge: their model performs no probabilistic inference, so a local conflict cannot propagate; it retains only forward-order pairs ($i < j$), so a backward relation

is unreachable, which is the same structural cap we measure in §3.5; and it produces no per-claim attribution, so an incoherence is detected but not located.

F.2 The shared interface, and what "the same input" means

Every baseline implements one method, `score(atoms, response)`, returning a single scalar in $[0, 1]$ with higher meaning more coherent, so the ladder scorer reads a baseline column exactly as it reads an LCS column. The record it returns carries four fields beyond the score, and two of them are load-bearing for the claim that this is an ablation.

- `score`: the value, or *nothing at all*. An abstention is represented distinctly from 0.0 throughout. This is not fastidiousness: 0.0 is a real score meaning "maximally incoherent", so collapsing the two would let an infrastructure failure — a throttled endpoint, an unparseable generation — read as a confident verdict of incoherence, and a run against a degraded endpoint would report systematically lower coherence rather than reporting that it failed.
- `atoms_scored` and `pairs_scored`: how many atoms the baseline saw and how many claim pairs it actually evaluated. These are recorded on every row so a report can *prove* the arms shared one decomposition. Without them the comparison is an assertion; with them it is checkable, and a coverage difference — the thing most likely to differ silently between two quadratic-cost arms — shows up as a number rather than as a discrepancy nobody looks for.
- `diagnostics`: per-baseline extras (the raw unnormalized quantity, per-pair labels, the argmax pair, the call-failure count, wall-clock). Never required to read the table; the first place to look when a number surprises.

The abstention rules, collected in one place, are: fewer than two atoms, for any pairwise baseline (a contradiction rate over zero pairs is undefined, and returning 1.0 there would score a single-claim response as perfectly coherent — the same vacuity trap §D identifies for the "all relations satisfied" readout); every call in the batch failed to parse; no rating found in a judge's completion; an empty noun grid; an empty response. In each case the row records a reason string, and the report counts abstentions separately rather than imputing a value.

One asymmetry, stated plainly. The two judges and the three discourse metrics accept the atom list and *ignore* it: they read the response prose. So for five of the ten published columns the input is not held fixed with the LCS's decomposition, and those comparisons are correspondingly less controlled — a judge could in principle be reading something the atomizer discarded. The five NLI-based and count-based columns do consume the atoms, and for those the ablation claim is exact. We prefer stating this to quietly describing all ten columns as seeing the same input.

F.3 Ladder-blind controls

Three controls are implemented, computed with no model at all.

Claim count is $\min(1,\, n/64)$, where n counts atoms with non-blank text. *Response length* is $\min(1,\, |\mathrm{tok}(r)|/1024)$, where tok splits on runs of non-word characters and drops empties. The two saturation constants are set well above the corpus — the ladder items carry sixteen claims and a few hundred tokens — so that the normalization is never clipping in the range where the comparison happens; a sixteen-claim, 646-token response scores 0.250 and 0.631 respectively, both comfortably interior. *Edit distance* is Python's `SequenceMatcher` similarity ratio against a reference response, and abstains when no reference is supplied.

Edit distance deserves a word because it is the sharpest control available for a perturbation ladder and yet is absent from Table 8. A ladder family is built by applying operators to a base response, so similarity to that base is very nearly a count of the operators applied, and a measure tracking it would reproduce a declared ordering while knowing nothing about coherence. It is implemented and exported; the ladder driver instantiates only the two controls that need no reference, so the published table carries claim count and length. We report this rather than leaving the reader to infer that only two controls exist.

Why these scores carry no direction. A control has to be mapped into "higher = more coherent" to sit in the same table as the other columns, and for these quantities no such mapping is defensible: there is no reason a longer response should be more coherent, or less. Rather than pick a direction and quietly encode a hypothesis, each control reports its raw quantity in diagnostics and a normalized score whose only guarantee is *monotonicity* in that quantity. A control's ordering cell is therefore a test of whether the control *tracks* the declared ordering, in either direction — not of whether it agrees with it. This is what makes a passing control bad news.

Example 2 (Controls are blind by construction)**.** *Take a three-claim,* 40*-token response and an expanded paraphrase asserting the same content in six claims and* 80 *tokens. Claim count moves from* $3/64 = 0.047$ *to* $6/64 = 0.094$ *and length from* $40/1024 = 0.039$ *to* $80/1024 = 0.078$*: both double. Neither response is more coherent than the other, and no operation on the claim set has occurred beyond restating it. Conversely, a perturbation that reorders two sentences leaves both controls exactly fixed, whatever it does to the argument. The ladder corpus is built so that the second case is the norm: every rung of a family shares one claim set, so claim count is flat within a family by construction, which is why it scores* $20/20$ *on invariance and* $0/50$ *on ordering.*

F.4 Local contradiction detection

The score is one minus the fraction of claim pairs the NLI extractor labels contradictory,

$$\text{CONTRADICTION RATE} \;=\; 1 \;-\; \frac{\big|\{(i,j) : \hat{y}_{ij} = \text{CONTRADICTION}\}\big|}{\big|\{(i,j) : \hat{y}_{ij} \text{ parsed}\}\big|}, \tag{11}$$

taken over all $\binom{n}{2}$ unordered pairs $i < j$. Only one arc per pair is scored: contradiction is symmetric in the sense this baseline cares about — if a_i contradicts a_j then a_j contradicts a_i — so scoring both would double the cost and change nothing. Baselines that are deliberately directional do their own enumeration (§F.5).

Three implementation points determine what the number means.

The denominator counts parsed verdicts, not issued calls. A call that failed, or returned a verdict with no readable label, is counted in a separate failure tally and *excluded* from both numerator and denominator. An unparseable verdict is a missing measurement, not evidence of consistency, and §F.8 explains why the alternative is actively dangerous for this particular estimator.

A soft variant is available and is not the headline. Accumulating each flagged pair's contradiction *probability* instead of counting labels is the obvious refinement, and it is implemented, because it separates two different failures: mislabelling pairs, versus labelling them correctly but aggregating badly. The label-counting form is what we report, because it is the form the literature actually runs, and beating a refinement nobody uses would be a straw man.

It is ungrounded on purpose. The extractor can be given the response as context, prepended to the premise, matching the relation miner's response-grounded regime (§3.5). That switch is off. The point of this column is to be the ungrounded *local* comparison; grounding it would quietly turn it into a different and better method, and the gap we want to measure is the one attributable to typing and propagation alone.

Example 3 (The rate accumulates; the maximum will not)**.** *Four claims give* $\binom{4}{2} = 6$ *pairs. Plant one contradictory pair: the score is* $1 - \frac{1}{6} = 0.833$*. Plant a second, elsewhere in the response:* $1 - \frac{2}{6} = 0.667$*. The measure falls again, as it should — two independent conflicts are worse than one. Hold this example; the same two responses are what separate ROSCOE's two aggregations in Ex. 4. Now suppose instead that the defect is a* missing *entailment: claim* a_4 *is asserted but nothing in the response supports it, and no pair anywhere contradicts. Every pair is labelled neutral, the rate is* 0*, and the score is exactly* 1.0 *— a perfect verdict on a response whose support chain is broken. This is the predicted failure of §F.1, and it is structural rather than an accuracy problem: there is no contradictory pair to find.*

F.5 ROSCOE Self-Consistency, and its ablation arms

The published measure is

$$\text{SC} \;=\; 1 \;-\; \max_{i=2..n} \max_{j<i} p_{\text{contr}}(h_i, h_j), \tag{12}$$

with our atoms substituted for ROSCOE's reasoning steps. The arcs are enumerated exactly as written: for each i from 1 to $n-1$ and each $j < i$, one call with a_i as premise and a_j as hypothesis — $\binom{n}{2}$ backward-directed questions. The symmetric arm additionally enqueues (j, i), giving $n(n-1)$ calls and removing the structural blindness to backward relations.

Reading a contradiction probability, and why non-contradictions contribute exactly zero. Our extractor reports the probability of *the label it emitted*, not a distribution over all three labels (§B). So when the emitted label is entailment or neutral there is no contradiction mass to read, and we take $p_{\text{contr}} = 0$ exactly. This is the only sound reading available: the model's stated verdict is "not a contradiction", and manufacturing a residual such as $1 - p(\text{entailment})$ would fabricate a number the extractor never produced. A verdict with no label at all is a failure, counted and excluded. The consequence matters for interpreting the mean arm: it is a mean over a vector that is mostly exact zeros, so it behaves like a contradiction *density* rather than like an average severity.

Example 4 (The maximum saturates where the mean accumulates)**.** *Take the two responses of Ex. 3, with* $p_{\text{contr}} = 0.9$ *on each planted pair and* 0 *on the rest of the six. With one planted pair,* $\max = 0.9$ *and the score is* 0.100*; the mean is* $0.9/6 = 0.15$ *and the score is* 0.850*. Add the second planted pair: the maximum is* still 0.9*, so the score is* still 0.100*, while the mean rises to* $1.8/6 = 0.3$ *and the score falls to* 0.700*. One arrangement of the same evidence registers the second conflict and the other cannot. On the seventy ladder items this is not a hypothetical: the published max arm is pinned at the floor — exactly constant at zero under* `llama-3.3-70b`*, and under* `gpt-oss-120b` *taking* 18 *distinct values every one of which lies below* 5×10^{-7}*, so that at the* 10^{-6} *resolution the ordering axis uses it is a single value there too — while the mean arm, over the* same *pairwise probabilities, takes* 57 *and* 63 *distinct values at that same resolution. The ablation's answer is that aggregation, not direction, accounts for the gap.*

The aggregation and the direction are exposed as options precisely so the three differences can be separated rather than merely described. Table 8 carries the faithful max arm and the mean arm; the symmetric arm is implemented and instantiated by the fixture driver, and is not in the published table because the ladder driver builds the two aggregation arms only.

F.6 LLM JUDGES

Both judges read the response as prose and return a 1–5 rating rescaled by $(k-1)/4$. The G-Eval prompt reproduces the SummEval coherence rubric rather than paraphrasing it — a rubric we invented would be a judge we tuned — and the direct prompt differs from it only in carrying no rubric and asking for no reasoning, keeping the same scale and the same answer format so that the two differ on one axis. Both are reproduced verbatim in App. K.

Parsing. The rating is read with the pattern `\[ ([1-5]) \]` and the *last* match is taken. This is not a detail: both prompts end by giving an example ("… as a single digit in square brackets, for example [3]"), models on this stack routinely echo the instruction before answering, and taking the first match would read the prompt's own example as the judgement — silently returning 3 for every item. A bare standalone digit is accepted as a fallback, and anything else is an abstention rather than a low score.

G-Eval's probability weighting. Where the backend exposes logprobs we report $\sum_{k=1}^{5} k\, p(k)$ rather than the emitted integer. The distribution is read off the rating token's alternatives: we walk the per-token logprobs, keep the *last* standalone token in $\{1, \ldots, 5\}$ — the same last-match rule as the regex, and for the same reason, since reasoning-channel tokens precede the answer on this stack — exponentiate its top-k alternatives, keep those that are digits in range, and renormalize over them. Absent a rating token or usable alternatives, the judge falls back to the emitted integer and records that the fallback happened, so a weighted column cannot be confused with a half-weighted one. The weighting matters more here than in G-Eval's original setting: a 1–5 integer scale is coarse, consecutive ladder rungs differ by a few words, and an unweighted judge returns the *same* integer for every rung of a family — its ordering agreement would then be decided by tie-breaking rather than by judgement.

What the variance measurement actually found. Every judge number is a mean over five runs (five is the floor; fewer cannot show a spread), with the standard deviation, minimum, maximum and the list of emitted integers travelling alongside it; a judge abstains only if all five runs abstain. The measurement is worth reporting because it came out differently on the two backends, and the difference changes what the spread can be used for. Under `llama-3.3-70b` the five runs return the *same* integer on all seventy items for G-Eval and on sixty-nine of seventy for the direct judge, and the median recorded standard deviation is of order 10^{-10}: what it measures there is jitter in the logprob-weighted expectation, not disagreement between runs. Under `gpt-oss-120b` the same judges are genuinely unstable — only 20 and 19 of seventy items return five identical integers, and the median standard deviation is 0.100 on a $[0, 1]$ scale, wider than many of the gaps the table invites a reader to compare. So the spread bounds the comparison substantially on one backend and barely at all on the other, and a single reported standard deviation should not be read as a property of judges in general. Either way the determinism argument of Rem. 2 rests on more than instability: a judge cannot say *which* claim is responsible, which is the axis §F.1 puts weight on.

F.7 Discourse representations

Which of DiscoScore's metrics are usable, and why. DiscoScore's scorer exposes every metric as `(sys, ref)`, but its methods differ in whether the reference is consulted. RC, LC and EntityGraph forward only `sys` and ignore `ref` entirely, so they are genuine single-document coherence measures and are directly comparable to the LCS. LexicalGraph requires external word vectors. The DS_Focus_* and DS_SENT_* families consume the reference, which a standalone response does not have. Hence three columns, the last of which *is* the Barzilay–Lapata grid projected onto a sentence graph, so the classical entity-grid baseline needs no separate reimplementation.

The three definitions, over the noun-lemma-by-sentence grid: RC is repeated noun mentions divided by distinct nouns; LC is the count of nouns occurring in more than one sentence divided by distinct nouns; EntityGraph is $\big(\sum_{\text{lemmas}} \sum_{i<j\in S(\ell)} 1/(j-i)\big)/|S|$, summing an adjacency weight over every pair of sentences sharing a lemma and averaging over sentences. RC and EntityGraph are *not* bounded above by one.

Which implementation produced the reported numbers. Ours. `disco_score` is not installable as a dependency: it is absent from PyPI, installs from a git URL, and calls `spacy_udpipe.download("en")` at *import* time, so importing it requires network access and makes the module unimportable offline. We therefore import it lazily and compute these three metrics from their definitions ourselves, and every reported discourse row records which path produced it; all of them record the local implementation. The approximation is in what counts as a noun: with no POS tagger, we take alphabetic tokens of at least four characters that are not in a 54-word function-word list, split sentences on punctuation followed by whitespace, and fold plurals by stripping a trailing `s` unless the word ends in `ss`. This over-counts nouns. It over-counts them *equally in both members of any compared pair*, which is what the argument needs: the claim being tested is about what these definitions can represent, not about reproducing DiscoScore's decimals, and the pair in §F.9 is noun-matched by construction.

Clamping, and what it does to the entity-graph column. Scores are clamped into $[0, 1]$ for reporting, and both the raw value and a clamped flag are recorded so the clamping is visible rather than silent. For EntityGraph on the ladder corpus it is not a cosmetic step: the raw metric takes 62 distinct values spanning 1.36 to 4.93 across the seventy items, and every one of them is clamped to 1.0. Its perfect invariance score is therefore saturation above the reporting ceiling, not an inability to distinguish responses — the mirror image of a maximum saturated at the floor, and a distinction §5.4 now draws. Because a clamped tie can hide a real ordering, the direction checks in our drivers compare on the *raw* value wherever one is recorded; reporting the clamped tie instead would understate how wrong a cohesion metric gets the pair in §G.3, where all three discourse metrics rank the adversarial summary above the faithful one.

Abstention. RC and LC divide by the number of distinct nouns and EntityGraph by the sentence count, so noun-free text divides by zero. That is "no cohesion signal", not "incoherent", and the baseline abstains.

F.8 COST, THROTTLING, AND HOW FAILURES ARE COUNTED

The pairwise baselines issue $O(n^2)$ NLI calls per response: 120 at the ladder's sixteen atoms, 528 at the thirty-three of the longest fixture. That is the cost to keep in mind when comparing against the LCS, which issues one call per *candidate pair* under a windowed policy and then performs inference over the mined relations only (§D.10).

All of them route through one throttled batch path, sharing the pipeline's limiter — a 1500-requests-per-minute token bucket and a bounded in-flight ceiling — rather than issuing $\binom{n}{2}$ sequential calls. Two properties of that path are worth recording.

Each call is built from the extractor's own prompt and options, not a fixed prompt. The NLI extractor's prompt varies with the estimation method (§B), and under the direct few-shot prompt it is precisely the prompt that makes the label probability informative rather than saturated. Hardcoding one prompt in the batch path would silently score the baselines under a different estimator than the LCS they are being compared against, which would invalidate the ablation while leaving every number looking plausible.

A failed call is preserved as a failure. The extractor's own batch helper maps a failed call onto a neutral verdict with probability one — byte-identical to a genuine neutral. For a contradiction-*rate* baseline that substitution is not neutral at all: a throttled call becomes evidence of "no contradiction here", so Eq. 11 moves *up* when the endpoint degrades, and a rate-limited run reports a *more* coherent response than a healthy one. We therefore do not use it. The batch path returns a distinct failure sentinel that no caller can mistake for a label, every baseline excludes failures from its denominator and counts them, and a row whose calls all failed abstains. This is the same hazard the relation miner documents for mining, where a rate-limited call parsed as "no relation" costs recall silently, and it is why §E.4's insistence that failures be counted rather than absorbed applies on the baseline side too.

Verdicts are cached across arms. The NLI-based columns ask the *same* claim pairs: on a 26-claim response the identical 325 pairs would otherwise be scored once per arm. A verdict cache keyed on the model, the estimation method and the pair text makes the model see each distinct pair once. Failures are never written to it — caching one would promote a transient throttle into a permanent stored verdict of "no contradiction here", which is the first hazard made durable.

How a baseline column is scored against the ladder. The corpus states its constraints per readout, because the LCS reports several; a baseline has one score, so only the readout-independent content transfers. C1 (strict increase between consecutive rungs) and C3 (endpoint separation, rung 0 below rung 4) apply directly, deduplicated across readouts, over the increase-type families CONFLICT and CHAIN: 50 assertions. The invariance axis takes every declared rung pair of the invariance-type families ORDER and CONTROL, satisfied when the score does not move: 20 pairs. Two scores closer than 10^{-6} count as flat, so floating-point noise cannot pass for a strict increase — and, on the invariance axis, so that jitter beneath that threshold counts as unchanged. C2 is *excluded*: it predicts the internal behaviour of a particular readout (§D), which a single-score baseline makes no claim about, and scoring a baseline against it would credit or punish it for a property it never asserted. These are the same definitions used for the LCS arms, which is what makes the baseline rows and the LCS rows of Table 8 commensurable.

F.9 WHAT THE BASELINES SHOW BEFORE ANY LADDER SWEEP

Two properties of these baselines are worth establishing before any ladder sweep, because both bear on what the comparison in §5.5 can support.

First, *cohesion is not coherence, and the discourse floor cannot tell the difference.* On a pair of two-sentence responses with identical nouns and identical repetition structure — "The weld passed the tensile test. The weld held under load." against "… The weld *failed* under load." — all three single-document DiscoScore metrics return *exactly* equal values, while the pairwise-NLI and ROSCOE baselines separate the pair. The metrics are defined over noun repetition, so this is a property of their definitions rather than an accuracy failure; it is also the cleanest available demonstration that the axis we measure is not the one entity-based coherence models measure.

Second, *a controlled contrast has to be checked, not assumed.* A measure can reproduce a declared ordering for reasons that have nothing to do with coherence, so a contrast is only informative once the model-free proxies have been shown *not* to reproduce it. The claim-identical ordering pair of §G.3 is the clean case: claim count is exactly flat there by construction, response length differs by four tokens (265 against 269) in the direction *opposite* to the declared ordering, and all three discourse metrics rank the adversarial summary *above* the faithful one, the self-serving-first ordering carrying more noun repetition. A cohesion metric prefers the less coherent response of the two. The ladder corpus of §5 generalizes exactly this property: every rung of a family shares one claim set, so a claim-count proxy is flat by construction across all seventy items.

F.10 Mean readout values per source

Table 7 of §5 reports only the constraint counts, since averaging a readout across the rungs of a ladder collapses the ordering the ladder exists to test. The mean values are recorded here as a level comparison. The BIDIRECTIONAL density cost is visible quantitatively: $\mathrm{LCS}_{\mathrm{lp}}$ collapses to 0.037 and 0.033 against gold's 0.410 on the same seventy responses, from adding roughly two thousand spurious factors — the log-partition readout registers added factor mass most directly.

Source	Policy	$\mathrm{LCS}_{\mathrm{mm}}$	$\mathrm{LCS}_{\mathrm{cn}}$	$\mathrm{LCS}_{\mathrm{re}}$	$\mathrm{LCS}_{\mathrm{lp}}$
GOLD	—	0.795	0.515	0.238	0.410
GOLD_VALID	—	0.794	0.576	0.316	0.485
mined by `llama-3.3-70b`					
	WINDOWED	0.796	0.676	0.384	0.224
	BIDIRECTIONAL	0.723	0.542	0.205	0.037
mined by `gpt-oss-120b`					
	WINDOWED	0.825	0.500	0.425	0.129
	BIDIRECTIONAL	0.808	0.462	0.251	0.033

Two further measurements belong with these. The cohesion baselines drift by at most 0.0096 across a permuted family, which is why they score $16/20$ rather than $20/20$ on invariance. And the 0.014 run-to-run F1 movement of §5.1 supersedes the ± 0.06 we previously estimated on a ten-item corpus: the larger corpus tightens reproducibility rather than merely enlarging the denominator.

F.11 Judge saturation, measured

The two judges of §5.5 fail in opposite ways, and the mechanism is saturation rather than noise. `llama-3.3-70b`'s G-Eval is astonishingly stable and useless: mean per-item standard deviation 0.0002, only three distinct values across the seventy items, and a mean score of 0.996 — it rates almost everything at ceiling, so it cannot order anything. `gpt-oss-120b`'s G-Eval uses its range (fourteen distinct values, mean 0.749) and pays in stability (mean sd 0.0800, max 0.187). Reporting judge variance alone would have called the first judge the reliable one, which is exactly backwards for the task. Both judges also violate invariance freely: `gpt-oss`'s G-Eval moves by up to 0.55 across rungs that differ only in sentence order.

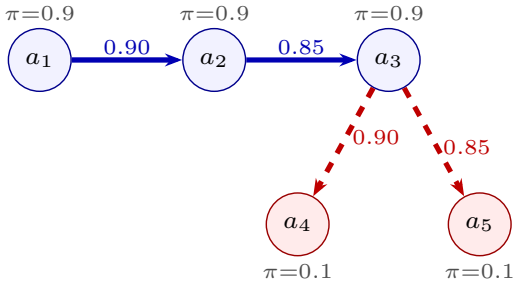
F.12 Per-measure ladder agreement

G Worked examples in full

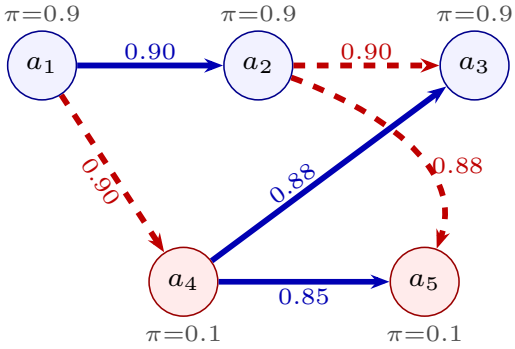
We work four examples. The first is the smallest complete instance of the two-stage model: one claim set, two responses, exact inference. The second is a richer sixteen-claim graph that exercises the conflict taxonomy. The third is a claim-identical pair that isolates ordering. The first three are hand-authored, which bounds what they can show: they demonstrate that the model reads a graph correctly, not that the estimator finds such graphs in text it has never seen. The fourth is therefore a corpus rather than an example — one hundred machine-generated responses whose incoherence is planted by construction, scored end to end with no hand-drawn relations anywhere.

Table 31: Declared ordering assertions satisfied on the ladder corpus, under two different judge models. Only the readout-independent content of a constraint is scorable by a single-score baseline: the four C1 consecutive-increase pairs and the C3 endpoint separation, giving five per family. C2 is excluded because it predicts the internal behaviour of particular readouts (§D.7), which a baseline makes no claim about. The ORDER and CONTROL families assert invariance rather than increase and so contribute *no* scorable assertion, leaving ten families and 50 assertions; the LCS rows are scored on *exactly* those 50, so they are comparable to the rows below them, while the 82.7% of Table 7 is computed over the fuller 202-assertion set including invariance and is not. Judges are the mean of five runs. Model-free baselines are identical under both judges, as they must be; the two judge rows are the entire difference.

Measure	Assertions satisfied				
	`llama-3.3-70b`		`gpt-oss-120b`		Uses LLM
$\mathrm{LCS}_{\mathrm{bl}}$ (gold relations)	49/50		**98.0**		no
$\mathrm{LCS}_{\mathrm{mm}}$ (gold relations)	37/50		74.0		no
$\mathrm{LCS}_{\mathrm{lp}}$ (gold relations)	36/50		72.0		no
$\mathrm{LCS}_{\mathrm{cn}}$ (gold relations)	34/50		68.0		no
RC (cohesion)	30/50	60.0	30/50	60.0	no
LC (cohesion)	28/50	56.0	28/50	56.0	no
LENGTH (control)	20/50	40.0	20/50	40.0	no
ROSCOE-SC (mean)	16/50	32.0	17/50	34.0	yes
CONTRADICTION RATE	12/50	24.0	12/50	24.0	yes
JUDGE, direct rating	4/50	8.0	18/50	36.0	yes
JUDGE, G-Eval	0/50	0.0	21/50	42.0	yes
ROSCOE-SC (max)	0/50	0.0	0/50	0.0	yes
ENTITY GRAPH	0/50	0.0	0/50	0.0	no
CLAIM COUNT (control)	0/50	0.0	0/50	0.0	no



(a) **Response A, coherent.** The three true claims form a support chain ($a_1 \to a_2 \to a_3$) and each false claim is *attributed and rejected* by a CONTRADICTION edge from a_3. Every claim ends on the correct side of its own prior: $\mathrm{LCS}_{\mathrm{mm}} = 0.590$, $\mathrm{LCS}_{\mathrm{cn}} = 0.963$, $\log Z = -0.95$.

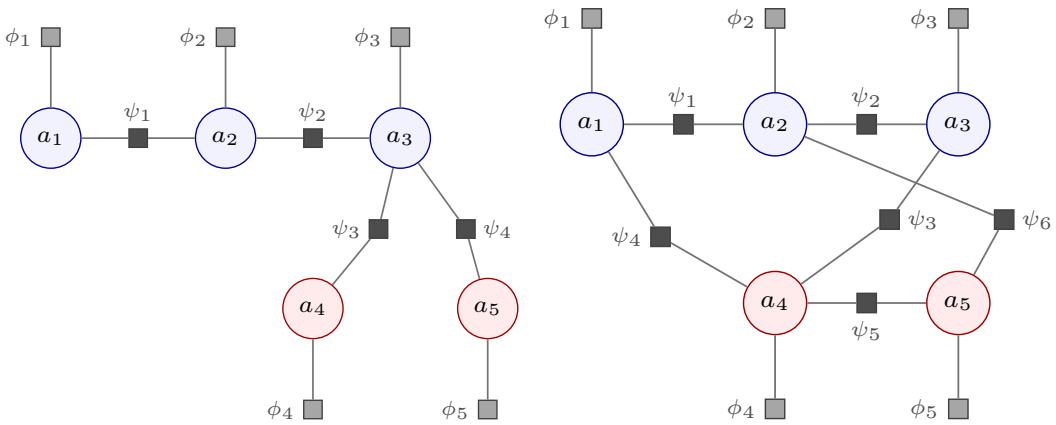


(b) **Response B, incoherent.** The same five claims. The response contradicts its own chain ($a_2 \not\to a_3$) and makes the false a_4 a load-bearing premise for the true a_3, while a_1 and a_2 deny a_4 and a_5. The *true* claim a_3 is dragged to 0.513 — far below its own 0.9 prior: $\mathrm{LCS}_{\mathrm{mm}} = 0.494$, $\mathrm{LCS}_{\mathrm{cn}} = 0.414$, $\log Z = -3.78$.

Figure 5: Relation graphs for the five-claim worked pair (§B). Blue nodes are claims true of the world ($\pi_i = 0.9$), red nodes false ($\pi_i = 0.1$); the *same* five claims appear in both panels. Solid blue: ENTAILMENT; dashed red: CONTRADICTION. Line width grows with the relation weight p. The panels differ only in *how the response arranges* the claims, which is the whole of what coherence measures — no per-claim factuality check can separate them.

G.1 A FIVE-CLAIM PAIR, END TO END

The point of this example is that factuality and coherence are separable, so we hold the first fixed and vary only the second. Five claims about the Voyager 1 mission are fixed once and for all, three of



(a) **Response A.** Five unaries and four pairwise factors.

(b) **Response B.** The same five unaries, but six pairwise factors wired so that ψ_3 makes a false claim a premise for a true one.

Figure 6: The coherence MRF of Eq. 1 for the two responses, as an explicit factor graph. Circles are the binary claim variables; grey squares the unary factors $\phi_i = [1 - \pi_i, \pi_i]$ that carry the priors; black squares the pairwise factors ψ_{τ_r,p_r} of Table 26, one per mined relation. **The priors enter only through the** ϕ_i: the claim-level potentials are identical across the two panels, and every difference in score comes from the ψ_r — their number, their endpoints, and their couplings.

them true of the world and two false; stage one is taken as given, contributing the priors $\pi = 0.9$ for each true claim and $\pi = 0.1$ for each false one through the composition of §3.6:

		π_i	Claim
a_1	true	0.9	Voyager 1 crossed the heliopause in August 2012.
a_2	true	0.9	After the crossing, Voyager 1's instruments measured a sharp rise in plasma density.
a_3	true	0.9	Voyager 1 is therefore in interstellar space.
a_4	false	0.1	Voyager 1's plasma wave instrument failed permanently before the crossing.
a_5	false	0.1	Voyager 1 has left the Solar System entirely.

Two responses then assert *these same five claims* and differ only in how they arrange them. Because the claim set is identical, every per-claim factuality check returns the same verdict on both; the relation graphs (Figure 5) and the MRFs they induce (Figure 6) are what differ. Table 32 gives both responses in full, with the span asserting each claim marked $[a_i]$ at its first mention.

The italicized spans are the ones the estimator reads as relations, and they are the whole difference between the two. Table 33 lays the two relation sets side by side, one row per claim pair, with each relation's discourse sense and the coupling it compiles to. In A the false claims are named, attributed, and then refused on a stated ground, which compiles to a CONTRADICTION edge from a_3. In B the same falsehoods are named and then *used* — a_4 is offered as the reason a_3 holds — while the response also denies a_3 outright and, in its last sentence, denies a_4 and a_5 too. Mentioning a falsehood is not what costs B its score; asserting it and reasoning from it is. All values below are exact, by $2^5 = 32$-world enumeration.

Example 5 (Coherent)**.** *Response A's four relations (Table 33, column A) are a two-edge support chain over the true claims and one rejection of each false claim. The model returns* $\mathrm{LCS}_{\mathrm{mm}} = 0.590$ *and* $\mathrm{LCS}_{\mathrm{cn}} = 0.963$ *with* $\log Z = -0.95$*. Every claim finishes on the correct side of its own prior: the true claims sit at* 0.939*,* 0.989 *and* 0.992 *— each* above *its* 0.9 *prior, because the chain supplies mutual support — and the false ones at* 0.013 *and* 0.020*, pushed below* 0.1 *by the rejections. Nothing in the response fights anything else, so no claim is penalized.*

Example 6 (Incoherent)**.** *Response B keeps only* $a_1 \to a_2$ *and replaces the rest with six relations that pull against each other (Table 33, column B): it denies its own conclusion, promotes the false* a_4 *to a premise for that same conclusion, chains* a_4 *into* a_5*, and then denies both falsehoods in its closing sentence. Three of its six relations are conflicts, and two of those point at claims the response*

Response A (coherent) — $\mathrm{LCS}_{\mathrm{mm}} = 0.590$, $\mathrm{LCS}_{\mathrm{cn}} = 0.963$, $\log Z = -0.95$

Voyager 1 crossed the heliopause in August 2012, $[a_1]$ and in the days that followed its instruments measured a sharp rise in plasma density $[a_2]$ — the signature of the denser medium beyond, and *the measurement that established the boundary had been passed.* Voyager 1 is therefore in interstellar space. $[a_3]$ Two persistent misconceptions are worth clearing up. The first is that the probe's plasma wave instrument had failed permanently before the crossing; $[a_4]$ *it cannot have, since it is precisely that instrument which returned the density profile the conclusion rests on.* The second is that Voyager 1 has left the Solar System altogether. $[a_5]$ *Being in interstellar space is not the same thing: the Sun's gravitational reach extends through the Oort cloud, far beyond the probe's present distance.*

Response B (incoherent) — $\mathrm{LCS}_{\mathrm{mm}} = 0.494$, $\mathrm{LCS}_{\mathrm{cn}} = 0.414$, $\log Z = -3.78$

Voyager 1 crossed the heliopause in August 2012, $[a_1]$ and after the crossing its instruments measured a sharp rise in plasma density. $[a_2]$ *That said, the probe cannot really be counted as being in interstellar space.* $[a_3]$ The plasma wave instrument had in fact failed permanently before the crossing, $[a_4]$ *and it is precisely that failure which tells us the probe reached interstellar space.* The instrument failure also means Voyager 1 has left the Solar System entirely. $[a_5]$ *Of course, the August 2012 crossing rules out any such permanent failure, and the density measurement is flatly inconsistent with the probe having left the Solar System.*

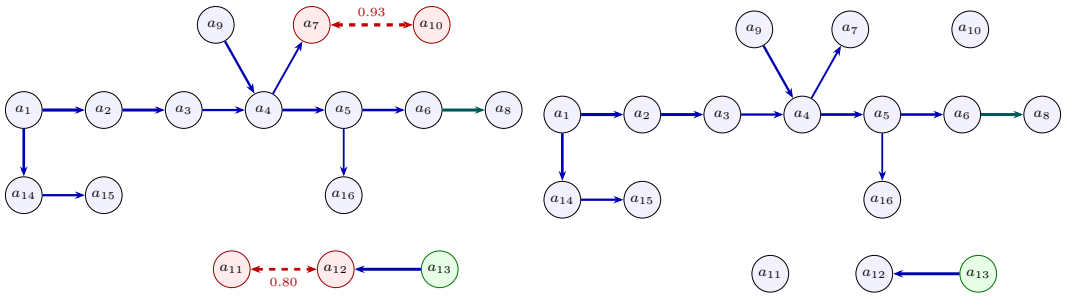
Table 32: The two responses in full. They assert the *same* five claims of §G.1; each claim's first mention is marked $[a_i]$. Italicized spans are the ones the estimator reads as relations. In A they discharge the two false claims; in B they reason *from* them.

Table 33: The mined relations of both responses, one row per claim pair. S is the Level-2 discourse sense, τ the Level-1 coupling it compiles to (Table 24), p the weight. E abbreviates ENTAILMENT, C CONTRADICTION. These are exactly the edges of Figure 5 and the factors ψ_r of Figure 6; a dash means the response draws no relation over that pair.

	Response A (coherent)				**Response B** (incoherent)				
Pair	Edge	S	τ	p	Edge	S	τ	p	What differs
$\{a_1, a_2\}$	$a_1 \to a_2$	Cau-Eff	E	.90	$a_1 \to a_2$	Cau-Eff	E	.90	The one relation both agree on.
$\{a_2, a_3\}$	$a_2 \to a_3$	Evidence	E	.85	$a_2 \to a_3$	Contrast	C	.90	A supports its conclusion; B denies it. Support becomes conflict.
$\{a_3, a_4\}$	$a_3 \to a_4$	Contrast	C	.90	$a_4 \to a_3$	Evidence	E	.88	**The arrow reverses.** A rejects a_4; B makes it a premise.
$\{a_3, a_5\}$	$a_3 \to a_5$	Contrast	C	.85		—			A refuses a_5; B never draws the distinction.
$\{a_4, a_5\}$		—			$a_4 \to a_5$	Cau-Eff	E	.85	B alone chains one falsehood into the other.
$\{a_1, a_4\}$		—			$a_1 \to a_4$	Contrast	C	.90	B denies a_4, which it had just used as a premise — this strands a_3.
$\{a_2, a_5\}$		—			$a_2 \to a_5$	Contrast	C	.88	Likewise for a_5: asserted, then denied.
Totals		2 E, 2 C				3 E, 3 C			A's conflicts point *outward*, at the false claims; half of B's point *inward*.

itself asserts. The score falls to $\mathrm{LCS}_{\mathrm{mm}} = 0.494$ *and* $\mathrm{LCS}_{\mathrm{cn}} = 0.414$*, with* $\log Z = -3.78$*. The diagnostic that localizes the damage is per-claim: the* true a_3 *is dragged to* 0.513*, far below its own* 0.9 *prior, because its only support arrives from* a_4*, which the response itself denies — and a premise the text refuses cannot transmit support. Three claims finish below their own priors, against two in Response A, and* a_3 *is the one that matters: a claim true of the world, demoted to a coin flip by the argument built around it.*

Two things are worth reading off this pair. First, the priors enter *only* through the unary factors ϕ_i of Figure 6, which are identical in both panels; the entire $0.590 \to 0.494$ drop is produced by the ψ_r, so it is attributable to arrangement alone. Second, a claim being true of the world does not protect it:



(a) **Incoherent** (base). Two EXCLUSIVE conflicts (dashed): $a_7 \leftrightarrow a_{10}$ is *unresolved*; $a_{11} \leftrightarrow a_{12}$ is a concession the holding a_{13} (green) resolves. $\mathrm{mm} = 0.607$, $\log Z = -9.64$.

(b) **Coherent**. The identical support structure with both conflict edges removed; a_{10} and a_7 are no longer suppressed. $\mathrm{mm} = 0.620$, $\log Z = -8.25$, $\mathrm{lp} = 1.00$.

Figure 7: Relation graphs for the AeroParts recall report (§B). Solid blue: ENTAILMENT; teal: EQUIVALENCE; dashed red, double-headed: EXCLUSIVE. Line width grows with the relation weight p. **The two panels differ only in the two conflict edges**, so the $0.607 \to 0.620$ lift and the 1.39 nats of recovered mass are attributable to conflict structure alone.

coherence is a property of the joint distribution, and a_3 is suppressed by *how* it is argued for, not by what it says. This is the mechanism Remark 2 anticipates and the ordering pair of §G.3 exhibits at scale.

A caution on readouts. $\mathrm{LCS}_{\mathrm{lp}}$ is 0.194 for Response A and 0.267 for Response B, which inverts the order — correctly, and for a reason Eq. 8 makes plain: its references $\log Z_{\max}$ and $\log Z_{\min}$ are recomputed *per response* (-0.49 and -1.06 for A, -1.65 and -4.55 for B), so it reports how close a response comes to the ceiling of *its own* edge skeleton, not how the two responses compare. It is a within-response diagnostic. Raw $\log Z$ is comparable across responses and moves the right way, by 2.83 nats.

G.2 A SIXTEEN-CLAIM RECALL REPORT

The second example is a sixteen-claim aviation recall report, which exercises the conflict taxonomy more fully than the five-claim pair: it carries both an unresolved conflict and a conceded one. Its relation graph is twelve support edges — an entailment spine $a_1 \to a_2 \to \ldots \to a_6$ (shipment certified to spec, installation, entry into service, in-flight failures, regulatory probe, grounding order), branches to the pulled-from-service claim, the maintenance-log evidence, the tribunal's damages award and the regulator's bulletin, and one EQUIVALENCE edge — plus two conflicts. Every value below is exact, by 2^{16} enumeration, and the graph is reproduced edge-by-edge in Figure 7 with each edge's weight shown by line thickness.

Example 7 (Incoherent)**.** *The report asserts both that* no passengers or crew were harmed *and that a wire report* claimed three passengers died. *These are exhaustive alternatives rather than mere opposition: they cannot both hold and cannot both fail, so the relation is* EXCLUSIVE *with* $p = 0.93$ *— and nothing in the response resolves it. A second tension, between the airline's pilot-error argument and the final report's metallurgical-defect finding* ($p = 0.80$)*, is a* Concession *that the tribunal's holding does resolve, and Eq. 2 discounts it accordingly. The model returns* $\mathrm{LCS}_{\mathrm{mm}} = 0.607$ *with* $\log Z = -9.64$*. The diagnostic that localizes the incoherence is per-claim: the marginal of the claim on the losing side of the* unresolved *conflict is dragged* below its own prior*, which no aggregate score would reveal. Applying the concession discount alone — resolving the second tension while leaving the first — raises the score to* 0.612*, as C2 requires.*

Example 8 (Coherent)**.** *Removing the two conflict edges and changing nothing else leaves the identical support structure (Figure 7b). The score rises to* $\mathrm{LCS}_{\mathrm{mm}} = 0.620$ *with* $\log Z = -8.25$ *and* $\mathrm{LCS}_{\mathrm{lp}} = 1.00$*: the response is now as coherent as its own edge skeleton permits, which is what the ceiling of Eq. 8 means. The suppressed claim recovers to its prior.*

Because the two panels of Figure 7 differ *only* in the two conflict edges, the $0.607 \to 0.620$ lift and the 1.39 nats of recovered probability mass are attributable to conflict structure alone — which is the

property the ladder construction of §5.1 generalizes from one pair to a family. The full ladder over this family is monotone throughout: $0.586 < 0.607 < 0.612 < 0.613 < 0.620$ over the rungs *worse*, *base*, *concession-resolved*, *one-conflict-fixed*, *coherent*.[1]

G.3 A CLAIM-IDENTICAL ORDERING PAIR

This pair makes the order-sensitivity claim concretely. Two summaries of one appellate judgment are built from an *identical* set of eighteen claims — the same atom texts, one a permutation of the other — differing only in whether self-serving assertions precede or follow the findings that qualify them. The faithful ordering scores $\mathrm{LCS}_{\mathrm{mm}} = 0.95$ and the self-serving-first ordering 0.90. No per-claim factuality check can separate them, because every claim is byte-identical; what differs is which claims the response leaves standing unqualified at the point where it asserts them.

It is worth being exact about *how* the two scores differ, because Remark 2 might seem to forbid it: nothing in Eq. 1 references position, so an edit preserving the relation set cannot move any readout. The pair does not preserve the relation set. Mining is response-grounded (§3.5), so the relations the estimator draws are a function of the prose, not of the claim list: the faithful summary states each tension beside the truth or holding that settles it, which the estimator reads as a Concession the text resolves and Eq. 2 discounts; the adversarial summary leaves the same claims standing unqualified until a later holding overrides them, which reads as a live conflict. The claim set is what factuality sees and it is identical; the relation set is what coherence sees and it is not.

H k-ARY REPRESENTABILITY

This appendix proves Prop. 2 of §5.10 and quantifies the gap it describes.

Proof of Prop. 2. By Eq. 1, $\mathbb{P}(1,1,1) = Z^{-1} \prod_i \phi_i(1) \prod_{i<j} \psi_{ij}(1,1)$, a product of non-negative terms, which vanishes only if some term does. If $\phi_i(1) = 0$ then a_i is false almost surely and both pairs containing i have probability zero. If $\psi_{ij}(1,1) = 0$ then every world with $a_i = a_j = 1$ carries that factor, so $\mathbb{P}(a_i{=}1, a_j{=}1) = 0$ directly. A ternary factor that vanishes only at $(1,1,1)$ has neither consequence. □

The gap is quantitative as well as qualitative. With uniform priors, a ternary factor supported off $(1,1,1)$ gives $\mathbb{P}(1,1,1) = 0$ while leaving each pair jointly true with probability 0.143; the best pairwise attempt that drives $\mathbb{P}(1,1,1)$ below 10^{-4} leaves the pairs at 0.028, and a random search over 2×10^5 pairwise parameterizations found nothing better. The pairwise model can only suppress the bad world by also suppressing the good ones that share a $(1,1)$ cell with it. Every figure here is exact by 2^3 enumeration and reproduced by a released script and regression test, so the proposition is checked and not merely asserted.

On the three-claim arithmetic instance of §5.10 — a league of 356 teams forming 32 conferences of exactly 10 members — the readouts report the passage as coherent at $\mathrm{LCS}_{\mathrm{mm}} = 0.68$ and $\mathrm{LCS}_{\mathrm{lp}} = 1.00$. Prop. 2 says no reweighting of the pairwise factors could have helped. Of the two milder consequences named in §5.10, transitivity is the more visible: the model will happily hold $a \Rightarrow b$ and $b \Rightarrow c$ at high weight while leaving a and c unrelated, since it has no rule tying them.

I CORRESPONDENCE WITH MARKOV LOGIC

The coherence MRF has a natural first-order encoding (Richardson & Domingos, 2006), with $\mathbb{P}(\mathbf{a}) = \frac{1}{Z} \exp\big(\sum_k w_k n_k(\mathbf{a})\big)$ for $n_k(\mathbf{a})$ the number of true groundings of formula F_k. Evidence predicates $\mathit{Entail}(i,j)$, $\mathit{Contradict}(i,j)$, $\mathit{Equiv}(i,j)$ and $\mathit{Resolves}(h,i,j)$ carry the mined graph; $\mathit{Holds}(i)$ is the query predicate. Working out the correspondence is instructive in both directions.

[1] Treating both tensions as plain CONTRADICTION rather than EXCLUSIVE — the three-coupling model that omits the exhaustiveness distinction — gives $\mathrm{LCS}_{\mathrm{mm}} = 0.587$ and $\log Z = -9.75$ for Example 7, with Example 8 unchanged at 0.620 and -8.25. The distinction of §3.2 is worth 0.020 of score here, and the reason it matters more than that figure suggests is Table 27: with EXCLUSIVE conflicts, readout (b)'s conflict term is pinned at 1.000 and the readout fails outright.

I.1 THE NAIVE ENCODING IS A DIFFERENT MODEL

The obvious encoding gives each relation one weighted clause, e.g. $\mathit{Entail}(i,j) \wedge a_i \Rightarrow a_j$ with $w = \mathrm{logit}(p)$. This is *not* equivalent to Definition 1. Its induced pairwise factor is $[e^w, e^w, 1, e^w]$, which is not row-stochastic in the sense of Prop. 1: it rewards the vacuous $a_s = 0$ rows with e^w on *both* target states, so a clause is best satisfied by making its antecedent false. Brute force over the 2^{16} worlds of the recall report of §G.2 gives mean marginal support 0.455 under the naive MLN against 0.587 under the corresponding MRF, with per-claim marginals differing by up to 0.31. These are two different models, not two encodings of one.

The failure is sharpest at the mode. The all-false world satisfies every implication vacuously, so under the naive encoding the most probable configuration of any purely-implicational graph is the one in which the response asserts nothing — the maximum-a-posteriori reading of a response is that none of it holds. This is Prop. 3 appearing in a second guise, and it is the same lesson: an encoding that scores rule satisfaction rather than belief can be maximized by having nothing to satisfy.

I.2 A THREE-CLAUSE EXPANSION IS EXACTLY EQUIVALENT

Writing the log-potential in the general pairwise form $\ln \psi(a_s, a_t) = \alpha + \beta a_s + \gamma a_t + \delta\, a_s a_t$ and matching it against the with-priors tables of Table 26 gives, for π_s the family's fixed source-prior parameter and $\mathrm{L}(x) = \mathrm{logit}(x)$:

τ	α (unit)	β (source)	γ (target)	δ (conjunction)
ENTAILMENT	$\ln(1-\pi_s)$	$\ln \frac{1-p}{1-\pi_s}$	$\mathrm{L}(\pi_s)$	$+\mathrm{L}(p) - \mathrm{L}(\pi_s)$
CONTRADICTION	$\ln(1-\pi_s)$	$\ln \frac{p}{1-\pi_s}$	$\mathrm{L}(\pi_s)$	$-\mathrm{L}(p) - \mathrm{L}(\pi_s)$
EQUIVALENCE	$\ln p$	$\ln \frac{1-p}{p}$	$\ln \frac{1-p}{p}$	$+2\,\mathrm{L}(p)$

At the atom prior $\pi_s = \frac{1}{2}$ the $\mathrm{L}(\pi_s)$ terms vanish and the table reduces to $\gamma = 0$ with $\delta = \pm\mathrm{L}(p)$: there the conjunction weight *is* exactly $\pm\mathrm{logit}(p)$, the interaction the implication was meant to carry all along, and the unit and source clauses are the correction that makes each row a conditional. Away from $\frac{1}{2}$ the $\mathrm{L}(\pi_s)$ terms are load-bearing, which matters for the two-stage composition of §3.6, whose priors are factuality posteriors rather than uniform. With this expansion the ground Markov logic network reproduces the MRF exactly: at $\pi_s = \frac{1}{2}$, mean support 0.5866 and $\log Z = -9.751$ for both, with zero per-claim difference to machine precision; the agreement is verified against the brute-force oracle at priors on and off $\frac{1}{2}$.

We establish the equivalence for the three original couplings only. EXCLUSIVE and CO_NECESSITY are not tabulated here, so the claim "the MLN encoding reproduces the MRF" should be read as scoped to the three-coupling fragment. That is a real limitation for the recall report of §G.2, whose conflicts are EXCLUSIVE; the five-claim pair of §G.1 uses CONTRADICTION throughout and so falls inside the tabulated fragment.

I.3 WHICH VIEW TO KEEP, AND WHY BOTH

The reason to prefer the MRF for the measure itself is Remark 1: its pairwise factor family is closed at three parameters, so the inferential vocabulary cannot quietly grow as senses are added, and every relation type is guaranteed to be a 2×2 potential. The reason to keep the Markov logic view is that the rules our model cannot express are naturally written there — a transitivity rule $\mathit{Entail}(i,j) \wedge \mathit{Entail}(j,k) \Rightarrow \mathit{Entail}(i,k)$, a concession-cancellation rule $\mathit{Resolves}(h,i,j) \wedge a_h \Rightarrow \neg \mathit{Penalize}(i,j)$ replacing the fixed discount of Eq. 2, and a double-conflict rule — and their weights could be *learned* from labelled ladders rather than set by hand. That is the route past the two limitations §5.10 names, and it requires an MLN solver (MC-SAT for marginals, MaxWalkSAT for the mode) rather than the bucket-elimination stack used here.

J THE REIFIED COHERENCE NODE, WORKED

To make Eq. 7 concrete, take the three-claim subnet consisting of the unresolved casualty conflict of Example 7 together with one entailment feeding it: claims a_7, a_{10} joined by a conflict at $p = 0.93$,

and $a_4 \to a_7$ an entailment at $p = 0.65$, all priors $\frac{1}{2}$, with $\rho = \frac{1}{2}$. Writing out the eight-cell vote factor for each relation from Eq. 7 and enumerating the 2^4 worlds of (a_4, a_7, a_{10}, R) gives $\mathbb{P}(R{=}1) = 0.453$: below its prior, because both relations have violating worlds that carry appreciable mass, and the conflict's violating world $(1, 1)$ carries the most.

Two properties of the construction are worth noting. The $R = 0$ branch is flat, so $R = 0$ asserts nothing and the node cannot be driven up merely by adding relations that happen to be satisfied. And the factor is a *noisy* AND rather than a hard one: a relation with $p_r < 1$ leaves some mass on its violating worlds even when $R = 1$, so a single strongly-violated relation reduces $\mathbb{P}(R{=}1)$ without pinning it to zero. The reason we do not adopt the readout despite this reasonable behaviour is Table 27: it dips at the concession variant for the reason of Prop. 4, and it requires the free parameter ρ, violating R4.

K THE ESTIMATION PROMPTS, VERBATIM

The four prompts below are reproduced exactly as they are issued, extracted programmatically from the released source so that this appendix cannot drift from the code. Placeholders in double braces are substituted at call time; `{{sense_menu}}` is substituted at build time from one of the two menus in §K.5.

K.1 PROMPT A, FIRST GENERATION — DISCOURSE SENSE AND COUPLING

6,148 characters. Placeholders: `{{sense_menu}}`, `{{response}}`, `{{atom_a}}`, `{{atom_b}}`.

```

Instructions:
You are given the full RESPONSE a model produced, and two atomic claims, A and B, both taken FROM THAT RESPONSE, in their order of appearance (A comes before B). Your task is to decide the discourse/logical relation FROM A TO B, following the steps below.

IMPORTANT -- ground your decision in the response. Assert a coupling ONLY if the response ITSELF draws that connection between A and B (as written, or as a clear step in the author's argument/narrative). Do NOT assert a relation that is merely plausible in general but that the response does not actually make. If A and B both appear in the response yet the response draws no logical or discourse dependence between them, the answer is None.

1. Reason step by step, referring to the response: does the response present A as causing, enabling, providing evidence for, restating, elaborating, temporally preceding, contrasting with, or contradicting B? Is B a claim the response later withdraws or that a holding resolves? Consider the direction (A to B). If the response links A and B only indirectly through other claims, or not at all, that is None.

2. Name the DISCOURSE SENSE, one of: {{sense_menu}}.
   - Cause-Effect: A causes/leads to B. Effect-Cause: A is the effect, B its cause.
   - Evidence: A provides evidence for B. Condition: A is a condition for B.
   - Restatement: A and B assert the same thing. Instantiation: A is a general claim, B a specific instance (or vice versa).
   - Contrast: A and B are in opposition but NOT exhaustive (they need not cover all possibilities; both could conceivably be false).
   - Concession: A and B are in tension but the text concedes/resolves it ("although A, still B", or a holding settles it).
   - Alternative: A and B are EXHAUSTIVE competing options -- EXACTLY ONE holds (they are mutually exclusive AND together cover the possibilities: not both, and not neither). E.g. "no one was harmed" vs "three people died"; "the cause was pilot error" vs "the cause was a metallurgical defect".
   - Disjunction: AT LEAST ONE of A and B holds (they may both hold, but the response rules out neither being true) -- e.g. two supporting findings at least one of which must be present.
   - Precedence/Succession: A and B are ordered in time with no truth dependence.
   - None: the response draws no logical or discourse dependence between A and B.

3. Map the sense to a COUPLING, one of: entailment, contradiction, equivalence, exclusive, co_necessity, none.
   - Cause-Effect, Effect-Cause, Evidence, Condition, Instantiation -> entailment
   - Restatement -> equivalence
   - Contrast, Concession -> contradiction
   - Alternative -> exclusive       (exactly one of A, B is true)
   - Disjunction -> co_necessity     (at least one of A, B is true)
   - Precedence, Succession, None -> none
```

```
   Prefer "exclusive" over "contradiction" when the two claims are not just incompatible but
   ↪  EXHAUSTIVE (one of them must be true); prefer "contradiction" when they merely cannot
   ↪  both hold but could both be false.

4. Give your final answer as two bracketed tags on ONE line, sense first:
[sense=Cause-Effect] [coupling=entailment]
A JSON object {"sense":"Cause-Effect","coupling":"entailment"} is also acceptable.

Use the following examples to better understand your task.

Example 1 (the response makes the causal link):
RESPONSE: The company launched a flawed product last quarter. Reviewers panned it, returns
↪  spiked, and the company's stock price fell 15 percent over the same period.
A: The company launched a flawed product last quarter.
B: The company's stock price fell 15 percent last quarter.
1. Reasoning: the response presents the flawed launch as the head of a chain (panning,
↪  returns) that ends in the stock decline, so the response itself draws a causal link from
↪  A to B.
2. Discourse sense: Cause-Effect.
3. Coupling: A causing B is a positive inferential link, i.e. entailment.
4. Final answer:
[sense=Cause-Effect] [coupling=entailment]

Example 2 (both claims present, but the response draws NO connection -> None):
RESPONSE: The quarterly report was published in April. Separately, the annual audit was
↪  scheduled for December. The two processes are run by different teams and were not
↪  related this year.
A: The quarterly report was published in April.
B: The annual audit was scheduled for December.
1. Reasoning: both claims appear in the response, and one might imagine a reporting-to-audit
↪  link in general, but this response explicitly treats them as separate and unrelated. The
↪  response draws no dependence from A to B.
2. Discourse sense: None.
3. Coupling: no dependence the response asserts, i.e. none.
4. Final answer:
[sense=None] [coupling=none]

Example 3 (the response states an EXHAUSTIVE alternative -> exclusive):
RESPONSE: The official statement said no one was harmed in the incident. However, the
↪  coroner's report confirmed that three people died in the incident.
A: No one was harmed in the incident.
B: Three people died in the incident.
1. Reasoning: the response sets A and B against each other ("However, ...") and they cannot
↪  both be true; but they also cannot both be false -- either people were harmed or they
↪  were not -- so exactly one holds. This is exhaustive, not a mere contrast. No holding
↪  resolves it.
2. Discourse sense: Alternative.
3. Coupling: exactly one of A, B is true, i.e. exclusive.
4. Final answer:
[sense=Alternative] [coupling=exclusive]

Example 4 (at least one must hold -> co_necessity):
RESPONSE: The defect was caught in review: at least one of the two independent checks -- the
↪  vibration analysis or the metallurgical assay -- flagged it.
A: The vibration analysis flagged the defect.
B: The metallurgical assay flagged the defect.
1. Reasoning: the response asserts the defect WAS caught by at least one check, so A and B
↪  cannot both be false; but both could hold (both checks may have flagged it). This is a
↪  disjunction, not an exclusion.
2. Discourse sense: Disjunction.
3. Coupling: at least one of A, B is true, i.e. co_necessity.
4. Final answer:
[sense=Disjunction] [coupling=co_necessity]

Your task:
RESPONSE: {{response}}
A: {{atom_a}}
B: {{atom_b}}
```

K.2 PROMPT A, SECOND GENERATION — REBALANCED FOR PRECISION

6,361 characters. Same placeholders. The five changes relative to the first generation, and their motivation, are given in §3.5: few-shot balance from 3:1 related to 2:3; negative exemplars showing silence rather than explicit disavowal; a verifiable evidence span; the discriminating question asked first; and NONE given first position and equal prose. Transitivity is named as a non-relation.

```

```

```
Instructions:
You are given the full RESPONSE a model produced, and two atomic claims, A and B, both taken FROM THAT RESPONSE. Decide the discourse/logical relation FROM A TO B.

Most pairs of claims in a text are NOT related. They are simply near each other. Your default answer is None, and you should depart from it only when the response itself draws a dependence between A and B.

Three things that are NOT relations, however plausible they look:
  - Topical adjacency. A and B being about the same subject, or sitting in neighbouring sentences, is not a relation.
  - World knowledge. If the link is true in general but the response does not draw it, the answer is None.
  - Transitivity. If the response links A to some third claim C, and C to B, that does NOT make A related to B. Report only the links the text draws directly.

1. First ask the discriminating question: does the response ITSELF assert a dependence from A to B -- as written, or as an explicit step in the author's argument? Identify the words in the response that do it. If you cannot point to such words, the answer is None.

2. If and only if you can, name the DISCOURSE SENSE, one of: {{sense_menu}}.
   - None: the response draws no dependence between A and B. This is the common case: unrelated, merely adjacent, related only by world knowledge, or related only through a third claim.
   - Cause-Effect: the response says A causes/leads to B. Effect-Cause: A is the effect, B its cause.
   - Evidence: the response offers A as evidence/support for B.
   - Restatement: A and B say the same thing -- a paraphrase, a converse, or a restatement the text marks ("in other words", "equivalently", "that is").
   - Contrast: A and B are opposed but NOT exhaustive (both could be false).
   - Concession: A and B are in tension and the text concedes or resolves it ("although A, still B"; a holding settles it).
   - Alternative: A and B are EXHAUSTIVE competing options -- exactly one holds (not both, not neither).
   - Disjunction: AT LEAST ONE of A and B holds; both may.
   - Precedence: A and B are ordered in time with no truth dependence.

3. Map the sense to a COUPLING, one of: entailment, contradiction, equivalence, exclusive, co_necessity, none.
   - Cause-Effect, Effect-Cause, Evidence -> entailment
   - Restatement -> equivalence
   - Contrast, Concession -> contradiction
   - Alternative -> exclusive       (exactly one of A, B is true)
   - Disjunction -> co_necessity    (at least one of A, B is true)
   - Precedence, None -> none
   Prefer "exclusive" when the two claims are incompatible AND exhaustive; "contradiction" when they merely cannot both hold but could both be false.

4. Answer on ONE line, with three bracketed tags. The evidence must be a SHORT VERBATIM QUOTE COPIED FROM THE RESPONSE -- the words that draw the link. Do not quote claim A or claim B; quote the response. Use [evidence=none] when the sense is None.
[sense=Evidence] [coupling=entailment] [evidence="which is why the panel held"]

Use the following examples to better understand your task.

Example 1 (adjacent and on-topic, but the response draws no link -> None):
RESPONSE: The survey was administered in March to a sample of 400 households. Response rates in rural districts have declined over the past decade. The analysis weighted results by district population.
A: The survey was administered in March to a sample of 400 households.
B: Response rates in rural districts have declined over the past decade.
1. The two sentences are adjacent and both concern the survey, but the response never says the March administration bears on rural response rates. There are no words that draw a dependence.
2. Discourse sense: None.
3. Coupling: none.
4. [sense=None] [coupling=none] [evidence=none]

Example 2 (the response makes the causal link explicitly):
RESPONSE: The company launched a flawed product last quarter. Reviewers panned it, returns spiked, and as a direct result the company's stock price fell 15 percent over the same period.
A: The company launched a flawed product last quarter.
B: The company's stock price fell 15 percent last quarter.
1. The response presents the launch as the head of a chain ending in the stock decline, and marks it: "as a direct result".
2. Discourse sense: Cause-Effect.
3. Coupling: entailment.
4. [sense=Cause-Effect] [coupling=entailment] [evidence="as a direct result"]

Example 3 (true in general, but this response does not draw it -> None):
RESPONSE: Higher interest rates raise the cost of corporate borrowing. The firm opened three distribution centres last year. Its logistics costs fell by eight percent.
```

```
A: Higher interest rates raise the cost of corporate borrowing.
B: The firm opened three distribution centres last year.
1. One could argue borrowing costs affect expansion decisions, but that is world knowledge. This response states the two facts side by side and links neither to the other.
2. Discourse sense: None.
3. Coupling: none.
4. [sense=None] [coupling=none] [evidence=none]

Example 4 (the response states an exhaustive alternative -> exclusive):
RESPONSE: The two readings are mutually exclusive and exactly one must be right: either no one was harmed in the incident, or three people died in it.
A: No one was harmed in the incident.
B: Three people died in the incident.
1. The response sets A and B against each other and says exactly one must be right: "mutually exclusive and exactly one must be right: either ... or".
2. Discourse sense: Alternative.
3. Coupling: exclusive.
4. [sense=Alternative] [coupling=exclusive] [evidence="mutually exclusive and exactly one must be right"]

Example 5 (linked only through a third claim -> None):
RESPONSE: The alloy is chemically identical to the certified reference. It therefore meets the reference specification. Meeting that specification is a precondition for airframe use.
A: The alloy is chemically identical to the certified reference.
B: Meeting that specification is a precondition for airframe use.
1. The response links A to the middle claim (it meets the specification), and the middle claim to B. It does not link A to B directly; that would be transitivity, which is not a relation to report.
2. Discourse sense: None.
3. Coupling: none.
4. [sense=None] [coupling=none] [evidence=none]

Your task:
RESPONSE: {{response}}
A: {{atom_a}}
B: {{atom_b}}
```

K.3 PROMPT B, DEFAULT — SURROGATE-TOKEN CONDITIONAL STRENGTH

2,182 characters. Placeholders: `{{response}}`, `{{atom_a}}`, `{{atom_b}}`, `{{coupling}}` (twice). This is the one prompt that does not end in a newline: it terminates with `Answer:` including the trailing space, so that the model's next token is the surrogate whose logprobs are read. Reproducing it without that trailing space changes the tokenization of the answer position and therefore the estimate.

```

Instructions:
You are given the full RESPONSE, two claims A and B drawn from it, and a {{coupling}} relation that holds from A to B. Assuming A is TRUE, judge -- IN THE CONTEXT OF THIS RESPONSE -- whether the relation's implication about B is credible: at least plausible / more likely than not, NOT whether it is certain.

- entailment or equivalence: given A and how the response uses it, is B at least plausibly TRUE (more likely than not)?
- contradiction or exclusive: given A and how the response uses it, is B at least plausibly FALSE (more likely than not)? (For "exclusive", A and B are exhaustive alternatives, so A being true makes B false.)
- co_necessity: A and B are a pair of which at least one holds. Given the response rules out "neither", is it at least plausible that B holds when A does NOT?

Answer with a SINGLE WORD, the very first word of your reply: Yes or No.
- Answer "Yes" if the implication is credible/plausible (even if not certain).
- Answer "No" only if the implication is implausible or the response does not actually support it.
Do not output anything before the word Yes or No.

Example 1 (response supports a near-certain entailment):
RESPONSE: The new alloy is chemically identical to the certified reference alloy, so it meets the certified reference specification.
A: The new alloy is chemically identical to the certified reference alloy.
B: The new alloy meets the certified reference specification.
coupling: entailment
Answer: Yes

Example 2 (weak but plausible, and the response draws the link -- still Yes):
```

```
RESPONSE: The company launched a flawed product last quarter, and its stock price fell 15 percent over the same period.
A: The company launched a flawed product last quarter.
B: The company's stock price fell 15 percent last quarter.
coupling: entailment
Answer: Yes

Example 3 (clear contradiction stated by the response):
RESPONSE: The official statement said no one was harmed, but three people died in the incident.
A: No one was harmed in the incident.
B: Three people died in the incident.
coupling: contradiction
Answer: Yes

Your task:
RESPONSE: {{response}}
A: {{atom_a}}
B: {{atom_b}}
coupling: {{coupling}}
Answer:
```

K.4 PROMPT B, BASELINE — VERBALIZED CONDITIONAL STRENGTH

2,112 characters. Retained only for the comparison reported in §5; verbalized confidence is weakly calibrated (Xiong et al., 2024).

```


Instructions:
You are given the full RESPONSE and two claims A and B drawn from it. Assume claim A is TRUE and, IN THE CONTEXT OF THIS RESPONSE, estimate how strongly A determines B under a {{coupling}} relation from A to B.

- For an entailment or equivalence coupling: how likely is B to be TRUE given A?
- For a contradiction or exclusive coupling: how likely is B to be FALSE given A? (exclusive = A and B are exhaustive alternatives, so A true forces B false.)
- For a co_necessity coupling (at least one of A, B holds): how likely is B to be TRUE when A is FALSE?

Answer with a single probability in [0, 1] to two decimals, after one short sentence of justification. A near-certain link is close to 1.00; a merely plausible link is around 0.60-0.70. End your answer with the probability wrapped in brackets on its own, exactly in the form: [p=0.NN]

Example 1:
RESPONSE: The new alloy is chemically identical to the certified reference alloy, so it meets the certified reference specification.
A: The new alloy is chemically identical to the certified reference alloy.
B: The new alloy meets the certified reference specification.
coupling: entailment
Justification: chemical identity to a certified reference almost guarantees the specification is met, so B follows very strongly from A.
[p=0.95]

Example 2:
RESPONSE: The company launched a flawed product last quarter, and its stock price fell 15 percent over the same period.
A: The company launched a flawed product last quarter.
B: The company's stock price fell 15 percent last quarter.
coupling: entailment
Justification: a flawed product can plausibly drive a stock decline, but many other factors affect price, so the link is only moderate.
[p=0.65]

Example 3:
RESPONSE: The official statement said no one was harmed, but three people died in the incident.
A: No one was harmed in the incident.
B: Three people died in the incident.
coupling: contradiction
Justification: if no one was harmed, it is almost certain that B (three deaths) is false.
[p=0.93]

Your task:
RESPONSE: {{response}}
A: {{atom_a}}
B: {{atom_b}}
coupling: {{coupling}}
```

K.5 THE TWO SENSE MENUS

The full menu offers all twelve senses of Table 24 plus NONE:

```
Cause-Effect, Effect-Cause, Evidence, Condition, Restatement, Instantiation, Contrast,
↪  Concession, Alternative, Disjunction, Precedence, Succession, None
```

The restricted menu offers only the senses the evaluation corpus labels, which is the admissibility filter of §3.5 applied at elicitation time rather than after it:

```
Cause-Effect, Effect-Cause, Evidence, Restatement, Contrast, Concession, Alternative,
↪  Disjunction, Precedence, None
```

Note that the restricted menu omits Succession as well as Condition and Instantiation, leaving nine relation-bearing senses. Succession is dropped because it is the mirror of Precedence and both compile to NONE, so offering both invites an arbitrary choice on a relation that produces no factor either way. The second-generation Prompt A's body is written against this restricted menu; when it is run with the full menu, three senses are offered with no gloss and no coupling mapping in the prompt body, and we report which menu each source used.

L SERIALIZATION AND THE VARIABLE-ORDERING TRAP

One implementation detail is worth recording because it is silent when wrong. Networks are serialized in UAI format, whose variables are identified by integer index; the index order is the sort order of $(\text{cardinality}, \text{name})$, and with uniformly binary claim variables that reduces to *lexicographic order by name*. So a_{10} sorts before a_2. Inference returns marginals by index, and mapping them back by any other convention — source order, numeric order — silently permutes marginals across claims. The score is unchanged, since Eq. 5 is a mean, but every per-claim diagnostic is then attributed to the wrong claim, and the "claim dragged below its prior" readout points at an innocent claim. The auxiliary variables of §D.4 are named with a prefix that sorts after all claim names for the same reason, so that adding them cannot shift the claim indices.